EUROPEAN COMMISSION
European Health and Digital Executive Agency
Director

AMENDMENT Reference No AMD-101016444-7

Grant Agreement number: 101016444 — FLEXIBLE MID-INFRARED PHOTONIC SOLUTIONS FOR RAPID FARM-TO-FORK SENSING OF FOOD CONTAMINANTS (PHOTONFOOD)

The parties agree to amend the Grant Agreement as follows ('**Amendment**'):

1. Change of Annex 1 (description of the action)

Annex 1 is changed and replaced by the Annex 1 attached to this Amendment.

All other provisions of the Grant Agreement and its Annexes remain unchanged.

This Amendment **enters into force** on the day of the last signature.

This Amendment **takes effect** on the date on which the amendment enters into force, except where a different date has been agreed by the parties (for one or more changes).

Please inform the other members of the consortium of the Amendment.

SIGNATURES

For the coordinator For the Agency

Enclosures:

Annex 1

EUROPEAN COMMISSION
European Health and Digital Executive Agency

The Director

ANNEX 1 (part A)

Research and Innovation action

NUMBER — 101016444 — PHOTONFOOD

Table of Contents

1.1. The project summary....3
1.2. The list of beneficiaries....4
1.3. Workplan Tables - Detailed implementation....5
1.3.1. WT1 List of work packages....5
1.3.2. WT2 List of deliverables....6
1.3.3. WT3 Work package descriptions....11
Work package 1....11
Work package 2....15
Work package 3....18
Work package 4....21
Work package 5....24
Work package 6....28
Work package 7....32
Work package 8....37
1.3.4. WT4 List of milestones....42
1.3.5. WT5 Critical Implementation risks and mitigation actions....43
1.3.6 WT6 Summary of project effort in person-months....45
1.3.7. WT7 Tentative schedule of project reviews....46

1.1. The project summary

Project Number [1]	101016444	Project Acronym [2]	PHOTONFOOD

One form per project

General information	
Project title [3]	FLEXIBLE MID-INFRARED PHOTONIC SOLUTIONS FOR RAPID FARM-TO-FORK SENSING OF FOOD CONTAMINANTS
Starting date [4]	01/01/2021
Duration in months [5]	48
Call (part) identifier [6]	H2020-ICT-2020-2
Topic	ICT-37-2020 Advancing photonics technologies and application driven photonics components and the innovation ecosystem
Fixed EC Keywords	Photonics
Free keywords	interband cascade light emitting diodes interband cascade lasers quantum cascade lasers infrared waveguides

Abstract [7]

Analytical techniques for the measurements of chemical and microbial contaminations along the food chain require detection levels in the ppb range. A direct measurement of contaminants at theses concentrations in a food matrix is not possible by any photonics principle. PHOTONFOOD aims to overcome this barrier by developing an integrated solution that combines innovations in smart paper-based sample treatment, mid-infrared (MIR) sensing and advanced data analysis. Mid-infrared (MIR) spectroscopy has proven to be the most reliable and broadly applicable spectroscopic method for detection, characterization and quantification of chemical and microbial contamination.
To transform MIR sensing from existing lab solutions into a portable solution to broad usage in the food chain, PHTONFOOD aims to develop novel infrared light sources, specifically interband cascade light emitting diode (IC-LED), and interband and quantum cascade lasers (ICL/QCL). The light sources will be combined with sophisticated waveguide technology and 3D-paper microfluidics. The project will develop a solution consisting of: (1) A mid-fidelity (MI-FI) device with an envisioned low-price range that can be used for daily monitoring, and (2) a medium-price range high-fidelity (HI-FI) device for reference analysis and accreditation. The solution will be validated and demonstrated in real scenarios for fungal and water mould (oomycetes) contamination, mycotoxins in wheat, nuts, dried fruits and aquaponic-based herbs, as well as pesticides and antibiotics in aquaponic-based herbs. The research will be performed by a multidisciplinary and highly acknowledged research team in collaboration with leading mycotoxin analysis laboratories, an aquaponics farm, a distributor of dry foods and farms with on-site food processing and vending. Strategies for transferring the obtained data and integrating it into data platforms for tracing food contamination, improving food quality and reducing waste will be developed.

1.2. List of Beneficiaries

Project Number [1]	101016444	Project Acronym [2]	PHOTONFOOD

List of Beneficiaries

No	Name	Short name	Country	Project entry date[8]	Project exit date
1	NORGES MILJO-OG BIOVITENSKAPLIGE UNIVERSITET	NMBU	Norway		
2	UNIVERSITAET ULM	UULM	Germany		
3	UNIVERSITAET FUER BODENKULTUR WIEN	BOKU	Austria		
4	WAGENINGEN UNIVERSITY	WU	Netherlands		
5	NEMZETI ELELMISZERLANC-BIZTONSAGI HIVATAL	NEBIH	Hungary		
6	NANOPLUS NANOSYSTEMS AND TECHNOLOGIES GMBH	NANOPLUS	Germany		
7	ROMER LABS DIVISION HOLDING GMBH	ROMER	Austria		
8	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	IRIS	Spain		
9	ACCELOPMENT SCHWEIZ AG	accelCH	Switzerland		
10	BIGH ANDERLECHT	BIGH	Belgium		
11	SEEBERGER GMBH	Seeberger GmbH	Germany		
12	BAMA GRUPPEN AS	BAMA Gruppen AS	Norway		
13	HAHN-SCHICKARD-GESELLSCHAFT FUER ANGEWANDTE FORSCHUNG E.v.	HS	Germany		

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]		Start month[12]	End month[13]
WP1	Technology acceptance and techno-economic evaluation	5 - NEBIH		1	48
WP2	Paper microfluids	4 - WU		1	48
WP3	Novel MIR light sources	6 - NANOPLUS		1	30
WP4	Modular sensor solution	2 - UULM		1	30
WP5	Data analysis and software development	1 - NMBU		1	48
WP6	Demonstration and validation	3 - BOKU		1	48
WP7	Communication, dissemination and exploitation	9 - accelCH		1	48
WP8	Coordination and management	1 - NMBU		1	48
		Total			

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Report on stakeholder and user requirements	WP1	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D1.2	Overview on technical and user experiences	WP1	5 - NEBIH	Report	Confidential, only for members of the consortium (including the Commission Services)	40
D1.3	Report on cost assessment	WP1	7 - ROMER	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D1.4	Report final user requirements, technology accept., market potential	WP1	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D2.1	Deliverable report on sampling, extraction and selective purification approaches for one target contaminant (DON)	WP2	4 - WU	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.2	Deliverable report on the development of an interface between a fully integrated paper microfluidic device and optical components	WP2	4 - WU	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D2.3	Diversification of the paper-based microfluidics to at least a second target contaminant in a second commodity finalized.	WP2	4 - WU	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D3.1	Advanced IC-LEDs for sensor integration (hardware)	WP3	6 - NANOPLUS	Demonstrator	Confidential, only for members of the consortium (including the	18

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D3.2	Monolithic tuneable lasers for sensor integration (hardware, report)	WP3	6 - NANOPLUS	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	26
D4.1	Thin-film waveguide-based sensing interface for HI-FI analyser	WP4	2 - UULM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	18
D4.2	Performance test report for MI-FI analyzer	WP4	2 - UULM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	30
D4.3	Performance test report for HI-FI analyzer	WP4	2 - UULM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24
D5.1	Pre-processing algorithms for HI-FI and MI-FI spectra	WP5	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D5.2	Classification and calibration models for MI-FI and HI-FI instruments ready for fungi and mycotoxins in all matrices	WP5	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D5.3	Calibration and classification models validated	WP5	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	48
D5.4	Digital Global Control System (GCS) Technical Manual	WP5	8 - IRIS	Other	Confidential, only for members of the consortium (including the Commission Services)	36

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.5	Digital Global Control System (GCS) User Manual	WP5	8 - IRIS	Other	Confidential, only for members of the consortium (including the Commission Services)	40
D6.1	Report on preliminary comparability study	WP6	3 - BOKU	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D6.2	Report on completed validation studies	WP6	3 - BOKU	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D6.3	Report on completed demonstration exercises	WP6	5 - NEBIH	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D6.4	Evaluation of collected feedback data (report)	WP6	5 - NEBIH	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D6.5	Report on ring trial (interlaboratory comparison study)	WP6	3 - BOKU	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D7.1	PHOTONFOOD website	WP7	9 - accelCH	Websites, patents filing, etc.	Public	2
D7.2	Communication and Dissemination Plan (CDP)	WP7	9 - accelCH	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D7.3	Interim communication and dissemination activities report	WP7	9 - accelCH	Report	Confidential, only for members of the consortium (including the	24

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D7.4	Policy guidelines	WP7	5 - NEBIH	Report	Public	36
D7.5	Live demonstrations	WP7	6 - NANOPLUS	Demonstrator	Public	42
D7.6	Roadmap for exploitation	WP7	13 - HS	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D7.7	Final Scientific Symposium	WP7	9 - accelCH	Other	Public	48
D8.1	Next stage plans and internal progress reports	WP8	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.2	Interim Status Report	WP8	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D8.3	Communication framework between partners	WP8	1 - NMBU	Other	Public	24
D8.4	Six month progress reports, mid term reports and final report	WP8	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D8.5	Organize initial meetings, mid term meetings and final meetings	WP8	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	48
D8.6	Report on gender, societal and ethical issues of exploitation	WP8	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D8.7	Data management plan	WP8	1 - NMBU	ORDP: Open	Confidential, only for members of the consortium	6

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				Research Data Pilot	(including the Commission Services)	

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	5 - NEBIH
Work package title	Technology acceptance and techno-economic evaluation		
Start month	1	**End month**	48

Objectives

Objective: To conduct comprehensive research on user expectations, user experience and technology acceptance for a portable, easy-to-use, broadly adaptable photonics device. The specific objectives are:
• Identify relevant stakeholders, explore user requirements and expectations towards the innovative multi-purpose photonic sensor solution.
• Organize the collection of user experience during the validation and demonstration process in WP6.
• Conduct comparative cost assessment for the MI-FI and HI-FI devices and reference methods considering realistic conditions.
• Assess user experience and technology acceptance among target users and stakeholders.

Description of work and role of partners

WP1 - Technology acceptance and techno-economic evaluation [Months: 1-48]
NEBIH, NMBU, UULM, BOKU, WU, NANOPLUS, ROMER, IRIS, Seeberger GmbH, BAMA Gruppen AS
Task 1.1 Stakeholder expectations and user requirements survey (M1-M42) (lead partner: NMBU; participants: BOKU, ROMER, NEBIH, Seeberger, BAMA, WU). Since the exact apparatus of the portable PHOTONFOOD platform cannot be fully defined before the beginning of the project, we will follow an agile approach to user requirements. A kick-off workshop involving (a) the PHOTONFOOD WP leaders and (b) representatives of the prospective user organisations involved in the project (ROMER, NEBIH, Seeberger) will be organised at the beginning of the project (M1) to define a questionnaire for a stakeholder survey. The stakeholder survey will consist of an online questionnaire and a structured follow-up telephone interview. It will be conducted in Austria, Germany, Hungary, the Netherlands and Norway, covering 15 organisations per country from different levels of the food chain (3 from agriculture, 3 processors, 3 retailers, 4 labs and 2 border inspection points). At the project kick-off meeting, the PHOTONFOOD task leaders will develop an initial architecture envisioning. The process will then be iterated throughout the lifetime of the project on an annual basis. The component-specific requirements will be continuously updated within the tasks responsible for component development (WP2 to WP5). The system-level and platform-level requirements will be updated at workshops held during the annual project meetings in M8, M18, M25 and M35. Based on the stakeholder and user requirements, deliverable D1.1: Report on stakeholder and user requirements (M24) will be compiled.
Task 1.2 Collection and analysis of technical and user experience during validation and demonstration (M24-M34) (lead partner: NEBIH; participants: BOKU, NMBU, ROMER, Seeberger, BAMA). During the demonstration phase (when WP6 is active), system-level and platform-level requirements and actual experience will be collected on a continuous basis in close collaboration between the members of the WP6 team and the users and stakeholders involved in the testing and validation. The parameters measured in WP1 during the validation phase of the innovative photonic device (and in parallel with the reference methods) will include: time requirements, cost requirements, level of skill needed for operation, ergonomics, and user experience. User experience data will be gathered from the three main user groups involved in the demonstration activities (industry, retail and food inspection services), using qualitative (observations and interviews) and quantitative methods (user surveys). A standard usability framework model will be adopted (UTAUT: Unified theory of acceptance and use of technology) which allows a highly structured analysis of such mixed-methods data. The experiences will be analysed in a systematic way, presented to the WPs engaged in technology development (WP2 to WP5).
Task 1.3 Cost assessment (M24-M34) (lead partner: ROMER; participants: BOKU, NEBIH, NMBU, Seeberger). Based on the analysis of the cost elements recorded during the demonstration phase in Task 1.2, a comparative unit cost assessment will be conducted against the currently available reference methods. This part is of critical importance for the feasibility of the whole technology development. Historical data will be used to judge cost reduction effect of production scaling up, but the initial costs should remain within a reasonable threshold (the unit cost of one sample should remain under 3€ for mycotoxin testing even with the prototype). The calculation will include a Monte Carlo simulation of cost schemes for the IPD and the best available reference method. Results of this task will be discussed with the WP leaders dealing with technology development (WP2 to WP5).

Task 1.4 Technology acceptance and market potential (M36-M48) (lead partner: NMBU; participants: NEBIH, ROMER, WU). In the final task of WP1, the technology acceptance and future market potential will be assessed in key customer segments. Like in the previous tasks, a combination of online questionnaires and telephone follow-up interviews will be used for data collection. The target respondents will include: (a) senior executives in the industry, commodity distribution and retail who have overall responsibility for quality and safety in their respective organizations, and (b) senior representatives of food safety authorities with overall responsibility for standards and detection methods. The target sectors will not only include the value chains covered in the demonstration in WP6 but also other value chains where related quality and safety assessment problems occur. This will enable us to assess the further market and business development potential of the PHOTONFOOD technologies in other value chains. Key assessment indicators will include technology acceptance, customer expectations, upskilling/training requirements, probability of adoption and willingness to pay. Again, we will combine qualitative and quantitative methodologies in the assessment.
Role of partners: NEBIH will be leading this WP and is responsible for conducting a stakeholder survey in Hungary, collecting and analysing user experience, supporting the cost assessment and contributing to the assessment of technology acceptance. NMBU will be responsible for conducting the same activities in Norway, and will lead the assessment of technology acceptance and market potential. ROMER will be responsible for leading the cost assessment and will contribute to the collection of user requirements and experience and will support the assessment of market potential. BOKU will be responsible for conducting a stakeholder survey in Austria and will contribute to the collection of user requirements and experience, also to the cost assessment task. Seeberger and BAMA will contribute to the collection of user requirements and experience, and Seeberger will also play a part in the cost assessment task. NP, IRIS, WU and UULM will provide technical and cost related information on the photonic devices, as well as ensuring the efficient information flow and feedback loop between WP1 and the technology WPs (WP2 to WP5) through the whole project. Furthermore, UULM will be responsible for conducting a stakeholder survey in Germany, and WU in the Netherlands.

Participation per Partner

Partner number and short name	WP1 effort
1 - NMBU	
2 - UULM	
3 - BOKU	
4 - WU	
5 - NEBIH	
6 - NANOPLUS	
7 - ROMER	
8 - IRIS	
11 - Seeberger GmbH	
12 - BAMA Gruppen AS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Report on stakeholder and user requirements	1 - NMBU	Report	Confidential, only for members of the consortium (including	24

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	
D1.2	Overview on technical and user experiences	5 - NEBIH	Report	Confidential, only for members of the consortium (including the Commission Services)	40
D1.3	Report on cost assessment	7 - ROMER	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D1.4	Report final user requirements, technology accept., market potential	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D1.1 Report on stakeholder and user requirements (NMBU, M24);
D1.2 Overview on technical and user experiences (NEBIH, M40);
D1.3 Report on cost assessment (ROMER, M36);
D1.4 Overview on final user requirements, technology acceptance and market potential (NMBU, M48)

D1.1 : Report on stakeholder and user requirements [24]

Stakeholders and users will be identified, an report on stakeholder and user requirements will be given.

D1.2 : Overview on technical and user experiences [40]

Technical and user experiences will be collected during the validation and demonstration process and an overview on technical and user experiences will be given.

D1.3 : Report on cost assessment [36]

A comparative cost assessment for the MI-FI and HI-FI devices and reference methods will be performed and reported.

D1.4 : Report final user requirements, technology accept., market potential [48]

The technology acceptance, market potential and final user requirements will be summarized in a final report.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.

Work package number [9]	WP2	Lead beneficiary [10]	4 - WU
Work package title	Paper microfluids		
Start month	1	End month	48

Objectives

To develop an integrated sample handling tool that will allow the user-friendly analysis of food contaminants using MIR detection. The specific objectives are:

• To develop paper microfluidic sampling, extraction and pre-treatment for food contaminants from hard and soft commodities.

• To develop a functionalized cartridge for the paper microfluidic devices that allows robust and user-friendly interfacing with spectroscopic detection.

Description of work and role of partners

WP2 - Paper microfluids [Months: 1-48]

WU, NMBU, UULM, BOKU, NEBIH, IRIS

Task 2.1 Paper microfluidic sampling and extraction (M1-M34) (lead partner: WU, participants: NMBU, UULM, BOKU, NEBIH, IRIS). In this task, solutions will be developed for allowing the entire sample-handling chain to be executed not only on-site, but by non-experts as well. This requires first of all a strategy towards the acquisition of sample. In the case of hard samples, such as nuts, an extract needs to be prepared. In previous work (FoodSmartphone), an extraction module was developed for consumer testing in a cookie matrix. Further development in this WP will be required to adapt such a strategy towards different commodities. A dip-stick type sampler will then be developed to bring the sample onto the paper microfluidic device. On the other hand, when soft or liquid samples are tested, a different sampling strategy is required, where a swab type sampler can be used, which then directly feeds into the paper microfluidic device. To improve the extraction efficiency of the contaminants, while reducing the co-extraction of unwanted matrix constituents, the cellulose fibers can be chemically modified to allow specific chemical interaction with those contaminants.

Task 2.2 Paper microfluidic pre-treatment/preconcentration (M1-M48) (lead partner: WU, participants: NMBU, UULM, BOKU, NEBIH, IRIS). Once the contaminants are extracted from their matrix through the approaches in task 2.1, further purification and concentration will be required before MIR spectroscopic analysis. In order to achieve selective purification, several routes will be considered. Firstly, the use of immunochemistry will be explored. Combining immunocapture with paper microfluidic functionality is an attractive approach, as the location for the preconcentration can be precisely chosen to match the required dimensions for readout. Secondly, the use of chemical surface modification will be further explored. Paper can readily be modified with a plethora of functional groups, which allows for solid-phase extraction on paper. While this is less specific than immunocapture, it does allow for the devices to be more easily adapted for different applications. Thirdly, an innovative method in paper microfluidics will be developed through adapting a liquid-liquid extraction on a paper-based platform, as demonstrated in the initial work published by members of the consortium43. Through application of multilayered paper microfluidic devices, which are specifically modified, immiscible solvents can be preferentially retained in selected layers, which allows for the construction of a multilayered liquid-liquid extraction system43. At the end of this task, the most appropriate strategy will be selected for further developments in PHOTONFOOD.

Task 2.3 Integration and interfacing with 3D printing (M1-M48) (lead partner: WU, participants: NMBU, UULM, BOKU, NEBIH, IRIS). While it is crucial for any truly user-friendly on-site analysis to address both sample extraction and enrichment, it is equally important to consider operational simplicity and system integration. Therefore, this task focuses on the design and development of a cartridge that will contain all the elements developed in tasks 2.1 and 2.2. Here, 3D printing is exceptionally useful since it allows very fast prototyping, leading to iterative design of devices in a short period of time. Moreover, it will facilitate complex paper microfluidic device layout, such as stacking of several layers for the multilayer liquid-liquid extraction. In addition, the cartridge will shield vulnerable parts of the device, by exposing only the sample acquisition and read-out parts of the device. This is beneficial since it designates these parts as such for the end-users. Finally, protocols for operational simplicity and clarity will be delivered.

Role of partners: WU will be leading this work package and is responsible for the realization of the integrated sample handling tool. BOKU will provide assistance for developing appropriate sample pre-treatment. NMBU and IRIS will support the data acquisition and analysis. UULM will support the coupling of the sampling device with spectroscopic analysis. NEBIH will participate in the evaluation of the user acceptance of the paper-based microfluidics.

Participation per Partner

Partner number and short name	WP2 effort
1 - NMBU	
2 - UULM	
3 - BOKU	
4 - WU	
5 - NEBIH	
8 - IRIS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.1	Deliverable report on sampling, extraction and selective purification approaches for one target contaminant (DON)	4 - WU	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.2	Deliverable report on the development of an interface between a fully integrated paper microfluidic device and optical components	4 - WU	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D2.3	Diversification of the paper-based microfluidics to at least a second target contaminant in a second commodity finalized.	4 - WU	Report	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D 2.1 Deliverable report on paper microfluidic sampling, extraction and selective purification approaches for one target contaminant (DON) in a target commodity (wheat) (WU, M24);
D 2.2 Deliverable report on the development of an interface between a fully integrated paper microfluidic device and optical components (WU, M42);
D 2.3 Diversification of approach to at least a second target contaminant in a second soft sample commodity finalized (WU, M48).

D2.1 : Deliverable report on sampling, extraction and selective purification approaches for one target contaminant (DON) [24]

Deliverable report on paper microfluidic sampling, extraction and selective purification approaches for one target contaminant (DON) in a target commodity (wheat) .

D2.2 : Deliverable report on the development of an interface between a fully integrated paper microfluidic device and optical components [42]

Deliverable report on the development of an interface between a fully integrated paper microfluidic device and optical components.

D2.3 : Diversification of the paper-based microfluidics to at least a second target contaminant in a second commodity finalized. [48]

When the diversification of the paper-based microfluidics to at least a second target contaminant in a second commodity is finalized, a report will be prepared.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.
MS3	Portable Microfluidics Pre-prototype	4 - WU	24	Evaluation of specifications and design.
MS4	Portable Microfluidics Prototype	4 - WU	36	Validation with MI-FI & HI-FI prototypes in lab validation

Work package number [9]	WP3	Lead beneficiary [10]	6 - NANOPLUS
Work package title	Novel MIR light sources		
Start month	1	End month	30

Objectives

To develop dedicated interband cascade (IC) and quantum cascade (QC) gain materials forming the basis for advanced light emitting diode (IC-LEDs) and monolithically tunable laser light sources. Specifically, low-cost MIR LEDs will be developed emitting in the 1500-1800 cm-1 spectral window for detecting bio contaminants in a medium-fidelity (MI-FI) sensing device, whereas monolithic widely tunable lasers will be implemented in a high-fidelity (HI-FI) analyzer based on the same spectral range. The specific objectives are:

- To develop tunable ICL/QCL MIR lasers for high precision measurements.
- To develop IC-LED light sources for low cost contamination detection.

Description of work and role of partners

WP3 - Novel MIR light sources [Months: 1-30]
NANOPLUS, NMBU, UULM, IRIS
Task 3.1 Growth of cascaded epitaxy for low cost MIR ILEDs and high-performance tunable lasers (M1-M24) (lead partner: NP, participants: UULM, IRIS)
In this task, the required cascaded layer structures for IC-LEDs and cascaded tunable MIR lasers will be realized. In this context, dedicated band structure engineering is performed by means of specialized solvers of the Schrödinger equation using e.g. k·p methods. This band structure design is complemented by means of optical simulations of the resulting layer stacks with respect to the actual devices performance. Based on these simulations, and based on the input from WP1 and WP4, it will be decided if either interband cascade (ICL) or quantum cascade laser (QCL) structures are more suitable for the realization of the high performance monolithic tunable laser devices. The epitaxially grown multilayer structures will be routinely characterized by high-resolution X-ray diffraction, secondary ion mass spectroscopy and secondary electron microscopy images. Based on this analysis, the growth parameters will be optimized to fabricate high quality material that is required for the cascaded devices. In order to aid the design and to achieve the required broad band emission in the 1500-1800 cm-1 spectral range, gain measurements will be performed to iteratively optimize the performance of the epitaxy. Broad area lasers and related test structures will be processed to be able to rapidly forecast the device performance. One of the main objectives in this context will be the detailed band structure analysis versus the layer structure modifications to achieve fast progress.
Task 3.2 MIR IC-LED fabrication for MI-FI analyzers (M1-28) (lead partner: NP, participants: UULM, NMBU)
Task 3.2 is dedicated to the design and realization of low-cost IC-LEDs in the 1500-1800 cm-1 spectral region. This task is the main focus within this work package due to the intrinsic capabilities of scaling up the technology to the consumer market. Based on the results and layer structures from Task 3.1, optical simulations will be performed to investigate an optimal chip design and the simulations will be iteratively up-dated according to the obtained results. One key challenge will be to maximize the out-coupling of the light generated in the active zone which is confined to the chip by total internal reflection due to the high refractive index of the semiconductor material. In this context, specific out-coupling structures such as coatings, micro lenses and surface treatment schemes will be investigated and iteratively optimized. The chip layout will be transferred into the epitaxial material by means of suitable lithography, dielectric- and metal-layer deposition and removal steps. The readily processed chips will be pre-characterized on wafer level prior to singulation. In order to be used in a sensor, the IC-LED chips have to be appropriately packaged in a housing suitable for low-cost, high volume applications. Therefore, a SMD type of package is the first choice. Thermal simulations will be made for ensuring the best possible heat management of the chips and the SMD package to enable superior performance under application conditions. Application grade IC-LEDs will be characterized and delivered to WP4 for the integration into the MI-FI sensor. Special care will be taken with respect to the scalability of all steps, in particular the device characterization to ensure low cost devices. Lifetime and environmental tests that proof the robustness under operation conditions will complete the task.
Task 3.3 MIR tunable laser fabrication for HI-FI analyzers (M4-M30) (lead partner: NP, participants: UULM, NMBU).
The initial simulations results and the decision regarding the optimal epitaxy structure (interband or quantum cascade respectively) from Task 3.1 will be the starting point of this task. Further optical simulations will be performed to define the initial chip design that enables the required tuning range in the 1500-1800 cm-1 spectral range. Suitable semiconductor structuring processes, such as optical and high precision electron beam lithography processing steps, will be investigated and iteratively optimized to realize the tunable monolithic lasers in combination with dielectric- and

metal-layer deposition and removal steps. An extremely thorough optical characterization with respect to the achievable wavelength and side-mode suppression ratios at the pre-defined operation points will be required. This is due to the nature of the device design, based on the Vernier effect, that leads to a high degree of freedom in terms of laser operation. Those data will be analyzed in depth and fed back to the next iteration of chip design and processing. The integration of the lasers in a suitable housing and the definition of suitable electronics will complete the task.
Role of partners: NP will be leading this work package and is responsible for the realization of the IC-LED light source of the MI-FI device as well as the monolithic tunable MIR laser based on a cascaded laser scheme for the HI-FI device. UULM will contribute to the specifications and the assessment of the performance data of the light sources. NMBU will participate in the evaluation of the signal achieved by the light sources with respect to the signal to noise ratio required.

Participation per Partner

Partner number and short name	WP3 effort
1 - NMBU	
2 - UULM	
6 - NANOPLUS	
8 - IRIS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Advanced IC-LEDs for sensor integration (hardware)	6 - NANOPLUS	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	18
D3.2	Monolithic tuneable lasers for sensor integration (hardware, report)	6 - NANOPLUS	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	26

Description of deliverables

D3.1 Advanced IC-LEDs hardware ready for MI-FI integration (hardware, report) (NP, M18);
D3.2 Monolithic tunable hardware ready for HI-FI integration (hardware, report) (NP, M26)

D3.1 : Advanced IC-LEDs for sensor integration (hardware) [18]

When the advanced IC-LEDs for the MI-FI analysers are developed and when they are ready for sensor integration, a report will be prepared and the hardware will be presented.

D3.2 : Monolithic tuneable lasers for sensor integration (hardware, report) [26]

When the monolithic tuneable lasers for the HI-FI analysers are developed and when they are ready for sensor integration, a report will be given and the hardware will be presented.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.
MS5	IC-LEDs optical components	6 - NANOPLUS	18	Evaluation of the spectral target coverage (1800-1500 cm-1).
MS6	Monolithic tuneable lasers	6 - NANOPLUS	28	Evaluation of the spectral target coverage (1800-1500cm-1).
MS7	Five handheld MI-FI analyzer prototypes for demonstration	2 - UULM	30	Evaluation of MI-FI analyzer prototype design and performance on real samples.
MS8	One portable HI-FI analyzer prototype for demonstration	2 - UULM	24	Evaluation of HI-FI analyzer prototype design and performance on real samples

Work package number [9]	WP4	Lead beneficiary [10]	2 - UULM
Work package title	Modular sensor solution		
Start month	1	End month	30

Objectives

To develop a portable and field-deployable (prototype) MI-FI and HI-FI analyzers combining the MIR LEDs developed in WP 3 with single-bounce diamond ATR elements, a grating, and a line array detector (MI-FI device), and the ICL/QCL light source developed in WP3 with advanced thin-film nanocrystalline diamond (NCD) waveguides and a single-element detector (HI-FI device) for in-field fungal infection analysis. The specific objectives are:

- To prototype a HI-FI MIR photonic sensing device and integrate ICL/QCL light sources.
- To prototype MI-FI MIR photonic sensing devices and integrate MIR IC-LED light sources.

Description of work and role of partners

WP4 - Modular sensor solution [Months: 1-30]
UULM, NMBU, BOKU, WU, NEBIH, NANOPLUS, IRIS
Task 4.1 Thin-film nanocrystalline diamond waveguides (M1-M18) (lead partner: UULM; participants: NP). Task 4.1. is focused on the development of thin-film NCD chips (5x10 mm; actual waveguide thickness: 10-25 µm) readily fitting the MIR-GuideR ATR sensing system. The chips will be characterized and optimized in iterative cycles for maximum performance (i.e., MIR radiation throughput and intensity of the evanescent field). In addition, the MIR-GuideR system will be adapted to accommodate the smart paper strip developed in WP 2 for analyzing enriched/separated samples (i.e., ATR clip-on system pressing the detection zone of the paper strip (i.e., few mm2) onto the surface of the NCD chip).
Task 4.2 MI-FI analyzer based on MIR LEDs (M1-M30) (lead partner: UULM; participants: NP, NMBU, IRIS, BOKU, WU, NEBIH). In Task 4.2, the hand-held low-cost MI-FI analyzer platform will be designed, developed, and prototyped, The device will comprise a 3D-printed housing accommodating the MIR IC-LED developed in WP 3, a dispersion grating (plastic w/gold coating; potentially 3D printed), and a line array detector. If a line array detector with a surface-deposited gradient filter will be used (e.g., Pyreos), the dispersion grating will be replaced with appropriate beam-shaping optics (3D printed; gold-coated plastic optics). The sensing interface will be established via a hemispherical or prismatic diamond ATR crystal. Using MIR spectral ray tracing (via software packages SPRAY and SCOUT, M. Theiss Software), an optimized MI-FI analyzer prototype will be built and tested according to its analytical figures-of-merit and via spiked test samples. After optimization, five hand-held MI-FI prototype systems will be fabricated ready for field testing during the case studies lined out in WP6, and together with end users. The performance of the MI-FI device will also be benchmarked against commercially available near-infrared LED-based low-cost analyzers (e.g., Consumer Physics) and evaluated for its utility addressing additional quick-test application scenarios.
Task 4.3 HI-FI analyzer based on tunable MIR QCLs (M1-M24) (lead partner: UULM; participants: NP, NMBU, IRIS, BOKU, WU, NEBIH). Task 4.3. is focused on the design, development and optimization of a high-resolution HI-FI analyzer integrating tunable monolithic MIR ICL/QCL technology developed in WP3, a single element thermoelectrically cooled MCT detector (e.g., Vigo), and the MIR-GuideR ATR sensing interface with NCD waveguide chip for direct liquid (extract) analysis or for analyzing smart paper strips developed in WP 2. The HI-FI analyzer will be prototyped and performance optimized (i.e., analytical figures-of-merit, spiked test samples, etc.). As a final outcome, one portable prototype systems will be made available for field testing during the case studies specified in WP 6, and together with end users.
Role of partners:
UULM will be leading this work package and is responsible for the realization of the MI-FI and HI-FI devices. NP will provide the light sources. NP will further support the design of the MI-FI and HI-FI devices to ensure optimal integration of the light sources and help with the tunable laser operation. NMBU will evaluate the solutions with respect to data quality. IRIS will support the implementation of the software application. WU will help to optimize the photonic solutions for the interface with the microfluidics. NEBIH and BOKU will provide the necessary input for ensuring that the photonics solution is prepared for the end user.

Participation per Partner

Partner number and short name	WP4 effort
1 - NMBU	
2 - UULM	
3 - BOKU	
4 - WU	
5 - NEBIH	
6 - NANOPLUS	
8 - IRIS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Thin-film waveguide-based sensing interface for HI-FI analyser	2 - UULM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	18
D4.2	Performance test report for MI-FI analyzer	2 - UULM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	30
D4.3	Performance test report for HI-FI analyzer	2 - UULM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D 4.1 Thin-film waveguide-based sensing interface for HI-FI analyzer (UULM, M18);
D 4.2 Performance test report for portable HI-FI analyzer (UULM, M28);
D 4.3 Performance test report for handheld MI-FI analyzer (UULM, M22).

D4.1 : Thin-film waveguide-based sensing interface for HI-FI analyser [18]

When the thin-film waveguide-based sensing interface for the HI-FI analyser is developed it will be tested and demostrated and a report will be given.

D4.2 : Performance test report for MI-FI analyzer [30]

When the MI-FI analyser is developed and the IC-LED light sources are integratred in the MI-FI analyser, a performance test report for MI-FI analyzer will be conducted.

D4.3 : Performance test report for HI-FI analyzer [24]

When the HI-FI analyser is developed and the advanced light sources are integratred in the HI-FI analyser, a performance test report for HI-FI analyzer will be conducted.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.
MS7	Five handheld MI-FI analyzer prototypes for demonstration	2 - UULM	30	Evaluation of MI-FI analyzer prototype design and performance on real samples.
MS8	One portable HI-FI analyzer prototype for demonstration	2 - UULM	24	Evaluation of HI-FI analyzer prototype design and performance on real samples

Work package number [9]	WP5	Lead beneficiary [10]	1 - NMBU
Work package title	Data analysis and software development		
Start month	1	End month	48

Objectives

To develop a data analysis strategy for short wavelength range data from MI-FI and HI-FI data and to establish a software interface for mobile and laptop for field use. The specific objectives are:
• To use advanced data analysis to pre-process spectral data and calibrate MI-FI and HI-FI solution for detection of microbial and chemical contaminants in food.
• To provide a user-friendly interface, intuitive decision support tools, data management and data integration strategy.

Description of work and role of partners

WP5 - Data analysis and software development [Months: 1-48]
NMBU, UULM, BOKU, WU, NEBIH, IRIS
Task 5.1 Spectral pre-processing of HI-FI and MI-FI data (M1-M24) (lead partner: NMBU, participants: UULM)
Existing methods for spectral pre-processing take advantage of the availability of a broad spectral range for the estimation of baselines and multiplicative effects. Since IC-LED and ICL/QCL sources provide a limited bandwidth, the existing methods for the pre-processing have to be further developed and adapted. Our general spectral pre-processing strategy consists of the following steps: 1) baseline correction and normalization; 2) interferant removal and 3) signal enhancement. The pre-processing reduces complexity of classification models and results therefore often in more robust models. Model-based pre-processing toolbox developed in BioSpec based on Extended Multiplicative Signal Correction (EMSC) will be used for implementing all steps including removal of interferant signals from paper-based microfluidics and solvents used for extraction. As an alternative route, we will evaluate augmentation of data in combination with deep learning approaches. In the on-going MIRACLE project (see Table 4), where only a few laser wavelengths are available for calibration this approach has turned out to be very powerful. Especially, in cases where limited data sets are available, augmentation has turned out to be a powerful approach for calibration when it is used in combination with deep learning approaches developed for spectral data67
Task 5.2 Classification and calibration models for benchmark data, HI-FI and MI-FI data (M12-M36) (lead partner: NMBU, participants: WU, BOKU, UULM).
At the beginning of the project, broadband MIR spectrum benchmark instrumentation at NMBU and UULM will be used to establish benchmark data for calibration and classification model building (Task 6.1). Models will be built on the full available spectral range (4000-500 cm-1) and the short spectral range (1800-1500 cm-1) of the PHOTONFOOD solution. For calibration, samples will be prepared, MIR and reference data will be obtained for 100+100 (clean +contaminated) samples for each contaminant. All samples will be measured by benchmark MIR instrumentation at NMBU and UULM, and reference analysis will be performed at BOKU and NEBIH. Data sets will be augmented by techniques from the field of deep learning according to the same lines as in the on-going MIRACLE project (see Table 4). Calibration models using our BioSpec toolbox of machine learning techniques involving a broad range of methods such as PLS-DA, RF, SVM, ANN and DCNN will be established to calibrate against threshold values used for contaminants, while PLS-R, RF, ANN and DCNN will be used to calibrate against the exact values to detect the amount of contamination. The obtained data will be used as a benchmark in the development of the HI-FI and MI-FI data. When the HI-FI and MI-FI instrumentation is ready, the same samples that were used to establish benchmark data and calibration models will be probed by HI-FI and MI-FI instrumentation. The obtained data will be used to establish calibration and classification models for HI-FI and MI-FI data, and to compare with benchmark data.
Task 5.3 Model validation and demonstration (M36-M48) (lead partner: NMBU, participants: BOKU, UULM, NEBIH).
Models from Task 5.2 established on the short wavelength range will be validated in the field demonstration involving relevant stakeholders. The models will also be validated by a blind test set and further refined and optimized in order to accommodate the use of the IC-LED and ICL/QCL spectral data.
Task 5.4 Software application development and data management (M12-M40) (lead partner: IRIS, participants: NMBU, WU, BOKU, UULM). During this task, IRIS will design and build a digital Global Control System (GCS) to allow the processing of the data from the sensors and its integration with the data analysis models developed in the first part of this WP. Early in the project a data management plan will be developed (D8.7 M6). The GCS will centralize the data management. It will comprise the hardware to host data acquisition, data processing and data management, and a friendly user interface to easily manage the PHOTONFOOD platform. A decision support tool, with specific user-oriented modules to support decisions on-site will be developed. It will be accessible over the web and mobile devices,

such as smartphones and tablets. The hardware will consist of two main layers: an upper layer containing the processors and all the necessary peripherals (data storage units, terminals, external connectivity), and a lower layer comprising the communication with the analyser modules. Communications will take place through data buses. These will be mainly wireless whenever possible and wired when necessary. The GCS will have real-time capabilities and will be able to manage huge amounts of data. It will be able to integrate data delivered in standard industrial formats. The GCS will have a layered structure, and each layer will be accessible from the Graphical User Interface through the terminals. Firstly, Data Acquisition layer, which is the closest to the hardware, will manage the acquisition of data, as well as protocols, fault detection and hardware compatibilities. The valid acquired data will be stored in the general database to be accessible for higher layers. Secondly, the Data Processing Layer, which will convert the raw data into calibrated and filtered valid data to feed the calibration models developed in previous tasks. Finally, the Data Mining Layer, the highest data management layer, will produce the results of data analysis and provide decision support. The General Database will be designed and built to be as flexible and robust as possible, as compatible with multivariate nature of the data and real-time access as possible. The calibration models will have the capability to be updated with the feedback provided by the advanced AI algorithm executed in the cloud platform. The cloud application will be accessible over the web and mobile devices, such as smartphones and tablets. For the local application, there will be two versions: computer and mobile.

Task 5.5 Definition of the Cloud Architecture and communication with Food Trust™ (M36-M48) (lead partner: IRIS, participants: NMBU, UULM, advisory role for IBM). User requirements for the cloud platform will be defined. Different methodologies will be applied to set the user requirements, specifications, features, functionalities and the desired user experience. An in-depth analysis of existing technologies will be carried out to assess which are the most suitable ones to match the defined requirements. The in-depth architecture will be defined: backend, middleware, database, intelligent layer structures, etc. It will be a priority for the architecture to be flexible in terms of single-tenant or multi-tenant deployments, to guarantee its high performance to run intelligence layers. For that purpose, it will be designed how to tackle the envisaged computational workload and the interactions of the different modules. A Quality Analysis (QA) methodology will be in place (in keeping with software quality management: ISO 9126-3 and the subsequent ISO 25000:2005 quality model) to ensure consistent and standardized implementation. Communication with Food Trust™ platform for data integration will be ensured.

Role of partners: NMBU will be leading this work package and will be responsible for the algorithm development for the data pre-processing and the model building. IRIS will be responsible for the software development. BOKU and NEBIH will be involved providing reference analysis. IBM be involved in the development a strategy for transferring data into platforms such as the block chain-based IBM Food Trust™ data platform. UULM will assist in the interpretation of instrumental effects in the spectra and will be involved in activities related to data modelling and validation. WU will be participating in model development and software development by providing their feedback.

Participation per Partner

Partner number and short name	WP5 effort
1 - NMBU	
2 - UULM	
3 - BOKU	
4 - WU	
5 - NEBIH	
8 - IRIS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	Pre-processing algorithms for HI-FI and MI-FI spectra	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D5.2	Classification and calibration models for MI-FI and HI-FI instruments ready for fungi and mycotoxins in all matrices	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D5.3	Calibration and classification models validated	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	48
D5.4	Digital Global Control System (GCS) Technical Manual	8 - IRIS	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D5.5	Digital Global Control System (GCS) User Manual	8 - IRIS	Other	Confidential, only for members of the consortium (including the Commission Services)	40

Description of deliverables

D 5.1 Pre-processing algorithms for HI-FI and MI-FI spectra established. (NMBU, M36);
D 5.2 Classification and calibration models for MI-FI and HI-FI instruments ready for fungi and mycotoxins in all matrices (NMBU, M36);
D 5.3 MI-FI and HI-FI calibration models established and validated (NMBU, M48);
D5.4 Digital GCS technical manual report describing the installation at hardware level and architecture used to develop the software (IRIS, M36);
D5.5 Digital GCS user manual report analysing the overall functionalities of the software platform, capabilities, buttons, user interfaces, etc. (IRIS, M40)

D5.1 : Pre-processing algorithms for HI-FI and MI-FI spectra [36]

When pre-processing algorithms are developed for the short spectral range HI-FI and MI-FI spectra, they will be described in a report.

D5.2 : Classification and calibration models for MI-FI and HI-FI instruments ready for fungi and mycotoxins in all matrices [36]

Based on data from case studies, classification and calibration models for MI-FI and HI-FI instruments will be ready for fungi and mycotoxins in all matrices and reported.

D5.3 : Calibration and classification models validated [48]

The developed calibration and classification models will be validated in real settings and reported.

D5.4 : Digital Global Control System (GCS) Technical Manual [36]

A Digital GCS technical manual report describing the installation at hardware level and architecture used to develop the software will be given.

D5.5 : Digital Global Control System (GCS) User Manual [40]

A digital GCS user manual report analysing the overall functionalities of the software platform, capabilities, buttons, user interfaces, etc. will be given.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.
MS9	Software Alpha release	1 - NMBU	36	Pre-processing and calibration models validated in lab.
MS10	Software Beta release	1 - NMBU	48	Pre-processing and calibration models validated during demonstration

Work package number [9]	WP6	Lead beneficiary [10]	3 - BOKU
Work package title	Demonstration and validation		
Start month	1	End month	48

Objectives

To conduct 3 case studies where the project concept and vision for farm-to-fork sensing are implemented and demonstrated. In each case study, monitoring data acquired by portable photonics-based devices will enable quick on-site risk assessment by the end-users at different entry points of the food chains. The specific objectives are:
• Validation: Validate the developed solution considering the user-expectations explored in WP1 by comparing the performance characteristics of the innovative photonic devices to reference methods.
• Demonstration: Test the innovative photonic solution via on-site operation through the 3 case studies, which includes the measurement of mycotoxins and presence of fungi in (i) wheat and (ii) peanuts, also the examination of (iii) levels of pesticide, antibiotics and fungi in products from aquaponics.
• Give feedback to WP1 about the usability of the IPD in real settings and to the technology developers (WP2-5) about the modifications that are possibly needed based on the demonstration activities.

Description of work and role of partners

WP6 - Demonstration and validation [Months: 1-48]
BOKU, NMBU, UULM, WU, NEBIH, ROMER, IRIS, BIGH, Seeberger GmbH, BAMA Gruppen AS
Task 6.1 Building the spectral database (M1-24) (lead partner: NMBU, BOKU, participants: NEBIH, Seeberger, BIGH, WU). A database of spectra will be built for input to the model described in Task 5.1 and 5.2. In order to feed the spectral library with data and to validate our methods in a laboratory-based environment, well defined commodities will be provided by NEBIH, Seeberger and BIGH. It is important to note that separate sets of spectra are necessary for each specific matrix since MIR spectra of the latter are significantly different. Separate spectral databases for contaminated and clean samples are available for each matrix composed of nuts (such as peanuts, hazelnuts, and almonds), dried fruits, wheat and vegetables. It is critical that the ensemble of spectra is representative of the matrix and contamination. It is also critical that a large enough data set of clean and contaminated samples is available for training and testing of the chemometric model. An estimated 100+100 (clean + contaminated) samples need to be available for the creation of training and test sets in each matrix. With three measurement replicates, we will acquire approx. 600+600 spectra for each instrument (HI-FI, MI-FI, and benchtop benchmark instrumentation at NMBU and UULM). The final number depends on the repeatability of spectral measurements.
Task 6.2 Acquisition of reference data (M24-30) (lead partner: BOKU, participants: NEBIH, Seeberger, BIGH). For optimization and validation purposes, as well as for the preparation of matrix-calibrants of the developed flexible spectroscopic assays, 50 naturally contaminated samples for each matrix group will be analyzed by fully validated LC and LC-MS/MS methods developed by NEBIH and BOKU, respectively. These reference measures will be complemented by the determination of the concentration-related scope of the method. Therefore, an additional amount of 100 inoculated samples will be blended with non-contaminated sample material in order to prepare 10 different concentration levels. As non-contaminated sample material, bulk samples of 5-10 kg each of 3 different lots of the commodities wheat and peanuts will be used and mixed with the contaminated material to the appropriate concentration levels. Naturally contaminated, inoculated and non-contaminated sample material including different lots will be provided by NEBIH, Seeberger and BIGH.
Task 6.3 Performance and evaluation of a ring trial (interlaboratory comparison study) (M1-42) (lead partner: BOKU, participants: NEBIH, ROMER, UULM, NMBU). The trueness and the comparability of the measurement results obtained for mycotoxins in the naturally contaminated materials will additionally be ensured through the conduction of a ring-trial (interlaboratory comparison study) organized by BOKU. Natural contaminated sample material for each commodity will be tested by 8 different laboratories (BOKU, NEBIH, ROMER, UULM, NMBU) and finally subjected to a statistical evaluation. By comparing identical sample materials with different methods, a more accurate evaluation will be obtained regarding the measurement uncertainty in general, as well as the measurement quality of the new developed flexible spectroscopic assay in detail.
Task 6.4 On-site demonstration exercise at the end user level (M36-42) (lead partner: NEBIH, participants: BOKU, Seeberger, BIGH, BAMA). In order to test the applicability of the innovative and cross-validated spectroscopic approaches at end user level, for both farmers and official test laboratories, on-site measurements will be conducted. A prototype innovative photonic solution will be deployed to NEBIH, Seeberger, BIGH an BAMA. 1) At least 50 naturally contaminated samples for each commodity at different contamination levels from different sectors of the food

chain (wheat producing farms, food processors and retailers), will be collected by NEBIH and will be simultaneously examined by innovative photonic solution and the reference method used for official inspections under normal laboratory conditions at the authority. 2) Seeberger and BAMA will use the device for mycotoxin monitoring trial in a variety of nuts (peanut, walnut and almond). In addition, BAMA will use the device for monitoring trial in a variety of vegetables. 3) The objective of the demonstration at BIGH is to develop an easy-to-use, flexible, and broadly adaptable versatile analytic tool for mycotoxin, fungi and oomycetes detection that is also capable to analyse antibiotics and pesticide residues in end products. Technical supervision of the demonstration procedures will be provided by BOKU. Reference analyses for the collected samples will be carried out by BOKU and NEBIH.
Task 6.5 Customers feedback (M42-46) (lead partner: BOKU, participants: NEBIH, ROMER, Seeberger, BAMA). After the devices have undergone a practicability test by the end users, this knowledge will be incorporated in order to improve the terminal devices. Feedback about the technological feasibility will be obtained in form of a questionnaire and in form of direct communication with the end users. The best suited approach, including a full validation report (or a combination of approaches) and suitable analytical protocols, will be brought to the attention of various end users along the different food chains and to official control labs throughout the EU via seminars, on-site customer exercises, publications and direct communication with DG Health.

Role of partners: The spectral database for the contaminant/commodities combinations will be built by NMBU in cooperation with NEBIH and Seeberger. BOKU will organize the validation studies. NEBIH will take care of the demonstration exercise assisted by Seeberger, BIGH and BAMA, and will be responsible for the acquisition of reference data as integral part for the validation of the flexible spectroscopic assays. BOKU, NMBU and associated partner ROMER will assist NEBIH and will coordinate the conduction of a ring trial (interlaboratory comparison study) in order to verify the comparability of the measurement results. BOKU will additionally provide LC-MS/MS reference data and participate in on-site demonstration exercises. NEBIH will finally initiate and coordinate a feedback loop between customers and producer in cooperation with ROMER, Seeberger and BAMA. UULM, WU and IRIS will participate in the validation and demonstration to make sure that possible technical issues can be solved on site.

Participation per Partner

Partner number and short name	WP6 effort
1 - NMBU	
2 - UULM	
3 - BOKU	
4 - WU	
5 - NEBIH	
7 - ROMER	
8 - IRIS	
10 - BIGH	
11 - Seeberger GmbH	
12 - BAMA Gruppen AS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Report on preliminary comparability study	3 - BOKU	Report	Confidential, only for members of the	12

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				consortium (including the Commission Services)	
D6.2	Report on completed validation studies	3 - BOKU	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D6.3	Report on completed demonstration exercises	5 - NEBIH	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D6.4	Evaluation of collected feedback data (report)	5 - NEBIH	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D6.5	Report on ring trial (interlaboratory comparison study)	3 - BOKU	Report	Confidential, only for members of the consortium (including the Commission Services)	42

Description of deliverables

D6.1 Report on preliminary comparability study (BOKU, M12);
D6.2 Report on completed validation studies (BOKU, M42);
D6.3 Report on completed demonstration exercises (NEBIH, M46);
D6.4 Evaluation of collected feedback data (report) (NEBIH, M46)
D6.5 Report on full ring trial (interlaboratory comparison study) (M42)

D6.1 : Report on preliminary comparability study [12]

A report based on a preliminary comparability study involving the labs at BOKU, NEBIH and ROMER, with some participation of the spectroscopy lab at NMBU, using the technology available at this time point. The preliminary comparability study will provide samples for the first testing of pre-prototypes.

D6.2 : Report on completed validation studies [42]

A validation study will be performed including naturally contaminated, inoculated and non-contaminated sample material. A report on completed validation studies will be prepared.

D6.3 : Report on completed demonstration exercises [46]

On-site demonstration will be performed on the enduser level and a report on completed demonstration exercises will be prepared.

D6.4 : Evaluation of collected feedback data (report) [48]

Feedback about the technological feasibility will be obtained in form of a questionnaire and in form of direct communication with the end users. An evaluation of collected feedback data will be reported.

D6.5 : Report on ring trial (interlaboratory comparison study) [42]

A full ring trial (interlaboratory comparison study) involving eight laboratories: BOKU, NEBIH, ROMER (four locations), UULM and NMBU. Six will be used for reference analysis and two for spectroscopic analysis. The ring trial will provide the consortium very valuable samples and data to be used for the complete validation process.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.
MS11	Portable HI-FI solution validated	3 - BOKU	48	Performance validated in field and end users feedback implemented.
MS12	Handheld MI-FI solution validated	3 - BOKU	48	Performance validated in field and end users feedback implemented

Work package number [9]	WP7	Lead beneficiary [10]	9 - accelCH
Work package title	Communication, dissemination and exploitation		
Start month	1	End month	48

Objectives

To communicate about the project, raise awareness on food safety, and to disseminate results through publications, presentations and focused training. The specific objectives are:
• Define and plan strategy and measures in multi-stakeholders communication and dissemination plan.
• Deliver cross-media communication and public engagement activities to raise awareness about PHOTONFOOD and food safety in society.
• Disseminate the research and study results to the scientific community and the food industry.
• Provide guidelines and recommendations to decision- and policy-makers regarding food safety.
• Develop a plan for the future use of the PHOTONFOOD platform and the individual technologies and a roadmap for exploitation.

Description of work and role of partners

WP7 - Communication, dissemination and exploitation [Months: 1-48]
accelCH, NMBU, UULM, BOKU, WU, NEBIH, NANOPLUS, ROMER, IRIS, BIGH, Seeberger GmbH, BAMA Gruppen AS, HS
Task 7.1 Communication and Dissemination plan (M1-M12) (lead partner: accelCH, participants: all). The task will focus on setting up a communication and dissemination plan (CDP) with the support of all partners. The CDP will be also discussed with members of the advisory board. We will develop an outline key strategy for project dissemination and communications, for both internal (project partners) and external stakeholders. Further, we will identify dissemination tools and channels, define the target audience and evaluate the implementation and impact of activities and update the CDP.
Task 7.2 Cross-media communication to raise awareness in society (M1-M48) (lead partner: accelCH, participants: all). The following communication activities will be performed: (1). Project website: Set up and maintain the PHOTONFOOD website including 1) a web-based information resource for different target-groups, 2) integrated use of different media (e.g. blog, infographic) to enhance the outreach to society and, 3) an exhaustive list of the project results including all publications and presentation contributions. (2) Communication material: Digital and print material to support the outreach of the PHOTONFOOD project through existing and newly established networks will be developed. This will include a project flyer providing basic information (print and digital version), a poster and a roll-up for use at events organised by the partners or hosted by other relevant organisations to promote the project. All materials will follow the PHOTONFOOD design guidelines including the logo and visual elements to ensure a coherent style throughout all materials and channels. (3) Audio/visual material: Create interactive digital audio/visual material for is easy use and re-sharing through social media channels. This will include infographics to present complex information in a simple way to the general public as well as videos presenting the project activities and results including interviews with key partners will be created and distributed via the project website, social media channels and existing partner networks. (4) Press releases and articles: Publish press releases oriented towards the food industry and for mass media (general public) on outcomes from the project. In close collaboration with partner's media groups monitor and manage public relations related to the project and organise any requests for further information, statements and interviews.
Task 7.3 Disseminate research results to expert stakeholder groups (M13-M48) (lead partner: NMBU, participants: academic partners). The dissemination of the research results will be performed by using the following dissemination instruments: (1) Peer-reviewed publications: PHOTONFOOD partners will publish their findings and research advancements in high impact peer-reviewed journals, such as Journal of Biophotonics, Applied Spectroscopy, and Food Control. All partners are committed to following the Guidelines on Open Access to Scientific Publications and Research Data in Horizon 2020 that are in line with current trends in IP management and publication, and will establish a database of relevant journals for scientific dissemination. (2) Scientific conferences and presentations: All partners will present their findings through oral presentations, posters and conference papers at European and international scientific conferences, such as SciX, SPEC, Food Safety, PITTCON, and SPIE Photonics Europe. (3) Cluster event: accelCH will organise a cluster event in the first project year together with other projects funded under the ICT-37-2020 call to exchange knowledge and create synergies on how to handle key issues relevant for all projects to coordinate research developments. (4) Training: NEBIH, NP, WU and BOKU will integrate PHOTONFOOD results into training materials for already established training events that they organise for students on safety in the food chain and photonic technology.

This will include training through summer and winter schools, lab settings and integration into established bachelor and master programmes. (5) Policy brief: NEBIH will create a short, to the point policy brief based on outcomes from WP6 (validation). These will be directed towards policy makers or stakeholders who can influence policy. The aim is to highlight the relevance and benefits of the new PHOTONFOOD approach and its related research as well as offer recommendations for change to reduce regulatory barriers and identify drivers for the uptake of the PHOTONFOOD solutions to improve food safety.(6) White paper: NEBIH will develop a white paper in the final year on "photonics-based novel technologies for food safety applications" in the final year to provide recommendations to be considered for future guidelines and international regulations based on results from the validation WP (WP6). (7) Local demonstrations: NP will organise live local demonstration of the PHOTONFOOD platform, solutions and initial results derived from the validation in a realistic setting (WP6) to stakeholders along the food value at tradeshows, such as Analytica, Laser World of Photonics and local information events. UULM will organise live demonstration sessions of the devices locally in the lab as "Industry Information Days" and also hands-on 1-2 day workshops at UULM for industry with theoretical sessions and hands-on practical sessions. The local demonstrations will be held in the local language to ensure the wide dissemination of the PHOTONFOOD related results and offer a unique opportunity for networking with interested stakeholders. (8) Final event: To engage with members of parliament and policy- and decision makers, WU will organise a workshop as the final event. The workshop will consist of presentations of the PHOTONFOOD key results, the policy brief and white paper will be presented and a panel discussion with invited experts from our stakeholder networks covering ethical, legal, social and economic considerations will be organised.
Task 7.4 Future use and exploitation of the PHOTONFOOD platform and technologies (M1-M48) (lead partners: accelCH, HS, participants: all). The following measures will be taken for the future use and exploitation of the developed technology: (1) Business Model development: Potential business cases will be developed by the technology providers led by the Innovation Manager that will facilitate the future upscaling and uptake of PHOTONFOOD. The business model(s) will take into account results from the demonstration and validation (WP6) and outcomes and recommendations from all research WPs (WP1-5). Supported by all WP leaders, they will identify adoption barriers for the involved stakeholders and develop strategies to overcome these barriers by developing a PESTEL analysis to assess and monitor the macro-environmental factors that may impact our project and the developed technologies. (2) Promotion of exploitation and future funding plans: Detailed report to include the consortium's plan for commercialisation of the developed technology. Develop draft exploitation agreement to build on Consortium agreement. Actively seek licensees and future funding to ensure ease of commercialisation and funding during first year of exploitation. Subcontract services of company to carry out future funding and investment searches. Begin process of establishing the product in product range, draw up distribution agreement and supply chain agreements with partners. (3) Strategic grant planning: To facilitate the further non-commercial use of results and outcomes through follow-up projects, accelCH will pre-select funding programmes relevant to partners and their project results by identifying suitable funding instruments. accelCH will present an extensive list of funding opportunities within the consortium which will discuss all options and select the most suitable instruments. Based on partner feedback, accelCH will create a short list of the most suitable EU and national funding schemes, including requirements, timings and recommendations, develop a proposal preparation schedule and help with the initial set up of possible follow-up projects. (4) Standardisation and guidelines: The project results have the potential to be integrated into national food safety related guidelines. We will share our policy brief with decision and policy makers which will include recommendations for food safety guidelines. Our recommendations are not legally binding but could potentially influence regulations if our results are strong which could require countries to change their laws. (5) Roadmap for exploitation (RfE): the Exploitation Manager at HS will, together with the industry partners (especially NP) and the Innovation Manager at NMBU, develop a commercialisation strategy including an exploitation roadmap building upon the initial business models to examine the transferability of the PHOTONFOOD platform to the market based on outcomes from all research WPs. Sub-outcomes of the project such as individual technologies developed will also be closely monitored and documented to ensure their individual exploitable use. Thus, the RfE will comprise a plan for the future commercial and non-commercial use of results at Consortium level, group, and individual partner level. The RfE will cover the exploitation, starting during the lifetime of the project (short-term) and mid and long-term plans <4 years, and <10 years after the project ends).

Role of partners: accelCH will be leading this work package and in particular will be responsible for the development of the Communication and Dissemination Plan (Task 7.1), and for orchestrating all communication activities (Task 7.2). In addition, they will organise the Cluster Event in Task 7.3 and be in charge of all the documentation and reporting on WP7. NMBU will lead the partners in the dissemination of results to the target groups and will provide the Innovation Manager (Task 7.3). HS will lead the exploitation activities (Task 7.4), and in particular will be responsible for shaping the Roadmap for Exploitation. HS will furthermore pursue fabrication and commercialisation of the final HI-FI and MI-FI devices after the end of the project. NEBIH will be responsible for producing a white paper and policy briefs. NP and UULM will be in charge of the live demonstrations in Task 7.3. WU will organise the final event in Task 7.3. All

partners will contribute to WP7 by disseminating and communicating about the project and results and by supporting the other partners in the WP7 activities.

Participation per Partner

Partner number and short name	WP7 effort
1 - NMBU	
2 - UULM	
3 - BOKU	
4 - WU	
5 - NEBIH	
6 - NANOPLUS	
7 - ROMER	
8 - IRIS	
9 - accelCH	2
10 - BIGH	
11 - Seeberger GmbH	
12 - BAMA Gruppen AS	
13 - HS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	PHOTONFOOD website	9 - accelCH	Websites, patents filing, etc.	Public	2
D7.2	Communication and Dissemination Plan (CDP)	9 - accelCH	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D7.3	Interim communication and dissemination activities report	9 - accelCH	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D7.4	Policy guidelines	5 - NEBIH	Report	Public	36
D7.5	Live demonstrations	6 - NANOPLUS	Demonstrator	Public	42
D7.6	Roadmap for exploitation	13 - HS	Report	Confidential, only for members of the consortium (including	48

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	
D7.7	Final Scientific Symposium	9 - accelCH	Other	Public	48

Description of deliverables

D7.1 PHOTONFOOD website including audience specific areas, digital media and blog features and a secured area for central document storage (accelCH, M2);
D7.2 Communication and Dissemination Plan (CDP) developed defining key strategies and implementation process required to implement all communication and dissemination activities (accelCH, M9);
D7.3 Interim communication and dissemination activities report to provide an overview of the activities achieved to date and assess their impact by means of Key Performance Indicators. (accelCH, M24);
D7.4 Policy guidelines for the integration of the PHOTONFOOD platform into the standard approach in food safety (NEBIH, M36);
D7.5 Live demonstrations of the PHOTONFOOD solution (NP, M42);
D7.6 Roadmap for exploitation including strategies for the future commercial and non-commercial use of the outcomes and results (HS, M48);
D7.7 Final Scientific Symposium presenting the research results and the platform to scientists, food industry professionals and decision and policy makers (accelCH, M48)

D7.1 : PHOTONFOOD website [2]

The PHOTONFOOD website including audience specific areas, digital media and blog features and a secured area for central document storage will be published.

D7.2 : Communication and Dissemination Plan (CDP) [9]

A Communication and Dissemination Plan (CDP) will be developed defining key strategies and implementation process required to implement all communication and dissemination activities.

D7.3 : Interim communication and dissemination activities report [24]

An Interim communication and dissemination activities report will be delivered to provide an overview of the activities achieved to date and to assess their impact by means of Key Performance Indicators.

D7.4 : Policy guidelines [36]

Policy guidelines for the integration of the PHOTONFOOD platform into the standard approach in food safety will be provided.

D7.5 : Live demonstrations [42]

Live local demonstration of the PHOTONFOOD platform, solutions and initial results derived from the validation in a realistic setting to stakeholders along the food value at tradeshows.

D7.6 : Roadmap for exploitation [48]

Roadmap for exploitation including strategies for the future commercial and non-commercial use of the outcomes and results.

D7.7 : Final Scientific Symposium [48]

Final Scientific Symposium presenting the research results and the platform to scientists, food industry professionals and decision and policy makers .

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.

Work package number [9]	WP8	Lead beneficiary [10]	1 - NMBU
Work package title	Coordination and management		
Start month	1	End month	48

Objectives

Scientific, financial, administrative and ethical management of PHOTONFOOD with focus on the time and cost-efficient implementation of the work plan. The specific objectives are:
• Provide the scientific and technical leadership, as well as the administrative and financial management, to ensure that PHOTONFOOD delivers qualitatively high results within the time schedule and budget of the project description.
• Establish a single contact point with the EU and maintain continuously contact with EU representatives.
• Perform a quality assessment of the progress, results and impacts of the project.
• Deliver timely the progress reports and budget reports to the European commission.
• Make sure that the Description of Action (DoA) and the consortium agreement are followed.
• Establish a digital platform and intranet for the project to enable the exchange of data and information among the consortium partners.

Description of work and role of partners

WP8 - Coordination and management [Months: 1-48]
NMBU, BOKU, WU, NEBIH, NANOPLUS, ROMER, IRIS, accelCH, BIGH, Seeberger GmbH, BAMA Gruppen AS, HS
Task 8.1 Coordination (M1-M48) (lead partner: NMBU, participants: all). This task will perform the overall coordination of the scientific and technological developments of all WPs in order to make sure that the developed components and scientific findings are aligned. The Project Coordinator (NMBU) will establish a single contact point with the EU in order to communicate all results, scientific, financial, and related issues. A Project Manager (NMBU) and a Scientific Manager (Deputy Coordinator from NMBU) will assist the Coordinator (NMBU). The Executive Board (EB) and the General Assembly (GA) will meet biannually and annually, respectively, and arrange ad hoc meetings when required. The EB and the GA will monitor risks and project progress supported by the Coordinator and the Scientific manager. The coordinator will be supported by the project manager in all administrative issues and especially in reporting and finance. The milestones and the deliverables will be closely followed up by the scientific manager who will carefully review all progress plans. The Project Manager will be responsible for the on-time delivery of draft scientific and financial reports to the Project Coordinator. Gender equality will be monitored by the Coordinator and it will be made sure that the gender balance will be maintained throughout the whole project period.
Task 8.2 Financial management (M1-M48) (lead partner: NMBU, participants: all). The economy manager Christer Wulff-Olsen at NMBU will be responsible for the financial management of the project in close collaboration with the project manager. They will ensure the timely distribution of funds from the EC to the partners. During the kick-off meeting, the economy advisor will introduce all partners in payment routines and aspects of financial reporting. All partners will obtain templates and instructions for financial reporting. The economy manager will be responsible to prepare all financial reports and deliver them timely to the EC. All partners will be asked early to prepare their financial reports and each partner will be responsible to prepare the reports and deliver them to NMBU in time. The coordinator will monitor the financial management and inform the EU about all necessary financial transfers.
Task 8.3 Decision making, conflict management (M1-M48) (lead partner: NMBU). Each WP is led by one of the partners (WP leader) with a deputy WP leader from a different partner. This will ensure focus on the work that is agreed upon. The WP leaders take the routine decisions, communicate those with the Coordinator, monitor progress of output according to the work plan, and prepare all documents and documentations for their WP. The decision-making body will be the General Assembly, with 1 representative from each partner organization and chaired by the Coordinator (see details of composition in section 3.2), each partner has a vote and the ability to steer the project. Major strategic decisions will be made by the General Assembly, such as change of scope, IPR related decisions, conflicts, budget related decisions, etc. First line of day-to-day decision making will be up to the Executive board whose membership will comprise the Coordinator and the WP leaders. Matters of a strategic nature are floated to the General Assembly and will aim to be accompanied by a recommendation from the Executive board. The Project Coordinator will handle any cases of fraud, unethical behaviour, and breach of procedures, however, such incidences are expected to be highly unlikely. The Project Coordinator will distribute a list of decisions to be taken to all participants prior to each meeting. Partners may request in writing for additional topics to be discussed, and the updated agenda is distributed before the meeting. All meetings are documented, including the outcome of all decisions and distributed to all partners.

Task 8.4 Organization of reporting to the EC (M1-M48) (lead partner: NMBU, participants: all).
NMBU, the Project Coordinator, will present reporting procedures during the kick-off meeting. The Project Coordinator will provide templates for the formal reports and updates during the course of the project when necessary. The Project Coordinator will prepare a brief semi-annual report and distributes it to all participants. Formal project reports will be prepared by the Coordinator in keeping with the contractual reporting periods stipulated in the Grant Agreement. The Coordinator will present this formal report, including finance, IPR, ethics, and gender issues to the EC on time (and during formal review meetings as proposed by the EC). The Coordinator will also prepare a brief non-confidential version of the formal report for public access on the website.
Task 8.5 Day-to-day management (Internet, helpdesk and back-office) (M1-M48) (lead partner: NMBU, participants: all). A secure digital platform for networked information sharing will be established to minimize costs and environmental impact of travelling and to allow rapid access to collated data. The platform will have an option for open-source content management. The digital platform will be used for all internal communications. Access will be restricted to consortium members. Different access levels will be incorporated based on user roles. It will provide tools for knowledge management and the development of databases, to share preliminary results and internal documents amongst partners, to discuss and edit pre-publication documents and deliverables. Each time new data is available, it will be made sure that the data is shared in the project database according standard formats which will be agreed on at the beginning of the project. This will ensure a steady exchange and sharing of all spectroscopic and metadata between all partners. This intranet will be maintained by NMBU. Shared documents, contact data, templates, agreements and links to official guidelines for reference and reports and conclusions of meetings will be stored on the password protected part of the website which will be maintained by NMBU. All periodic reports and collated reports will be stored in the project database.
Task 8.6 Organization of consortium meetings (incl. telephone/video-conferences, web meetings) (M1-M48) (lead partner: NMBU, participants: all). A kick-off meeting for all project partners including the General Assembly (and the Executive board as part of the Committee), the Advisory Board and representatives from the EC will be organized by the Coordinator (NMBU. The back-office of NMBU will take care of all aspects of the consortium meetings, as presented in paragraph 3.2, such as agenda, invitations, preparation of materials (hand-outs, programmes, etc.), organisation of rooms and equipment, catering and dissemination activities to the consortium in cooperation with the host of the meeting, i.e. typically the location of one of the WP leaders. NMBU will also organize and host all consortium telephone/ videoconferences (regular according to schedule, and on demand). In total 4 General Assembly meetings, 6 Executive board meetings and several ad-hoc meetings, depending on the needs, will be organized. 2 plenary meetings will be organized in conjunction with an end user meeting in M35 and M45. The minutes of all meetings will be delivered in a timely manner by the Project Coordinator. In case of unexpected events (e.g. COVID-19) the procedures indicated by the EC will be followed.
Task 8.7 Maintenance of all legal agreements (M1-M48) (lead partner: NMBU, participants: all).
All partners will sign a Consortium Agreement, with detailed matters on access to know-how, intellectual property ownership, and liability in line with the information contained in Section 2.2b as the starting point. During the kick-off meeting, all contractual requirements on the Grant Agreement will be reviewed with all the partners to ensure that all partners recap on their commitments. All partners must report issues relevant to the legal agreements to the Coordinator in a timely manner. The Coordinator will decide on how to proceed with any deviation and will inform the Project Officer.
Task 8.8 Risk Management (M1-M48) (lead partner: NMBU, participants: all). Alterations to the work plan will be identified early by following partners tightly in consortium meetings, regular technical meetings and periodic reports. A control procedure will be established by the coordinator for handling identified alterations to the work plan. All partners will be asked to continuously monitor risk factors and to report problems timely to the coordinator. Changes will be monitored in order to respond to threats in a timely manner to ensure work packages continue smoothly. Areas of potential difficulties will be discussed with the respective partners and discussed in the consortium. The impact of problems will be considered thoroughly in the consortium and contingency plans will be activated and mitigation measures will be discussed. It is important to make sure that mitigation measures are thoroughly discussed in the consortium and the impact of the respective mitigation measures are understood for all affected partners and tasks. If necessary, resources will be reallocated, and objectives will be recovered and rescheduled.
Task 8.9 Quality management (M1-M48) (lead partner: NMBU, participants: all). The project will develop a quality handbook for quality assurance and control that contains all data format definitions, methods, guidelines for communications, conflict management, innovation management and project reporting. The quality handbook will be continuously overseen and revised in meetings of the GA (discussed biannually). The external advisory board will give input to the quality handbook.
Task 8.10 Innovation management (M1-M48) (lead partner: NMBU; participants: all). Freedom to operate: the IPR landscape will be mapped in the beginning of the project and continuously monitored throughout the project's lifetime. This will ensure that relevant background IP is duly integrated into the consortium agreement and that the necessary licenses are negotiated by the project consortium. IP strategy: the Innovation Manager will develop and negotiate a

coherent IP strategy with all project partners, balancing the different interests of the participants (secrecy clauses, patenting, trademarking, licensing, publishing). The IP strategy will be closely aligned with the exploitation strategy and roadmap (see WP7), ensuring that the IP strategy can leverage the further commercialisation and business development activities planned after the end of the project's lifetime and avoid undue conflicts between the consortium partners and with third parties. Ongoing IP management: the Innovation Manager will liaise continuously with the WP and Task Leaders to ensure that the IPR generated in the project is disclosed in a timely manner and protected in a way consistent with the IP strategy. Furthermore, the Innovation Manager will liaise continuously with the TTOs of the participating research organisations, and the legal departments/representatives of the commercial actors in the project, to ensure that all necessary agreements are made in time and potential conflicts (between consortium partners or with third parties) are proactively resolved or avoided.
Task 8.11 Data management plan (M1-M6) (lead partner: NMBU; participants: all). A data management plan (DMP) according to the Guidelines on Data Management in Horizon 2020 will be developed. The DMP will provide the procedures for efficient collection, safe storage, deep exploitation and evaluation of the data. The DMP will ensure that research data will be easily discoverable, accessible, assessable and intelligible, usable beyond the original purpose for which they were collected, interoperable to specific quality standards, whenever relevant. The database will be used as a first basis for exploitation of results and further exchange of knowledge and expertise.
Role of partners: NMBU will be leading this work package and is responsible.

Participation per Partner

Partner number and short name	WP8 effort
1 - NMBU	1
3 - BOKU	
4 - WU	
5 - NEBIH	
6 - NANOPLUS	
7 - ROMER	
8 - IRIS	
9 - accelCH	
10 - BIGH	
11 - Seeberger GmbH	
12 - BAMA Gruppen AS	
13 - HS	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	Next stage plans and internal progress reports	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.2	Interim Status Report	1 - NMBU	Report	Confidential, only for members of the	12

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				consortium (including the Commission Services)	
D8.3	Communication framework between partners	1 - NMBU	Other	Public	24
D8.4	Six month progress reports, mid term reports and final report	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D8.5	Organize initial meetings, mid term meetings and final meetings	1 - NMBU	Other	Confidential, only for members of the consortium (including the Commission Services)	48
D8.6	Report on gender, societal and ethical issues of exploitation	1 - NMBU	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D8.7	Data management plan	1 - NMBU	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6

Description of deliverables

D8.1 Next stage plans and internal progress reports (NMBU, M6);
D8.2 Interim Status Report (NMBU, M12);
D8.3 Communication framework between partners (NMBU, M24);
D8.4 Biannual progress reports, mid-term reports and final report (NMBU, M48);
D8.5 Organize initial, mid-term and final meetings (NMBU, M48);
D8.6 Report on gender, societal and ethical issues of exploitation (NMBU, M24);
D8.7 Data management plan (NMBU, M6).

D8.1 : Next stage plans and internal progress reports [6]

Next stage plans and internal progress reports

D8.2 : Interim Status Report [12]

Interim Status Report

D8.3 : Communication framework between partners [24]

A communication framework between partners will be established.

D8.4 : Six month progress reports, mid term reports and final report [48]

Six month progress reports, mid term reports and final report

D8.5 : Organize initial meetings, mid term meetings and final meetings [48]

Initial meetings, mid term meetings and final meetings will be organized.

D8.6 : Report on gender, societal and ethical issues of exploitation [24]

A report on gender, societal and ethical issues of exploitation will be prepared.

D8.7 : Data management plan [6]

A data management plan (DMP) according to the Guidelines on Data Management in Horizon 2020 will be developed.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Component-level user requirements update presented to WP leaders	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	5 - NEBIH	37	Report on system-level user requirements will be prepared.

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Component-level user requirements update presented to WP leaders	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	5 - NEBIH	25	Feedback to the development WPs (WP2-5) about the updated requirements will be given.
MS2	System-level user requirements update presented to WP leaders	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	5 - NEBIH	37	Report on system-level user requirements will be prepared.
MS3	Portable Microfluidics Pre-prototype	WP2	4 - WU	24	Evaluation of specifications and design.
MS4	Portable Microfluidics Prototype	WP2	4 - WU	36	Validation with MI-FI & HI-FI prototypes in lab validation
MS5	IC-LEDs optical components	WP3	6 - NANOPLUS	18	Evaluation of the spectral target coverage (1800-1500 cm-1).
MS6	Monolithic tuneable lasers	WP3	6 - NANOPLUS	28	Evaluation of the spectral target coverage (1800-1500cm-1).
MS7	Five handheld MI-FI analyzer prototypes for demonstration	WP3, WP4	2 - UULM	30	Evaluation of MI-FI analyzer prototype design and performance on real samples.
MS8	One portable HI-FI analyzer prototype for demonstration	WP3, WP4	2 - UULM	24	Evaluation of HI-FI analyzer prototype design and performance on real samples
MS9	Software Alpha release	WP5	1 - NMBU	36	Pre-processing and calibration models validated in lab.
MS10	Software Beta release	WP5	1 - NMBU	48	Pre-processing and calibration models validated during demonstration
MS11	Portable HI-FI solution validated	WP6	3 - BOKU	48	Performance validated in field and end users feedback implemented.
MS12	Handheld MI-FI solution validated	WP6	3 - BOKU	48	Performance validated in field and end users feedback implemented

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Time Planned project activities cannot be realized in time as planned / MEDIUM	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Proper planning and tight regular follow up of the project work during the implementation phase. Several activities will start in parallel.
2	Cost Project activities cannot be completed within the budget / LOW	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Careful planning and up-date follow up of the budget during the project period.
3	Management Key person(s) leave the partner organisations / MEDIUM	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Every key person has a competent named deputy in each WP.
4	Management Partners are reluctant to IP sharing / LOW	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Motivation of partners by allocating suitable roles for them in the consortium. Re-allocating the partner's tasks to the other partners, or if not possible, recruiting a replacing partner according to the rule
5	Management Partners are reluctant to IP sharing / LOW	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Carefully prepared Consortium Agreement that sets the IP sharing principles.
6	Technical User acceptance of the new photonic solution is low/LOW	WP1, WP2, WP3, WP4, WP5	User requirements and acceptance data from WP1 will be continuously feed to the WPs responsible for technology development to mitigate this risk.
7	Technical Investment cost and unit cost of laboratory tests will exceed the estimated threshold / LOW	WP1, WP2, WP3, WP4, WP5	Alternative technical scenarios have been identified during the project proposal development to mitigate this risk. The HI-FI and MI-FI solutions all bear marketing opportunities that could be exploited.
8	Technical WP2: The desirable preconcentration clean-up effect is insufficient for the application. /LOW WP2: The development of a fully integrated device could take long, which would result in delays in other WPs. / LOW	WP2	We have elaborated three strategies, based on existing research/concepts, for selective enrichment. In case of failure, there are back-up options readily available. The development of the different modules of the PHOTONFOOD solution is done in parallel. The modules will be combined in various stages of completion to ensure compatibility. Thus, if one modulus is delayed, the other WPs can still be achieved in time, whilst still assure that an integrated solution can be offered towards the end of the project.
9	Technical Delays due to unscheduled down-times of equipment/LOW Light sources do not meet expected out-put powers and wave-lengths/MEDIUM	WP3	Standardized preventive maintenance plans for all critical machines and tools to minimize and avoid unscheduled down-times. establishment of alternative processing routes where possible, e.g. on 2nd tools 1) Usage of established simulation tools to start with refined structures and early tests of gain structures by fast test processes e.g. via broadboard area lasers to enable quick feedback for improvement. 2) Careful time-planning to allow for 2-3 iterations minimum
10	Technical Calibration models having not sufficient	WP5	Mitigation measure: more measurements required in WP6.

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
	sensitivity/specificity./ MEDIUM		
11	Management Recurring COVID-19 outbreaks in European countries result in restrictions on travel and of events and project meetings / Medium	WP7, WP8	Should it be required, project meetings can be held via video conferencing. The live demonstrations (D7.5) could be postponed or turned into short videos published online. The Final Scientific Symposium (D7.7) would have to be turned into an online event.
12	Technical Access to labs is restricted or limited as result of recurring COVID-19 outbreaks. / MEDIUM	WP1, WP2, WP3, WP4, WP5, WP6	The possibility to carry out work in laboratories of another partner will be assessed. During periods of restricted access to labs, focus on theoretical work, documentation, reporting and communication tasks. If needed, seek solution with Project Officer, e.g. postponing tasks and extending the project.

community, including the broader public and potential end-users, and focus on expected outcomes and related socio-economic benefits for the EU. The professional communication kit will be updated *at mid-term and at project end* to reflect project progress achieved.

The communication activities will involve public relations experts to ensure the pertinence of messages to target audiences and high outreach at both national and EU level through the use of the relevant national and EU-wide media channels (e.g. newspapers and broadcast).

Project communications such as project press releases, workshops announcements, websites, brochures will respect the principle of fair visibility for all partners. If the logo of any individual beneficiary is included, then the logos of all beneficiaries will be included.

3. Implementation

3.1 Work plan — Work packages, deliverables

3.1.1. Overall structure of the work plan

The PHOTONFOOD project will be implemented by an interdisciplinary consortium of academic, authority and industry partners covering 14 organizations from eight European countries. The project duration will be 48 months and the breakdown of the planned work into work packages (WPs) is based on accomplishing related work, rather than on single tasks that occur at the same time. Therefore, several WPs are executed in parallel throughout the project. The project consists of eight work packages (WPs) (see Fig. 15), where the first WP1 is related to the evaluation of the technology acceptance and techno-economical evaluation, four WPs (WP2 – WP5) focused on the photonic technology developments and developing advanced data analysis tools and software applications, WP6 will perform validation and demonstration of PHOTONFOOD technology, WP7 and WP8 are related to dissemination, communication, exploitation and management activities, respectively (Fig. 15).

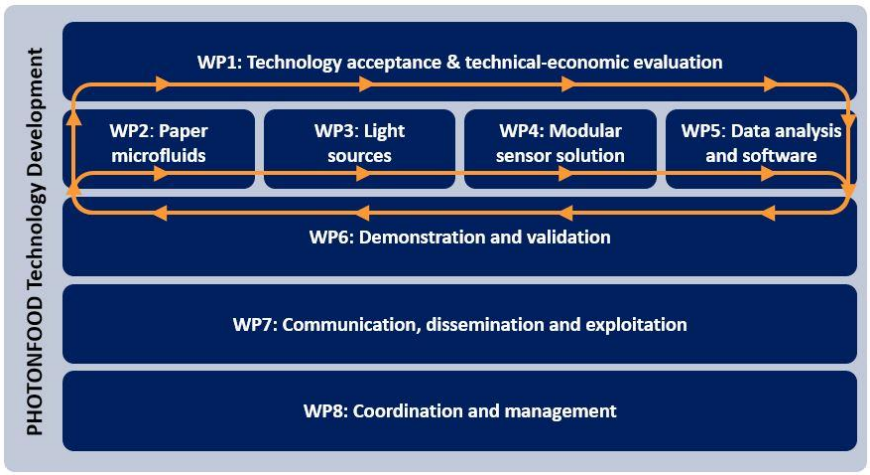

Figure 15: Overall structure of the workplan

The general overview of WPs is provided below, the list of the WPs and a detailed description of each WP are provided in Table 3.1a and 3.1b, respectively.

Assisting WPs: WP1 "Technology acceptance and techno-economical evaluation" will conduct comprehensive study on evaluating of user expectations, experience and technology acceptance of photonic-based technologies for identifying microbiological and chemical contaminations. WP1 will ensure that PHOTONFOOD technology is developed in accordance with user requirements. WP7 "Communication, dissemination and exploitation" will perform adequate dissemination and exploitation of PHOTONFOOD technology and will be responsible for the communication with the relevant audience. WP8 "Coordination and Management" will be responsible for the project coordination and management for the successful implementation of all project work activities.

Technological WPs: Four technological WPs will be responsible for the successful development of PHOTONFOOD technology. WP2 "Paper Microfluidics" focuses on the development of extraction /separation / up-concentration technology for separation of analyte markers of microbial and chemical contamination. WP2 will be strongly connected to photon-based device developing WPs, WP3 and WP4. WP3 "Novel MIR light sources" will focus on the development of cascaded gain materials forming the basis for advanced light emitting diode (IC-LEDs) and monolithically tunable laser light sources. WP4 "Modular sensor platform" will implement the results of WP3 and will perform development of portable and field-deployable (prototype) MI-FI and HI-FI analyzers. WP5 "Data analysis and software development" will focus on developing and utilizing advanced data analysis tools for the spectral-based detection and identification of microbiological and chemical contamination in different food matrices. In addition, WP5 will be focused on developing software application.

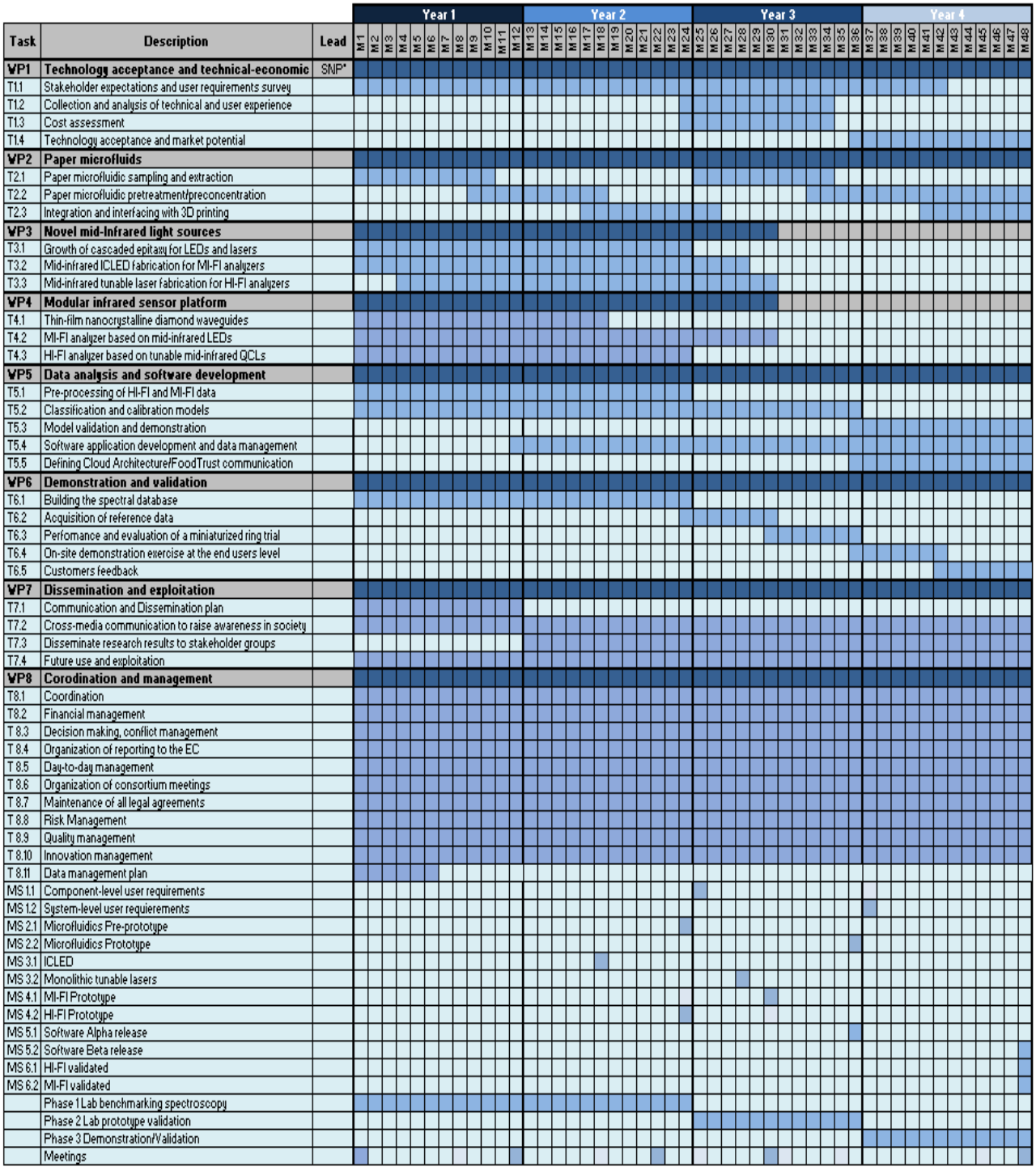

Figure 16: GANTT chart providing an overview over the timing of the project

Demonstration WP: WP6 "Demonstration and validation" will focus on demonstrating and validating the developed PHOTONFOOD technology and a vision for farm-to-fork sensing in three case studies. In each case study, monitoring data acquired by portable photonics-based devices will enable quick on-site risk assessment by the end-users at different entry points of the food chains.

The GANTT chart in Fig. 16 provides an overview of all work package's tasks and includes the milestones, validation phases and meetings (Table 3.2a).

Table 3.1a: List of work packages (WPs)

WP	WP Title	Lead Part. No	Lead Participant Short Name		Start Month	End Month
1	Technology acceptance and technical-economic evaluation	5	NEBIH		1	46

2	Paper microfluids	4	WU		1	48
3	Novel MIR light sources	6	NP		1	30
4	Modular MIR sensing platform	2	UULM		1	30
5	Data analysis and software development	1	NMBU		1	48
6	Demonstration and validation	3	BOKU		1	48
7	Communication, dissemination and exploitation	9	accelCH		1	48
8	Coordination and management	1	NMBU		1	48
		Total person- months				

Table 3.1b: Work package description.

WP number	**1**				**Start M1 – End M48**						**NEBIH**		
WP title	**Technology acceptance and techno-economic evaluation**												
Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	ccelCH	BIGH	eeberger	BAMA	HS
Person-months													-

Objective: To conduct comprehensive research on user expectations, user experience and technology acceptance for a portable, easy-to-use, broadly adaptable photonics device. The **specific objectives** are:

- Identify relevant stakeholders, explore user requirements and expectations towards the innovative multi-purpose photonic sensor solution.
- Organize the collection of user experience during the validation and demonstration process in WP6.
- Conduct comparative cost assessment for the MI-FI and HI-FI devices and reference methods considering realistic conditions.
- Assess user experience and technology acceptance among target users and stakeholders.

Task 1.1 Stakeholder expectations and user requirements survey (M1-M42) (lead partner: NMBU; participants: BOKU, ROMER, NEBIH, Seeberger, BAMA, WU). Since the exact apparatus of the portable PHOTONFOOD platform cannot be fully defined before the beginning of the project, we will follow an agile approach to user requirements. A kick-off workshop involving (a) the PHOTONFOOD WP leaders and (b) representatives of the prospective user organisations involved in the project (ROMER, NEBIH, Seeberger) will be organised at the beginning of the project (M1) to define a questionnaire for a stakeholder survey. The stakeholder survey will consist of an online questionnaire and a structured follow-up telephone interview. It will be conducted in Austria, Germany, Hungary, the Netherlands and Norway, covering 15 organisations per country from different levels of the food chain (3 from agriculture, 3 processors, 3 retailers, 4 labs and 2 border inspection points). At the project kick-off meeting, the PHOTONFOOD task leaders will develop an initial architecture envisioning. The process will then be iterated throughout the lifetime of the project on an annual basis. The component-specific requirements will be continuously updated within the tasks responsible for component development (WP2 to WP5). The system-level and platform-level requirements will be updated at workshops held during the annual project meetings in M8, M18, M25 and M35. Based on the stakeholder and user requirements, deliverable *D1.1: Report on stakeholder and user requirements* (M24) will be compiled.

Task 1.2 Collection and analysis of technical and user experience during validation and demonstration (M24-M34) (lead partner: NEBIH; participants: BOKU, NMBU, ROMER, Seeberger, BAMA). During the demonstration phase (when WP6 is active), system-level and platform-level requirements and actual experience will be collected on a continuous basis in close collaboration between the members of the WP6 team and the users and stakeholders involved in the testing and validation. The parameters measured in WP1 during the validation phase of the innovative photonic device (and in parallel with the reference methods) will include: time requirements, cost requirements, level of skill needed for operation, ergonomics, and user experience. User experience data will be gathered from the three main user groups involved in the demonstration activities (industry, retail and food inspection services), using qualitative (observations and interviews) and quantitative methods (user surveys). A standard usability framework model will be adopted (UTAUT: Unified theory of acceptance and use of technology) which allows a highly structured analysis of such mixed-methods data. The experiences will be analysed in a systematic way, presented to the WPs engaged in technology development (WP2 to WP5).

Task 1.3 Cost assessment (M24-M34) (lead partner: ROMER; participants: BOKU, NEBIH, NMBU, Seeberger). Based on the analysis of the cost elements recorded during the demonstration phase in Task 1.2, a comparative unit cost assessment will be conducted against the currently available reference methods. This part is of critical importance for the feasibility of the whole technology development. Historical data will be used to judge cost reduction effect of production scaling up, but the initial costs should remain within a reasonable threshold (the unit cost of one sample should remain under 3€ for mycotoxin testing even with the prototype). The calculation will include a Monte Carlo simulation of cost schemes for the IPD and the best available reference method. Results of this task will be discussed with the WP leaders dealing with technology development (WP2 to WP5).

Task 1.4 Technology acceptance and market potential (M36-M48) (lead partner: NMBU; participants: NEBIH, ROMER, WU). In the final task of WP1, the technology acceptance and future market potential will be assessed in key customer segments. Like in the previous tasks, a combination of online questionnaires and telephone follow-up interviews will be used for data collection. The target respondents will include: (a) senior executives in the industry, commodity distribution and retail who have overall responsibility for quality and safety in their respective organizations, and (b) senior representatives of food safety authorities with overall responsibility for standards and detection methods. The target sectors will not only include the value chains covered in the demonstration in WP6 but also other value chains where related quality and safety assessment problems occur. This will enable us to assess the further market and business development potential of the PHOTONFOOD technologies in other value chains. Key assessment indicators will include technology acceptance, customer expectations, upskilling/training requirements, probability of adoption and willingness to pay. Again, we will combine qualitative and quantitative methodologies in the assessment.

Role of partners: NEBIH will be leading this WP and is responsible for conducting a stakeholder survey in Hungary, collecting and analysing user experience, supporting the cost assessment and contributing to the assessment of technology acceptance. NMBU will be responsible for conducting the same activities in Norway, and will lead the assessment of technology acceptance and market potential. ROMER will be responsible for leading the cost assessment and will contribute to the collection of user requirements and experience and will support the assessment of market potential. BOKU will be responsible for conducting a stakeholder survey in Austria and will contribute to the collection of user requirements and experience, also to the cost assessment task. Seeberger and BAMA will contribute to the collection of user requirements and experience, and Seeberger will also play a part in the cost assessment task. NP, IRIS, WU and UULM will provide technical and cost related information on the photonic devices, as well as ensuring the efficient information flow and feedback loop between WP1 and the technology WPs (WP2 to WP5) through the whole project. Furthermore, UULM will be responsible for conducting a stakeholder survey in Germany, and WU in the Netherlands.

Deliverables: D1.1 Report on stakeholder and user requirements (NMBU, M24); **D1.2** Overview on technical and user experiences (NEBIH, M40); **D1.3** Report on cost assessment (ROMER, M36); **D1.4** Overview on final user requirements, technology acceptance and market potential (NMBU, M48)

WP number	**2**			**Start M1 – End M48**							**WU**		
WP title	**Paper microfluids**												
Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	accelCH	BIGH	Seeberger	BAMA	HS
Person-months													

Objective: To develop an integrated sample handling tool that will allow the user-friendly analysis of food contaminants using MIR detection. The **specific objectives** are:

- To develop paper microfluidic sampling, extraction and pre-treatment for food contaminants from hard and soft commodities.
- To develop a functionalized cartridge for the paper microfluidic devices that allows robust and user-friendly interfacing with spectroscopic detection.

Task 2.1 Paper microfluidic sampling and extraction (M1-M34) (lead partner: WU, participants: NMBU, UULM, BOKU, NEBIH, IRIS). In this task, solutions will be developed for allowing the entire sample-handling chain to be executed not only on-site, but by non-experts as well. This requires first of all a strategy

towards the acquisition of sample. In the case of hard samples, such as nuts, an extract needs to be prepared. In previous work (FoodSmartphone), an extraction module was developed for consumer testing in a cookie matrix. Further development in this WP will be required to adapt such a strategy towards different commodities. A dip-stick type sampler will then be developed to bring the sample onto the paper microfluidic device. On the other hand, when soft or liquid samples are tested, a different sampling strategy is required, where a swab type sampler can be used, which then directly feeds into the paper microfluidic device. To improve the extraction efficiency of the contaminants, while reducing the co-extraction of unwanted matrix constituents, the cellulose fibers can be chemically modified to allow specific chemical interaction with those contaminants.

Task 2.2 Paper microfluidic pre-treatment/preconcentration (M9-M48) (lead partner: WU, participants: NMBU, UULM, BOKU, NEBIH, IRIS). Once the contaminants are extracted from their matrix through the approaches in task 2.1, further purification and concentration will be required before MIR spectroscopic analysis. In order to achieve selective purification, several routes will be considered. Firstly, the use of immunochemistry will be explored. Combining immunocapture with paper microfluidic functionality is an attractive approach, as the location for the preconcentration can be precisely chosen to match the required dimensions for readout. Secondly, the use of chemical surface modification will be further explored. Paper can readily be modified with a plethora of functional groups, which allows for solid-phase extraction on paper. While this is less specific than immunocapture, it does allow for the devices to be more easily adapted for different applications. Thirdly, an innovative method in paper microfluidics will be developed through adapting a liquid-liquid extraction on a paper-based platform, as demonstrated in the initial work published by members of the consortium[43]. Through application of multilayered paper microfluidic devices, which are specifically modified, immiscible solvents can be preferentially retained in selected layers, which allows for the construction of a multilayered liquid-liquid extraction system[43]. At the end of this task, the most appropriate strategy will be selected for further developments in PHOTONFOOD.

Task 2.3 Integration and interfacing with 3D printing (M17-M48) (lead partner: WU, participants: NMBU, UULM, BOKU, NEBIH, IRIS). While it is crucial for any truly user-friendly on-site analysis to address both sample extraction and enrichment, it is equally important to consider operational simplicity and system integration. Therefore, this task focuses on the design and development of a cartridge that will contain all the elements developed in tasks 2.1 and 2.2. Here, 3D printing is exceptionally useful since it allows very fast prototyping, leading to iterative design of devices in a short period of time. Moreover, it will facilitate complex paper microfluidic device layout, such as stacking of several layers for the multilayer liquid-liquid extraction. In addition, the cartridge will shield vulnerable parts of the device, by exposing only the sample acquisition and read-out parts of the device. This is beneficial since it designates these parts as such for the end-users. Finally, protocols for operational simplicity and clarity will be delivered.

Role of partners: WU will be leading this work package and is responsible for the realization of the integrated sample handling tool. BOKU will provide assistance for developing appropriate sample pre-treatment. NMBU and IRIS will support the data acquisition and analysis. UULM will support the coupling of the sampling device with spectroscopic analysis. NEBIH will participate in the evaluation of the user acceptance of the paper-based microfluidics.

Deliverables: D 2.1 Deliverable report on paper microfluidic sampling, extraction and selective purification approaches for one target contaminant (DON) in a target commodity (wheat) (WU, M24); **D 2.2** Deliverable report on the development of an interface between a fully integrated paper microfluidic device and optical components (WU, M42); **D 2.3** Diversification of approach to at least a second target contaminant in a second soft sample commodity finalized (WU, M48).

WP number	**3**		**Start M1 – End M30**							**NP**			
WP title	**Novel MIR light sources**												
Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	accelCH	BIGH	Seeberger	BAMA	HS
Person-months													

Objective: To develop dedicated interband cascade (IC) and quantum cascade (QC) gain materials forming the basis for advanced light emitting diode (IC-LEDs) and monolithically tunable laser light sources.

Specifically, low-cost MIR LEDs will be developed emitting in the 1500-1800 cm^{-1} spectral window for detecting bio contaminants in a medium-fidelity (MI-FI) sensing device, whereas monolithic widely tunable lasers will be implemented in a high-fidelity (HI-FI) analyzer based on the same spectral range. The **specific objectives** are:

- To develop tunable ICL/QCL MIR lasers for high precision measurements.
- To develop IC-LED light sources for low cost contamination detection.

Task 3.1 Growth of cascaded epitaxy for low cost MIR ILEDs and high-performance tunable lasers (M1-M24) (lead partner: NP, participants: UULM, IRIS)

In this task, the required cascaded layer structures for IC-LEDs and cascaded tunable MIR lasers will be realized. In this context, dedicated band structure engineering is performed by means of specialized solvers of the Schrödinger equation using e.g. k·p methods. This band structure design is complemented by means of optical simulations of the resulting layer stacks with respect to the actual devices performance. Based on these simulations, and based on the input from WP1 and WP4, it will be decided if either interband cascade (ICL) or quantum cascade laser (QCL) structures are more suitable for the realization of the high performance monolithic tunable laser devices. The epitaxially grown multilayer structures will be routinely characterized by high-resolution X-ray diffraction, secondary ion mass spectroscopy and secondary electron microscopy images. Based on this analysis, the growth parameters will be optimized to fabricate high quality material that is required for the cascaded devices. In order to aid the design and to achieve the required broad band emission in the 1500-1800 cm^{-1} spectral range, gain measurements will be performed to iteratively optimize the performance of the epitaxy. Broad area lasers and related test structures will be processed to be able to rapidly forecast the device performance. One of the main objectives in this context will be the detailed band structure analysis versus the layer structure modifications to achieve fast progress.

Task 3.2 MIR IC-LED fabrication for MI-FI analyzers (M1-28) (lead partner: NP, participants: UULM, NMBU)

Task 3.2 is dedicated to the design and realization of low-cost IC-LEDs in the 1500-1800 cm^{-1} spectral region. This task is the main focus within this work package due to the intrinsic capabilities of scaling up the technology to the consumer market. Based on the results and layer structures from Task 3.1, optical simulations will be performed to investigate an optimal chip design and the simulations will be iteratively up-dated according to the obtained results. One key challenge will be to maximize the out-coupling of the light generated in the active zone which is confined to the chip by total internal reflection due to the high refractive index of the semiconductor material. In this context, specific out-coupling structures such as coatings, micro lenses and surface treatment schemes will be investigated and iteratively optimized. The chip layout will be transferred into the epitaxial material by means of suitable lithography, dielectric- and metal-layer deposition and removal steps. The readily processed chips will be pre-characterized on wafer level prior to singulation. In order to be used in a sensor, the IC-LED chips have to be appropriately packaged in a housing suitable for low-cost, high volume applications. Therefore, a SMD type of package is the first choice. Thermal simulations will be made for ensuring the best possible heat management of the chips and the SMD package to enable superior performance under application conditions. Application grade IC-LEDs will be characterized and delivered to WP4 for the integration into the MI-FI sensor. Special care will be taken with respect to the scalability of all steps, in particular the device characterization to ensure low cost devices. Lifetime and environmental tests that proof the robustness under operation conditions will complete the task.

Task 3.3 MIR tunable laser fabrication for HI-FI analyzers (M4-M30) (lead partner: NP, participants: UULM, NMBU). The initial simulations results and the decision regarding the optimal epitaxy structure (interband or quantum cascade respectively) from Task 3.1 will be the starting point of this task. Further optical simulations will be performed to define the initial chip design that enables the required tuning range in the 1500-1800 cm^{-1} spectral range. Suitable semiconductor structuring processes, such as optical and high precision electron beam lithography processing steps, will be investigated and iteratively optimized to realize the tunable monolithic lasers in combination with dielectric- and metal-layer deposition and removal steps. An extremely thorough optical characterization with respect to the achievable wavelength and side-mode suppression ratios at the pre-defined operation points will be required. This is due to the nature of the device design, based on the Vernier effect, that leads to a high degree of freedom in terms of laser operation. Those data will be analyzed in depth and fed back to the next iteration of chip design and processing. The integration of the lasers in a suitable housing and the definition of suitable electronics will complete the task.

Role of partners: NP will be leading this work package and is responsible for the realization of the IC-LED light source of the MI-FI device as well as the monolithic tunable MIR laser based on a cascaded laser scheme for the HI-FI device. UULM will contribute to the specifications and the assessment of the performance data

of the light sources. NMBU will participate in the evaluation of the signal achieved by the light sources with respect to the signal to noise ratio required.

Deliverables: D3.1 Advanced IC-LEDs hardware ready for MI-FI integration (hardware, report) (NP, M18); **D3.2** Monolithic tunable hardware ready for HI-FI integration (hardware, report) (NP, M26)

WP number	**4**		**Start M1 – End M30**									**UULM**	
WP title	**Modular sensor solution**												
Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	accelCH	BIGH	Seeberger	BAMA	HS
Person-months													

Objectives: To develop a portable and field-deployable (prototype) MI-FI and HI-FI analyzers combining the MIR LEDs developed in WP 3 with single-bounce diamond ATR elements, a grating, and a line array detector (MI-FI device), and the ICL/QCL light source developed in WP3 with advanced thin-film nanocrystalline diamond (NCD) waveguides and a single-element detector (HI-FI device) for in-field fungal infection analysis. The **specific objectives** are:

- To prototype a HI-FI MIR photonic sensing device and integrate ICL/QCL light sources.
- To prototype MI-FI MIR photonic sensing devices and integrate MIR IC-LED light sources.

Task 4.1 Thin-film nanocrystalline diamond waveguides (M1-M18) (lead partner: UULM; participants: NP). Task 4.1. is focused on the development of thin-film NCD chips (5x10 mm; actual waveguide thickness: 10-25 µm) readily fitting the MIR-Guide^R ATR sensing system. The chips will be characterized and optimized in iterative cycles for maximum performance (i.e., MIR radiation throughput and intensity of the evanescent field). In addition, the MIR-Guide^R system will be adapted to accommodate the smart paper strip developed in WP 2 for analyzing enriched/separated samples (i.e., ATR clip-on system pressing the detection zone of the paper strip (i.e., few mm^2) onto the surface of the NCD chip).

Task 4.2 MI-FI analyzer based on MIR LEDs (M1-M30) (lead partner: UULM; participants: NP, NMBU, IRIS, BOKU, WU, NEBIH). In Task 4.2, the hand-held low-cost MI-FI analyzer platform will be designed, developed, and prototyped, The device will comprise a 3D-printed housing accommodating the MIR IC-LED developed in WP 3, a dispersion grating (plastic w/gold coating; potentially 3D printed), and a line array detector. If a line array detector with a surface-deposited gradient filter will be used (e.g., Pyreos), the dispersion grating will be replaced with appropriate beam-shaping optics (3D printed; gold-coated plastic optics). The sensing interface will be established via a hemispherical or prismatic diamond ATR crystal. Using MIR spectral ray tracing (via software packages SPRAY and SCOUT, M. Theiss Software), an optimized MI-FI analyzer prototype will be built and tested according to its analytical figures-of-merit and via spiked test samples. After optimization, five hand-held MI-FI prototype systems will be fabricated ready for field testing during the case studies lined out in WP6, and together with end users. The performance of the MI-FI device will also be benchmarked against commercially available near-infrared LED-based low-cost analyzers (e.g., Consumer Physics) and evaluated for its utility addressing additional quick-test application scenarios.

Task 4.3 HI-FI analyzer based on tunable MIR QCLs (M1-M24) (lead partner: UULM; participants: NP, NMBU, IRIS, BOKU, WU, NEBIH). Task 4.3. is focused on the design, development and optimization of a high-resolution HI-FI analyzer integrating tunable monolithic MIR ICL/QCL technology developed in WP3, a single element thermoelectrically cooled MCT detector (e.g., Vigo), and the MIR-Guide^R ATR sensing interface with NCD waveguide chip for direct liquid (extract) analysis or for analyzing smart paper strips developed in WP 2. The HI-FI analyzer will be prototyped and performance optimized (i.e., analytical figures-of-merit, spiked test samples, etc.). As a final outcome, one portable prototype systems will be made available for field testing during the case studies specified in WP 6, and together with end users.

Role of partners:

UULM will be leading this work package and is responsible for the realization of the MI-FI and HI-FI devices. NP will provide the light sources. NP will further support the design of the MI-FI and HI-FI devices to ensure optimal integration of the light sources and help with the tunable laser operation. NMBU will evaluate the solutions with respect to data quality. IRIS will support the implementation of the software application. WU will help to optimize the photonic solutions for the interface with the microfluidics. NEBIH

and BOKU will provide the necessary input for ensuring that the photonics solution is prepared for the end user.

Deliverables: D 4.1 Thin-film waveguide-based sensing interface for HI-FI analyzer (UULM, M18); **D 4.2** Performance test report for portable HI-FI analyzer (UULM, M30); **D 4.3** Performance test report for handheld MI-FI analyzer (UULM, M24).

WP number	**5**		**Start M1 – End M48**								**NMBU**		
WP title	**Data analysis and software development**												
Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	accelCH	BIGH	eeberger	BAMA	HS
Person-months													

Objective: To develop a data analysis strategy for short wavelength range data from MI-FI and HI-FI data and to establish a software interface for mobile and laptop for field use. The **specific objectives** are:

- To use advanced data analysis to pre-process spectral data and calibrate MI-FI and HI-FI solution for detection of microbial and chemical contaminants in food.
- To provide a user-friendly interface, intuitive decision support tools, data management and data integration strategy.

Task 5.1 Spectral pre-processing of HI-FI and MI-FI data (M1-M24) (lead partner: NMBU, participants: UULM)

Existing methods for spectral pre-processing take advantage of the availability of a broad spectral range for the estimation of baselines and multiplicative effects. Since IC-LED and ICL/QCL sources provide a limited bandwidth, the existing methods for the pre-processing have to be further developed and adapted. Our general spectral pre-processing strategy consists of the following steps: 1) baseline correction and normalization; 2) interferant removal and 3) signal enhancement. The pre-processing reduces complexity of classification models and results therefore often in more robust models. Model-based pre-processing toolbox developed in BioSpec based on Extended Multiplicative Signal Correction (EMSC) will be used for implementing all steps including removal of interferant signals from paper-based microfluidics and solvents used for extraction. As an alternative route, we will evaluate augmentation of data in combination with deep learning approaches. In the on-going MIRACLE project (see Table 4), where only a few laser wavelengths are available for calibration this approach has turned out to be very powerful. Especially, in cases where limited data sets are available, augmentation has turned out to be a powerful approach for calibration when it is used in combination with deep learning approaches developed for spectral data[69]

Task 5.2 Classification and calibration models for benchmark data, HI-FI and MI-FI data (M12-M36) (lead partner: NMBU, participants: WU, BOKU, UULM).

At the beginning of the project, broadband MIR spectrum benchmark instrumentation at NMBU and UULM will be used to establish benchmark data for calibration and classification model building (Task 6.1). Models will be built on the full available spectral range (4000-500 cm^{-1}) and the short spectral range (1800-1500 cm^{-1}) of the PHOTONFOOD solution. For calibration, samples will be prepared, MIR and reference data will be obtained for 100+100 (clean+contaminated) samples for each contaminant. All samples will be measured by benchmark MIR instrumentation at NMBU and UULM, and reference analysis will be performed at BOKU and NEBIH. Data sets will be augmented by techniques from the field of deep learning according to the same lines as in the on-going MIRACLE project (see Table 4). Calibration models using our BioSpec toolbox of machine learning techniques involving a broad range of methods such as PLS-DA, RF, SVM, ANN and DCNN will be established to calibrate against threshold values used for contaminants, while PLS-R, RF, ANN and DCNN will be used to calibrate against the exact values to detect the amount of contamination. The obtained data will be used as a benchmark in the development of the HI-FI and MI-FI data. When the HI-FI and MI-FI instrumentation is ready, the same samples that were used to establish benchmark data and calibration models will be probed by HI-FI and MI-FI instrumentation. The obtained data will be used to establish calibration and classification models for HI-FI and MI-FI data, and to compare with benchmark data.

Task 5.3 Model validation and demonstration (M36-M48) (lead partner: NMBU, participants: BOKU, UULM, NEBIH). Models from Task 5.2 established on the short wavelength range will be validated in the

field demonstration involving relevant stakeholders. The models will also be validated by a blind test set and further refined and optimized in order to accommodate the use of the IC-LED and ICL/QCL spectral data.

Task 5.4 Software application development and data management (M12-M48) (lead partner: IRIS, participants: NMBU, WU, BOKU, UULM). During this task, IRIS will design and build a digital Global Control System (GCS) to allow the processing of the data from the sensors and its integration with the data analysis models developed in the first part of this WP. Early in the project a data management plan will be developed (D8.7 M6). The GCS will centralize the data management. It will comprise the hardware to host data acquisition, data processing and data management, and a friendly user interface to easily manage the PHOTONFOOD platform. A decision support tool, with specific user-oriented modules to support decisions on-site will be developed. It will be accessible over the web and mobile devices, such as smartphones and tablets. The hardware will consist of two main layers: an upper layer containing the processors and all the necessary peripherals (data storage units, terminals, external connectivity), and a lower layer comprising the communication with the analyser modules. Communications will take place through data buses. These will be mainly wireless whenever possible and wired when necessary. The GCS will have real-time capabilities and will be able to manage huge amounts of data. It will be able to integrate data delivered in standard industrial formats. The GCS will have a layered structure, and each layer will be accessible from the Graphical User Interface through the terminals. Firstly, Data Acquisition layer, which is the closest to the hardware, will manage the acquisition of data, as well as protocols, fault detection and hardware compatibilities. The valid acquired data will be stored in the general database to be accessible for higher layers. Secondly, the Data Processing Layer, which will convert the raw data into calibrated and filtered valid data to feed the calibration models developed in previous tasks. Finally, the Data Mining Layer, the highest data management layer, will produce the results of data analysis and provide decision support. The General Database will be designed and built to be as flexible and robust as possible, as compatible with multivariate nature of the data and real-time access as possible. The calibration models will have the capability to be updated with the feedback provided by the advanced AI algorithm executed in the cloud platform. The cloud application will be accessible over the web and mobile devices, such as smartphones and tablets. For the local application, there will be two versions: computer and mobile.

Task 5.5 Definition of the Cloud Architecture and communication with Food Trust™ (M36-M48) (lead partner: IRIS, participants: NMBU, UULM, advisory role for IBM). User requirements for the cloud platform will be defined. Different methodologies will be applied to set the user requirements, specifications, features, functionalities and the desired user experience. An in-depth analysis of existing technologies will be carried out to assess which are the most suitable ones to match the defined requirements. The in-depth architecture will be defined: backend, middleware, database, intelligent layer structures, etc. It will be a priority for the architecture to be flexible in terms of single-tenant or multi-tenant deployments, to guarantee its high performance to run intelligence layers. For that purpose, it will be designed how to tackle the envisaged computational workload and the interactions of the different modules. A Quality Analysis (QA) methodology will be in place (in keeping with software quality management: ISO 9126-3 and the subsequent ISO 25000:2005 quality model) to ensure consistent and standardized implementation. Communication with Food Trust™ platform for data integration will be ensured.

Role of partners: NMBU will be leading this work package and will be responsible for the algorithm development for the data pre-processing and the model building. IRIS will be responsible for the software development. BOKU and NEBIH will be involved providing reference analysis. IBM be involved in the development a strategy for transferring data into platforms such as the block chain-based IBM Food Trust™ data platform. UULM will assist in the interpretation of instrumental effects in the spectra and will be involved in activities related to data modelling and validation. WU will be participating in model development and software development by providing their feedback.

Deliverables: D 5.1 Pre-processing algorithms for HI-FI and MI-FI spectra established. (NMBU, M36); **D 5.2** Classification and calibration models for MI-FI and HI-FI instruments ready for fungi and mycotoxins in all matrices (NMBU, M36); **D 5.3** MI-FI and HI-FI calibration models established and validated (NMBU, M48); **D5.4** Digital GCS technical manual report describing the installation at hardware level and architecture used to develop the software (IRIS, M36); **D5.5** Digital GCS user manual report analysing the overall functionalities of the software platform, capabilities, buttons, user interfaces, etc. (IRIS, M40)

WP number	**6**	**Start M1 – End M48**	**BOKU**
WP title	**Demonstration and validation**		

Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	accelCH	BIGH	Seeberger	BAMA	HS
Person-months													

Objective: To conduct 3 case studies where the project concept and vision for farm-to-fork sensing are implemented and demonstrated. In each case study, monitoring data acquired by portable photonics-based devices will enable quick on-site risk assessment by the end-users at different entry points of the food chains. The **specific objectives** are:

- Validation: Validate the developed solution considering the user-expectations explored in WP1 by comparing the performance characteristics of the innovative photonic devices to reference methods.
- Demonstration: Test the innovative photonic solution via on-site operation through the 3 case studies, which includes the measurement of mycotoxins and presence of fungi in (i) wheat and (ii) peanuts, also the examination of (iii) levels of pesticide, antibiotics and fungi in products from aquaponics.
- Give feedback to WP1 about the usability of the IPD in real settings and to the technology developers (WP2-5) about the modifications that are possibly needed based on the demonstration activities.

Task 6.1 Building the spectral database (M1-24) (lead partner: NMBU, BOKU, participants: NEBIH, Seeberger, BIGH, WU). A database of spectra will be built for input to the model described in Task 5.1 and 5.2. In order to feed the spectral library with data and to validate our methods in a laboratory-based environment, well defined commodities will be provided by NEBIH, Seeberger and BIGH. It is important to note that separate sets of spectra are necessary for each specific matrix since MIR spectra of the latter are significantly different. Separate spectral databases for contaminated and clean samples are available for each matrix composed of nuts (such as peanuts, hazelnuts, and almonds), dried fruits, wheat and vegetables. It is critical that the ensemble of spectra is representative of the matrix and contamination. It is also critical that a large enough data set of clean and contaminated samples is available for training and testing of the chemometric model. An estimated 100+100 (clean + contaminated) samples need to be available for the creation of training and test sets in each matrix. With three measurement replicates, we will acquire approx. 600+600 spectra for each instrument (HI-FI, MI-FI, and benchtop benchmark instrumentation at NMBU and UULM). The final number depends on the repeatability of spectral measurements.

Task 6.2 Acquisition of reference data (M24-30) (lead partner: BOKU, participants: NEBIH, Seeberger, BIGH). For optimization and validation purposes, as well as for the preparation of matrix-calibrants of the developed flexible spectroscopic assays, 50 naturally contaminated samples for each matrix group will be analyzed by fully validated LC and LC-MS/MS methods developed by NEBIH and BOKU, respectively. These reference measures will be complemented by the determination of the concentration-related scope of the method. Therefore, an additional amount of 100 inoculated samples will be blended with non-contaminated sample material in order to prepare 10 different concentration levels. As non-contaminated sample material, bulk samples of 5-10 kg each of 3 different lots of the commodities wheat and peanuts will be used and mixed with the contaminated material to the appropriate concentration levels. Naturally contaminated, inoculated and non-contaminated sample material including different lots will be provided by NEBIH, Seeberger and BIGH.

Task 6.3 Performance and evaluation of a ring trial (interlaboratory comparison study) (M1-42) (lead partner: BOKU, participants: NEBIH, ROMER, UULM, NMBU). The trueness and the comparability of the measurement results obtained for mycotoxins in the naturally contaminated materials will additionally be ensured through the conduction of a ring-trial (interlaboratory comparison study) organized by BOKU. Natural contaminated sample material for each commodity will be tested by 8 different laboratories (BOKU, NEBIH, ROMER, UULM, NMBU) and finally subjected to a statistical evaluation. By comparing identical sample materials with different methods, a more accurate evaluation will be obtained regarding the measurement uncertainty in general, as well as the measurement quality of the new developed flexible spectroscopic assay in detail.

Task 6.4 On-site demonstration exercise at the end user level (M36-42) (lead partner: NEBIH, participants: BOKU, Seeberger, BIGH, BAMA). In order to test the applicability of the innovative and cross-validated spectroscopic approaches at end user level, for both farmers and official test laboratories, on-site measurements will be conducted. A prototype innovative photonic solution will be deployed to NEBIH, Seeberger, BIGH an BAMA. 1) At least 50 naturally contaminated samples for each commodity at different contamination levels from different sectors of the food chain (wheat producing farms, food

processors and retailers), will be collected by NEBIH and will be simultaneously examined by innovative photonic solution and the reference method used for official inspections under normal laboratory conditions at the authority. 2) Seeberger and BAMA will use the device for mycotoxin monitoring trial in a variety of nuts (peanut, walnut and almond). In addition, BAMA will use the device for monitoring trial in a variety of vegetables. 3) The objective of the demonstration at BIGH is to develop an easy-to-use, flexible, and broadly adaptable versatile analytic tool for mycotoxin, fungi and oomycetes detection that is also capable to analyse antibiotics and pesticide residues in end products. Technical supervision of the demonstration procedures will be provided by BOKU. Reference analyses for the collected samples will be carried out by BOKU and NEBIH.

Task 6.5 Customers feedback (M42-48) (lead partner: BOKU, participants: NEBIH, ROMER, Seeberger, BAMA)**.** After the devices have undergone a practicability test by the end users, this knowledge will be incorporated in order to improve the terminal devices. Feedback about the technological feasibility will be obtained in form of a questionnaire and in form of direct communication with the end users. The best suited approach, including a full validation report (or a combination of approaches) and suitable analytical protocols, will be brought to the attention of various end users along the different food chains and to official control labs throughout the EU via seminars, on-site customer exercises, publications and direct communication with DG Health.

Role of partners: The spectral database for the contaminant/commodities combinations will be built by NMBU in cooperation with NEBIH and Seeberger. BOKU will organize the validation studies. NEBIH will take care of the demonstration exercise assisted by Seeberger, BIGH and BAMA, and will be responsible for the acquisition of reference data as integral part for the validation of the flexible spectroscopic assays. BOKU, NMBU and associated partner ROMER will assist NEBIH and will coordinate the conduction of a ring trial (interlaboratory comparison study) in order to verify the comparability of the measurement results. BOKU will additionally provide LC-MS/MS reference data and participate in on-site demonstration exercises. NEBIH will finally initiate and coordinate a feedback loop between customers and producer in cooperation with ROMER, Seeberger and BAMA. UULM, WU and IRIS will participate in the validation and demonstration to make sure that possible technical issues can be solved on site.

Deliverables: D6.1 Report on preliminary comparability study (BOKU,M12); **D6.2** Report on completed validation studies (BOKU, M42); **D6.3** Report on completed demonstration exercises (NEBIH, M46); **D6.4** Evaluation of collected feedback data (report) (NEBIH, M48); **D6.5** Report on ring trial (interlaboratory comparison study) (BOKU, M42)

WP number	**7**			**Start M1 – End M48**							**accelCH**		
WP title	**Communication, dissemination and exploitation**												
Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	accelCH	BIGH	Seeberger	BAMA	HS
Person-months													

Objective: To communicate about the project, raise awareness on food safety, and to disseminate results through publications, presentations and focused training. The **specific objectives** are:

- Define and plan strategy and measures in multi-stakeholders communication and dissemination plan.
- Deliver cross-media communication and public engagement activities to raise awareness about PHOTONFOOD and food safety in society.
- Disseminate the research and study results to the scientific community and the food industry.
- Provide guidelines and recommendations to decision- and policy-makers regarding food safety.
- Develop a plan for the future use of the PHOTONFOOD platform and the individual technologies and a roadmap for exploitation.

Task 7.1 Communication and Dissemination plan (M1-M12) (lead partner: accelCH, participants: all). The task will focus on setting up a communication and dissemination plan (CDP) with the support of all partners. The CDP will be also discussed with members of the advisory board. We will develop an outline key strategy for project dissemination and communications, for both internal (project partners) and external

stakeholders. Further, we will identify dissemination tools and channels, define the target audience and evaluate the implementation and impact of activities and update the CDP.

Task 7.2 Cross-media communication to raise awareness in society (M1-M48) (lead partner: accelCH, participants: all). The following communication activities will be performed: (1). **Project website:** Set up and maintain the PHOTONFOOD website including 1) a web-based information resource for different target-groups, 2) integrated use of different media (e.g. blog, infographic) to enhance the outreach to society and, 3) an exhaustive list of the project results including all publications and presentation contributions. (2) **Communication material:** Digital and print material to support the outreach of the PHOTONFOOD project through existing and newly established networks will be developed. This will include a project flyer providing basic information (print and digital version), a poster and a roll-up for use at events organised by the partners or hosted by other relevant organisations to promote the project. All materials will follow the PHOTONFOOD design guidelines including the logo and visual elements to ensure a coherent style throughout all materials and channels. (3) **Audio/visual material:** Create interactive digital audio/visual material for is easy use and re-sharing through social media channels. This will include infographics to present complex information in a simple way to the general public as well as videos presenting the project activities and results including interviews with key partners will be created and distributed via the project website, social media channels and existing partner networks. (4) **Press releases and articles:** Publish press releases oriented towards the food industry and for mass media (general public) on outcomes from the project. In close collaboration with partner's media groups monitor and manage public relations related to the project and organise any requests for further information, statements and interviews.

Task 7.3 Disseminate research results to expert stakeholder groups (M13-M48) (lead partner: NMBU, participants: academic partners). The dissemination of the research results will be performed by using the following dissemination instruments: (1) **Peer-reviewed publications:** PHOTONFOOD partners will publish their findings and research advancements in high impact peer-reviewed journals, such as Journal of Biophotonics, Applied Spectroscopy, and Food Control. All partners are committed to following the Guidelines on Open Access to Scientific Publications and Research Data in Horizon 2020 that are in line with current trends in IP management and publication, and will establish a database of relevant journals for scientific dissemination. (2) **Scientific conferences and presentations:** All partners will present their findings through oral presentations, posters and conference papers at European and international scientific conferences, such as SciX, SPEC, Food Safety, PITTCON, and SPIE Photonics Europe. (3) **Cluster event:** accelCH will organise a cluster event in the first project year together with other projects funded under the ICT-37-2020 call to exchange knowledge and create synergies on how to handle key issues relevant for all projects to coordinate research developments. (4) **Training:** NEBIH, NP, WU and BOKU will integrate PHOTONFOOD results into training materials for already established training events that they organise for students on safety in the food chain and photonic technology. This will include training through summer and winter schools, lab settings and integration into established bachelor and master programmes. (5) **Policy brief:** NEBIH will create a short, to the point policy brief based on outcomes from WP6 (validation). These will be directed towards policy makers or stakeholders who can influence policy. The aim is to highlight the relevance and benefits of the new PHOTONFOOD approach and its related research as well as offer recommendations for change to reduce regulatory barriers and identify drivers for the uptake of the PHOTONFOOD solutions to improve food safety.(6) **White paper:** NEBIH will develop a white paper in the final year on "photonics-based novel technologies for food safety applications'' in the final year to provide recommendations to be considered for future guidelines and international regulations based on results from the validation WP (WP6). (7) **Local demonstrations:** NP will organise live local demonstration of the PHOTONFOOD platform, solutions and initial results derived from the validation in a realistic setting (WP6) to stakeholders along the food value at tradeshows, such as Analytica, Laser World of Photonics and local information events. UULM will organise live demonstration sessions of the devices locally in the lab as "Industry Information Days" and also hands-on 1-2 day workshops at UULM for industry with theoretical sessions and hands-on practical sessions. The local demonstrations will be held in the local language to ensure the wide dissemination of the PHOTONFOOD related results and offer a unique opportunity for networking with interested stakeholders. (8) **Final event:** To engage with members of parliament and policy- and decision makers, WU will organise a workshop as the final event. The workshop will consist of presentations of the PHOTONFOOD key results, the policy brief and white paper will be presented and a panel discussion with invited experts from our stakeholder networks covering ethical, legal, social and economic considerations will be organised.

Task 7.4 Future use and exploitation of the PHOTONFOOD platform and technologies (M1-M48) (lead partners: accelCH, HS, participants: all). The following measures will be taken for the future use and exploitation of the developed technology: (1) **Business Model development:** Potential business cases will be developed by the technology providers led by the Innovation Manager that will facilitate the future upscaling and uptake of PHOTONFOOD. The business model(s) will take into account results from the demonstration and validation (WP6) and outcomes and recommendations from all research WPs (WP1-5). Supported by all WP leaders, they will identify adoption barriers for the involved stakeholders and develop strategies to overcome these barriers by developing a PESTEL analysis to assess and monitor the macro-environmental factors that may impact our project and the developed technologies. (2) **Promotion of exploitation and future funding plans:** Detailed report to include the consortium's plan for commercialisation of the developed technology. Develop draft exploitation agreement to build on Consortium agreement. Actively seek licensees and future funding to ensure ease of commercialisation and funding during first year of exploitation. Subcontract services of company to carry out future funding and investment searches. Begin process of establishing the product in product range, draw up distribution agreement and supply chain agreements with partners. (3) **Strategic grant planning:** To facilitate the further non-commercial use of results and outcomes through follow-up projects, accelCH will pre-select funding programmes relevant to partners and their project results by identifying suitable funding instruments. accelCH will present an extensive list of funding opportunities within the consortium which will discuss all options and select the most suitable instruments. Based on partner feedback, accelCH will create a short list of the most suitable EU and national funding schemes, including requirements, timings and recommendations, develop a proposal preparation schedule and help with the initial set up of possible follow-up projects. (4) **Standardisation and guidelines:** The project results have the potential to be integrated into national food safety related guidelines. We will share our policy brief with decision and policy makers which will include recommendations for food safety guidelines. Our recommendations are not legally binding but could potentially influence regulations if our results are strong which could require countries to change their laws. (5) **Roadmap for exploitation** (RfE): the Exploitation Manager at HS will, together with the industry partners (especially NP) and the Innovation Manager at NMBU, develop a commercialisation strategy including an exploitation roadmap building upon the initial business models to examine the transferability of the PHOTONFOOD platform to the market based on outcomes from all research WPs. Sub-outcomes of the project such as individual technologies developed will also be closely monitored and documented to ensure their individual exploitable use. Thus, the RfE will comprise a plan for the future commercial and non-commercial use of results at Consortium level, group, and individual partner level. The RfE will cover the exploitation, starting during the lifetime of the project (short-term) and mid and long-term plans <4 years, and <10 years after the project ends).

Role of partners: accelCH will be leading this work package and in particular will be responsible for the development of the Communication and Dissemination Plan (Task 7.1), and for orchestrating all communication activities (Task 7.2). In addition, they will organise the Cluster Event in Task 7.3 and be in charge of all the documentation and reporting on WP7. NMBU will lead the partners in the dissemination of results to the target groups and will provide the Innovation Manager (Task 7.3). HS will lead the exploitation activities (Task 7.4), and in particular will be responsible for shaping the Roadmap for Exploitation. HS will furthermore pursue fabrication and commercialisation of the final HI-FI and MI-FI devices after the end of the project. NEBIH will be responsible for producing a white paper and policy briefs. NP and UULM will be in charge of the live demonstrations in Task 7.3. WU will organise the final event in Task 7.3. All partners will contribute to WP7 by disseminating and communicating about the project and results and by supporting the other partners in the WP7 activities.

Deliverables: (1) **D7.1 PHOTONFOOD website** including audience specific areas, digital media and blog features and a secured area for central document storage (accelCH, M2); (2) **D7.2 Communication and Dissemination Plan (CDP)** developed defining key strategies and implementation process required to implement all communication and dissemination activities (accelCH, M9); **D7.3 Interim communication and dissemination activities report** to provide an overview of the activities achieved to date and assess their impact by means of Key Performance Indicators. (accelCH, M24); **D7.4 Policy guidelines** for the integration of the PHOTONFOOD platform into the standard approach in food safety (NEBIH, M36); **D7.5 Live demonstrations** of the PHOTONFOOD solution (NP, M42); **D7.6 Roadmap for exploitation** including strategies for the future commercial and non-commercial use of the outcomes and results (HS,

M48); **D7.7 Final Scientific Symposium** presenting the research results and the platform to scientists, food industry professionals and decision and policy makers (accelCH, M48)

WP number	**8**		**Start M1 – End M48**									**NMBU**	
WP title	**Coordination and management**												
Participant no.	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	NMBU	UULM	BOKU	WU	NEBIH	NP	ROMER	IRIS	accelCH	BIGH	Seeberger	BAMA	HS
Person-months													

Objective: Scientific, financial, administrative and ethical management of PHOTONFOOD with focus on the time and cost-efficient implementation of the work plan. The **specific objectives** are:

- Provide the scientific and technical leadership, as well as the administrative and financial management, to ensure that PHOTONFOOD delivers qualitatively high results within the time schedule and budget of the project description.
- Establish a single contact point with the EU and maintain continuously contact with EU representatives.
- Perform a quality assessment of the progress, results and impacts of the project.
- Deliver timely the progress reports and budget reports to the European commission.
- Make sure that the Description of Action (DoA) and the consortium agreement are followed.
- Establish a digital platform and intranet for the project to enable the exchange of data and information among the consortium partners.

Task 8.1 Coordination (M1-M48) (lead partner: NMBU, participants: all). This task will perform the overall coordination of the scientific and technological developments of all WPs in order to make sure that the developed components and scientific findings are aligned. The Project Coordinator (NMBU) will establish a single contact point with the EU in order to communicate all results, scientific, financial, and related issues. A Project Manager (NMBU) and a Scientific Manager (Deputy Coordinator from NMBU) will assist the Coordinator (NMBU). The Executive Board (EB) and the General Assembly (GA) will meet biannually and annually, respectively, and arrange *ad hoc* meetings when required. The EB and the GA will monitor risks and project progress supported by the Coordinator and the Scientific manager. The coordinator will be supported by the project manager in all administrative issues and especially in reporting and finance. The milestones and the deliverables will be closely followed up by the scientific manager who will carefully review all progress plans. The Project Manager will be responsible for the on-time delivery of draft scientific and financial reports to the Project Coordinator. Gender equality will be monitored by the Coordinator and it will be made sure that the gender balance will be maintained throughout the whole project period.

Task 8.2 Financial management (M1-M48) (lead partner: NMBU, participants: all). The economy manager Christer Wulff-Olsen at NMBU will be responsible for the financial management of the project in close collaboration with the project manager. They will ensure the timely distribution of funds from the EC to the partners. During the kick-off meeting, the economy advisor will introduce all partners in payment routines and aspects of financial reporting. All partners will obtain templates and instructions for financial reporting. The economy manager will be responsible to prepare all financial reports and deliver them timely to the EC. All partners will be asked early to prepare their financial reports and each partner will be responsible to prepare the reports and deliver them to NMBU in time. The coordinator will monitor the financial management and inform the EU about all necessary financial transfers.

Task 8.3 Decision making, conflict management (M1-M48) (lead partner: NMBU). Each WP is led by one of the **partners** (WP leader) with a deputy WP leader from a different partner. This will ensure focus on the work that is agreed upon. The WP leaders take the routine decisions, communicate those with the Coordinator, monitor progress of output according to the work plan, and prepare all documents and documentations for their WP. The decision-making body will be the General Assembly, with 1 representative from each partner organization and chaired by the Coordinator (see details of composition in section 3.2), each partner has a vote and the ability to steer the project. Major strategic decisions will be made by the General Assembly, such as change of scope, IPR related decisions, conflicts, budget related decisions, etc. First line of day-to-day decision making will be up to the Executive board whose membership will comprise the Coordinator and the WP leaders. Matters of a strategic nature are floated to the General Assembly and will aim to be accompanied by a recommendation from the Executive board. The Project

Coordinator will handle any cases of fraud, unethical behaviour, and breach of procedures, however, such incidences are expected to be highly unlikely. The Project Coordinator will distribute a list of decisions to be taken to all participants prior to each meeting. Partners may request in writing for additional topics to be discussed, and the updated agenda is distributed before the meeting. All meetings are documented, including the outcome of all decisions and distributed to all partners.

Task 8.4 Organization of reporting to the EC (M1-M48) (lead partner: NMBU, participants: all).

NMBU, the Project Coordinator, will present reporting procedures during the kick-off meeting. The Project Coordinator will provide templates for the formal reports and updates during the course of the project when necessary. The Project Coordinator will prepare a brief semi-annual report and distributes it to all participants. Formal project reports will be prepared by the Coordinator in keeping with the contractual reporting periods stipulated in the Grant Agreement. The Coordinator will present this formal report, including finance, IPR, ethics, and gender issues to the EC on time (and during formal review meetings as proposed by the EC). The Coordinator will also prepare a brief non-confidential version of the formal report for public access on the website.

Task 8.5 Day-to-day management (Internet, helpdesk and back-office) (M1-M48) (lead partner: NMBU, participants: all). **A secure digital platform** for networked information sharing will be established to minimize costs and environmental impact of travelling and to allow rapid access to collated data. The platform will have an option for open-source content management. The digital platform will be used for all internal communications. Access will be restricted to consortium members. Different access levels will be incorporated based on user roles. It will provide tools for knowledge management and the development of databases, to share preliminary results and internal documents amongst partners, to discuss and edit pre-publication documents and deliverables. Each time new data is available, it will be made sure that the data is shared in the project database according standard formats which will be agreed on at the beginning of the project. This will ensure a steady exchange and sharing of all spectroscopic and metadata between all partners. This intranet will be maintained by NMBU. Shared documents, contact data, templates, agreements and links to official guidelines for reference and reports and conclusions of meetings will be stored on the password protected part of the website which will be maintained by NMBU. All periodic reports and collated reports will be stored in the project database.

Task 8.6 Organization of consortium meetings (incl. telephone/video-conferences, web meetings) (M1-M48) (lead partner: NMBU, participants: all). A kick-off meeting for all project partners including the General Assembly (and the Executive board as part of the Committee), the Advisory Board and representatives from the EC will be organized by the Coordinator (NMBU. The back-office of NMBU will take care of all aspects of the consortium meetings, as presented in paragraph 3.2, such as agenda, invitations, preparation of materials (hand-outs, programmes, etc.), organisation of rooms and equipment, catering and dissemination activities to the consortium in cooperation with the host of the meeting, i.e. typically the location of one of the WP leaders. NMBU will also organize and host all consortium telephone/videoconferences (regular according to schedule, and on demand). In total 4 General Assembly meetings, 6 Executive board meetings and several ad-hoc meetings, depending on the needs, will be organized. 2 plenary meetings will be organized in conjunction with an end user meeting in M35 and M45. The minutes of all meetings will be delivered in a timely manner by the Project Coordinator. In case of unexpected events (e.g. COVID-19) the procedures indicated by the EC will be followed.

Task 8.7 Maintenance of all legal agreements (M1-M48) (lead partner: NMBU, participants: all).

All partners will sign a Consortium Agreement, with detailed matters on access to know-how, intellectual property ownership, and liability in line with the information contained in Section 2.2b as the starting point. During the kick-off meeting, all contractual requirements on the Grant Agreement will be reviewed with all the partners to ensure that all partners recap on their commitments. All partners must report issues relevant to the legal agreements to the Coordinator in a timely manner. The Coordinator will decide on how to proceed with any deviation and will inform the Project Officer.

Task 8.8 Risk Management (M1-M48) (lead partner: NMBU, participants: all). Alterations to the work plan will be identified early by following partners tightly in consortium meetings, regular technical meetings and periodic reports. A control procedure will be established by the coordinator for handling identified alterations to the work plan. All partners will be asked to continuously monitor risk factors and to report problems timely to the coordinator. Changes will be monitored in order to respond to threats in a timely manner to ensure work packages continue smoothly. Areas of potential difficulties will be discussed with the respective partners and discussed in the consortium. The impact of problems will be considered thoroughly in the consortium and contingency plans will be activated and mitigation measures will be

discussed. It is important to make sure that mitigation measures are thoroughly discussed in the consortium and the impact of the respective mitigation measures are understood for all affected partners and tasks. If necessary, resources will be reallocated, and objectives will be recovered and rescheduled.

Task 8.9 Quality management (M1-M48) (lead partner: NMBU, participants: all). The project will develop a **quality handbook** for quality assurance and control that contains all data format definitions, methods, guidelines for communications, conflict management, innovation management and project reporting. The quality handbook will be continuously overseen and revised in meetings of the GA (discussed biannually). The external advisory board will give input to the quality handbook.

Task 8.10 Innovation management (M1-M48) (lead partner: NMBU; participants: all). **Freedom to operate:** the IPR landscape will be mapped in the beginning of the project and continuously monitored throughout the project's lifetime. This will ensure that relevant background IP is duly integrated into the consortium agreement and that the necessary licenses are negotiated by the project consortium. **IP strategy:** the Innovation Manager will develop and negotiate a coherent IP strategy with all project partners, balancing the different interests of the participants (secrecy clauses, patenting, trademarking, licensing, publishing). The IP strategy will be closely aligned with the exploitation strategy and roadmap (see WP7), ensuring that the IP strategy can leverage the further commercialisation and business development activities planned after the end of the project's lifetime and avoid undue conflicts between the consortium partners and with third parties. **Ongoing IP management:** the Innovation Manager will liaise continuously with the WP and Task Leaders to ensure that the IPR generated in the project is disclosed in a timely manner and protected in a way consistent with the IP strategy. Furthermore, the Innovation Manager will liaise continuously with the TTOs of the participating research organisations, and the legal departments/representatives of the commercial actors in the project, to ensure that all necessary agreements are made in time and potential conflicts (between consortium partners or with third parties) are proactively resolved or avoided.

Task 8.11 Data management plan (M1-M6) (lead partner: NMBU; participants: all). A data management plan (DMP) according to the Guidelines on Data Management in Horizon 2020 will be developed. The DMP will provide the procedures for efficient collection, safe storage, deep exploitation and evaluation of the data. The DMP will ensure that research data will be easily discoverable, accessible, assessable and intelligible, usable beyond the original purpose for which they were collected, interoperable to specific quality standards, whenever relevant. The database will be used as a first basis for exploitation of results and further exchange of knowledge and expertise.

Role of partners: NMBU will be leading this work package and is responsible.

Deliverables: D8.1 Next stage plans and internal progress reports (NMBU, M6); **D8.2** Interim Status Report (NMBU, M12); **D8.3** Communication framework between partners (NMBU, M24); **D8.4** Biannual progress reports, mid-term reports and final report (NMBU, M48); **D8.5** Organize initial, mid-term and final meetings (NMBU, M48); **D8.6** Report on gender, societal and ethical issues of exploitation (NMBU, M24); **D8.7** Data management plan (NMBU, M6).

Table 3.1c: List of Deliverables[75]

NO	Name	WP	Lead	Type	Diss. level	Date
D1.1	Report on stakeholder and user requirements	1	NMBU	R	CO	M24
D1.2	Overview on technical and user experiences	1	NEBIH	R	CO	M40
D1.3	Report on cost assessment	1	ROMER	R	CO	M36
D1.4	Report final user requirements, technology accept., market potential	1	NMBU	R	CO	M48
D 2.1	Deliverable report on sampling, extraction and selective purification approaches for one target contaminant (DON)	2	WU	R	CO	M24
D 2.2	Deliverable report on the development of an interface between a fully integrated paper microfluidic device and optical components	2	WU	R	CO	M42
D 2.3	Diversification of the paper-based microfluidics to at least a second target contaminant in a second commodity finalized.	2	WU	R	CO	M48
D3.1	Advanced IC-LEDs for sensor integration (hardware)	3	NP	DEM	CO	M18

[75]**Key:** DL Dissemination Level; R Document, Report; DEM Demonstrator, pilot, prototype, plan designs; DEC Websites, patents filing, press & media actions, videos, etc.; OTH: Software, technical diagram, etc.; PU Public, fully open; CO Confidential, restricted to consortium set out in Model Grant Agreement; CI = Classified information as referred to in Commission Decision 2001/844/EC.

D3.2	Monolithic tuneable lasers for sensor integration (hardware, report)	3	NP	DEM	CO	M26
D4.1	Thin-film waveguide-based sensing interface for HI-FI analyser	4	UULM	DEM	CO	M18
D4.2	Performance test report for MI-FI analyzer	4	UULM	DEM	CO	M30
D4.3	Performance test report for HI-FI analyzer	4	UULM	DEM	CO	M24
D5.1	Pre-processing algorithms for HI-FI and MI-FI spectra	5	NMBU	OTH	CO	M36
D5.2	Classification and calibration models for MI-FI and HI-FI instruments ready for fungi and mycotoxins in all matrices	5	NMBU	OTH	CO	M36
D5.3	Calibration and classification models validated	5	NMBU	OTH	CO	M48
D5.4	Digital Global Control System (GCS) Technical Manual	5	IRIS	OTH	CO	M36
D5.5	Digital Global Control System (GCS) User Manual	5	IRIS	OTH	CO	M40
D6.1	Report on preliminary comparability study	6	BOKU	R	CO	M12
D6.2	Report on completed validation studies	6	BOKU	R	CO	M42
D6.3	Report on completed demonstration exercises	6	NEBIH	R	CO	M46
D6.4	Evaluation of collected feedback data (report)	6	NEBIH	R	CO	M48
D6.5	Report on ring trial (interlaboratory comparison study)	6	BOKU	R	CO	M42
D7.1	PHOTONFOOD website	7	accelCH	DEC	PU	M2
D7.2	Communication and Dissemination Plan (CDP)	7	accelCH	R	CO	M9
D7.3	Interim communication and dissemination activities report	7	accelCH	R	CO	M24
D7.4	Policy guidelines	7	NEBIH	R	PU	M36
D7.5	Live demonstrations	7	NP	DEM	PU	M42
D7.6	Roadmap for exploitation	7	HS	R	CO	M48
D7.7	Final Scientific Symposium	7	accelCH	OTH	PU	M48
D8.1	Next stage plans and internal progress reports	8	NMBU	R	CO	M6
D8.2	Interim Status Report	8	NMBU	R	CO	M12
D8.3	Communication framework between partners	8	NMBU	OTH	PU	M24
D8.4	Six month progress reports, mid term reports and final report	8	NMBU	R	CO	M48
D8.5	Organize initial meetings, mid term meetings and final meetings	8	NMBU	OTH	CO	M48
D8.6	Report on gender, societal and ethical issues of exploitation	8	NMBU	R	CO	M24
D8.7	Data management plan	5	NMBU	R	CO	M6

3.2 Management structure, milestones and procedures

Project Management Organisation structure and decision-making procedures

The coordination tasks defined in WP8 concern the overall coordination of the technological and scientific developments within the different WPs to ensure alignment of the developed products and scientific findings. **The Coordinator, Prof. Dr. Achim Kohler** (NMBU), will be responsible for the robust coordination of the PHOTONFOOD consortium and ensuring that the project is within scope, is delivered on time, within budget and to a high quality to ensure impact. Prof. Kohler is a well-renowned and well-respected academic from the MIR spectroscopy and data analysis field globally. Prof. Dr. Kohler has been involved in many international projects and has been successful co-founder of three spin-off companies of his university. He is an ideal figure to lead such a project and he also has the organisation and leadership skill-sets to effectively perform this role, which is demonstrated through his role as the Head of the Biospectroscopy and Data Modelling group at NMBU with a team of 20 researchers under his leadership. Prof. Kohler will also be responsible for communicating all results, scientific and financial, and related issues with the EU representative.

The Coordinator will be assisted by a **Project Manager** and a **Scientific Manager** (Deputy Coordinator) from NMBU. The Project Manager will assist the Coordinator in administrative issues related to reporting and finance. The highly experienced Project Manager **Stine Brekke Vennemo** (NMBU) will be responsible for the timely delivery of draft scientific and financial reports to the Coordinator. Stine Brekke Vennemo has her background in data engineering and will further be responsible for the data management in the project. She will also act as data protection officer and be responsible for managing all personal data according to GDPR described in section 5.1 (https://gdpr.eu/). The Scientific Manager Assoc. Prof. Volha Shapaval (NMBU) will monitor the progress of the milestones (see table 3.2a) and deliverables (see table 3.1c) and will (pre-)review all project reports. The **Dissemination Manager** Dr Johannes Ripperger (accelCH) will be responsible for leading the consortium in further planning and implementation of the project communication and dissemination activities, and ensuring the project reaches out to different audiences in the most effective way.

The **Innovation Manager** Prof. Dr Scholderer (NMBU) will be responsible for knowledge management within the consortium. He will lead the consortium in protecting the project results where appropriate, and in maintaining confidentiality of the results until the IP is protected to safeguard the innovation impacts. Prof. Scholderer has vast experience in relation to knowledge management in collaborative projects. He has experience of leading patent applications, open access publications, etc. in past projects. The **Exploitation Manager** Dr Julian Haas (HS) will lead the exploitation of the project results. Dr. Haas has experience from other projects (e.g. H2020 MIRACLE, see Table 4) on how to identify stakeholder demands and market opportunities.

Each WP is led by a WP leader who take routine decisions communicate those with the Coordinator, monitor progress of output and budget according to the work plan, and prepare all documents related to their WP.

Table 3.2a: List of milestones

Milestone	Milestone name	Related WP	Date	Means of verification
MS1.1	Component-level user requirements update presented to WP leaders	All WPs	M25	Feedback to the development WPs (WP2-5) about the updated requirements
MS1.2	System-level user requirements update presented to WP leaders	All WP	M37	Report on system-level user requirements
MS2.1	Portable Microfluidics Pre-prototype	WP2	M24	Evaluation of specifications and design
MS2.2	Portable Microfluidics Prototype	WP2	M36	Validation with MI-FI & HI-FI prototypes in lab validation
MS3.1	IC-LEDs optical components	WP3	M18	Evaluation of the spectral target coverage (1800-1500 cm^{-1})
MS3.2	Monolithic tuneable lasers	WP3	M28	Evaluation of the spectral target coverage (1800-1500cm^{-1})
MS 4.1	Five handheld MI-FI analyzer prototypes for demonstration	WP4, WP3	M30	Evaluation of MI-FI analyzer prototype design and performance on real samples
MS 4.2	One portable HI-FI analyzer prototype for demonstration	WP4, WP3	M24	Evaluation of HI-FI analyzer prototype design and performance on real samples
MS 5.1	Software Alpha release	WP5	M36	Pre-processing and calibration models validated in lab
MS 5.2	Software Beta release	WP5	M48	Pre-processing and calibration models validated during demonstration
MS 6.1	Portable HI-FI solution validated	WP6	M48	Performance validated in field and end users feedback implemented
MS 6.2	Handheld MI-FI solution validated	WP6	M48	

Two project management bodies are envisaged (see Fig. 17): The **Executive Board (EB)**, which is an executive management body of the project, and the **General Assembly (GA)**, which is the strategic steering and decision-making body. Day-to-day decision making will be the responsibility of the Executive board, chaired by the Coordinator, whose membership will comprise the Coordinator (chair), Scientific Manager (Deputy Coordinator), Project manager and WP leaders. The Executive board will meet as frequently as required with a minimum of every 6 months. In general, occasions will be sought to combine activities to minimise travel costs. The decision-making body on major strategic decisions will be the General Assembly, chaired by the Coordinator, whereby each partner organisation will appoint one representative. The GA will in addition include the Coordinator (chair), the Scientific Manager, the Project Manager, the Exploitation Manager, the Innovation Manager, and the Dissemination Manager). Each partner is assigned one vote to ensure a democratically run project (2/3 majority voting). Major strategic decisions will be made by the General Assembly, such as change of scope, IPR related decisions, conflicts, budget related decisions, etc. All matters of a strategic nature will be floated to the General Assembly and will aim to be accompanied by a recommendation from the Executive board. The GA suggests refocusing of research directions, proposes the publication plan and decides on conflicting issues such as prioritizing samples in case of a temporary breakdown or other contingency situation and subsequent mitigating actions (see table 3.2b). The General Assembly will meet once per year.

In addition, a voluntary **Advisory Board** has been established that will provide guidance and advice to the consortium through its international experts in themes of high relevance for the project. To this end, the Advisory Board (AB) of PHOTONFOOD involves two experts from the field of photonics (Prof. Gottfried

Strasser from TU Vienna, Austria, expert in light sources and Prof. Goran Mashanovich from University of Southampton, an expert in waveguides). IBM as a global cloud platform and cognitive solutions company represented by Loek Vredenberg who will be part of Advisory Board to engage with their blockchain based Food Trust platform and supporting skills. IBM will contribute to developing an interface between the PHOTONFOOD software and the IBM Food Trust platform for integrating data and utilizing data for improving food safety and reducing waste along the food chain. The Directorate General for Health and Food Safety/European commission in Brussels represented by Frans Verstraete will participate in the advisory board as a policy maker organization. The Advisory Board will provide an important mechanism for providing guidance on the General Assembly. It is noteworthy that the Advisory Board is a voluntary body. The Advisory Board will sign a confidentiality agreement with the Coordinator in order to safeguard the confidential aspects of the project that they might be exposed to. The members of the Advisory Board will have no decision-making capacity in the project. The AB will meet three times with the GA. Two times by video meetings and two times in person. A budget for this is foreseen in the NMBU budget. All advisory board members have expressed their willingness to join the Advisory Board (LOIs available, see Annex) should the PHOTONFOOD project be selected for funding. The advisory board members will provide additional 'reality checks' from the outset of the project. Since the consortium of PHOTONFOOD has a broad coverage of stakeholders that are strongly involved in the user acceptance and validation and demonstration activities, the advisory board was chosen to cover and policy maker members and scientific members to have further reality checks for the technological developments in the project.

Meetings

There will be 5 meeting points in the project during which one or several groups meet (Meetings are indicated in the GANTT Chart). The first of five Executive board (EB) meetings will take place during the Kick-off meeting (M1) in Norway. NMBU will present the decision-making and financial procedures and will put the corresponding documentation on the internal website. The other four EB meetings will take place in months M12, M22, M30, M48. Comprehensive progress reports and plans for the next 4-6 months will be presented by the WP leaders. These meetings serve to solve practical and scientific issues of non-strategic nature and to discuss immediate actions to be taken case milestones or deliverables cannot be delivered on time.

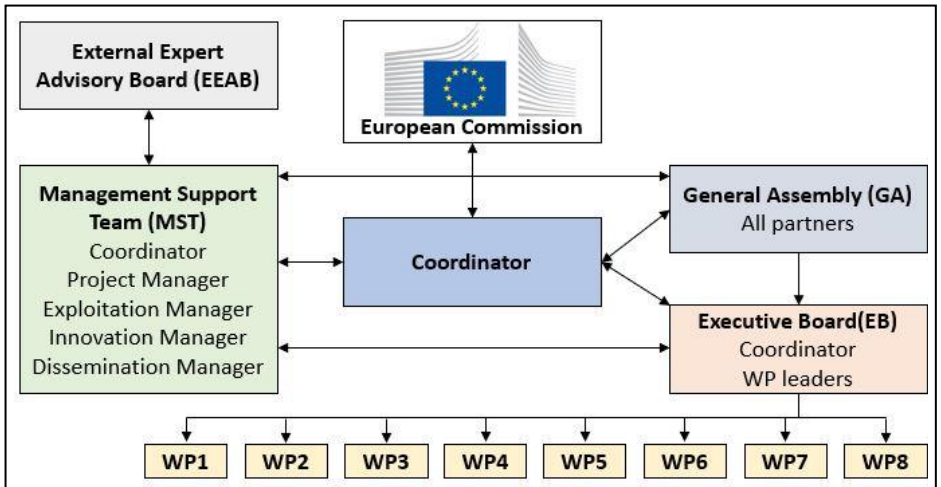

Figure 17: Management structure.

The four General Assembly (GA) meetings will take place during the kick-off meeting in M1 and in months M12, M30 and M48. At the GA meetings, major strategic decisions are made, the progress of the entire project is discussed in detail in cooperation with the Advisory Board (see below) and plans for the next period are reported. The General Assembly will verify that the project meets both deliverable and milestone targets. If there is a major decision point in a WP which could affect the scope of the work for example, the issue will be brought first to the Executive board and then to the General Assembly for decision making. The Project General Assembly will have an agenda which covers at least the following points: i) Update of progress against implementation plan; ii) Update on next stage of implementation; iii) Financial issues; iv) Contractual issues; v) Intellectual property issues; vi) Update on meetings stakeholders; vii) Gender implementation plan; viii) Dissemination of project results .

The Advisory Board (AB) meetings will take place in months M1, M22 and M48. In addition, annual videoconferences will be organised between the General Assembly and the Advisory Board. Each meeting will be held at the location of one of the WP leaders. The host will provide meeting rooms, lunches and one formal dinner for all attendants. The meeting in Month 12 will be held at UULM. The meetings at M30 and

M48 will be held in connection with end user meetings. Other meeting locations will be decided at the kick-off meeting. Videoconferencing will be used to address specific issues that might arise during the project or where specific decisions are required. WP leaders will also need to hold separate ad-hoc meetings of those partners involved in activities, and again these will be combined wherever possible with other meetings. The Coordinator will chair each meeting and will produce written minutes of each meeting, which shall be the formal record of all decisions taken. Minutes of each meeting will be distributed by e-mail to all project partners within a week. All Executive board Members and General Assembly members can suggest topics for the meeting agenda. Additionally, it is expected that WP leaders could be invited to specific stakeholder association meetings to describe specific commodity related aspects of the project. The Agenda will be a mixture of progress reports from WPs where there are non-confidential scientific and innovation achievements to report and items generated by the Project General Assembly that require end-user input. Reporting Procedures: In addition to formal deliverables identified in table 3.1c, WP leaders will provide the Coordinator with bi-annual and annual internal progress reports as follows: (i) bi-annual reports summarising progress on milestones and deliverables, unexpected results, and future strategies, and (ii) annual report on progress, highlighting any planning issues and also providing a summary of financial resources for the WP. The internal reports will feed into the Contractual reports required by the Grant Agreement.

In addition, all partners will contribute to the formal project progress reports and will submit formal financial statements as appropriate in line with the requirements of the Grant Agreement. Communication amongst the consortium will be enabled through a secure section of the PHOTONFOOD website (by password access), which will also serve as a structured repository for project documents and will be the main project platform for internal exchange of information.

Appropriateness of the organisational structure and decision-making mechanisms

This consortium includes European members that have a remit to address a pressing societal challenge in a 48-month work plan through the delivery of a series of ambitious scientific and technological advances and ensuring high impact. To this end, we are confident that a core project management unit made up of the Coordinator, Scientific Manager and Project Manager, each with key roles and responsibilities is a realistic and robust approach. This core unit and the supporting Executive board provide a sufficiently decentralised leadership to enable this ambitious project highly manageable. In the interest of a democratic and inclusive project for all partners, the existence of a General Assembly that can focus on the strategic direction of the project will ensure that all partners have a voice in how the project is run and all are afforded an opportunity to steer the project ensuring it delivers impacts for all partners and for the wider stakeholder communities through external engagements and interaction with a highly relevant and influential Advisory Board.

Innovation Management

Since PHOTONFOOD intends to deliver innovations with a real impact on food safety, it is crucial that the academic and industry partners in the consortium work in close collaboration with end-users and stakeholders. This has been taken carefully into account in the design of and the interaction between the project WPs. In WP1 and WP6, end-users are directly involved as project partners, defining user requirements, participating in the demonstrations, and providing access to facilities and materials. Ultimate success, however, will require uptake by much larger parts of the food and agribusiness sector. Therefore, WP1 will engage also customers, users and stakeholders from outside the project already at an early stage. This will ensure that the project can respond rapidly to changes in customer needs, user requirements, market conditions, regulatory initiatives, and complementary or competing technological developments. The PHOTONFOOD prototypes will be ready in M24 and M36, respectively. The validation, demonstration and systematic feedback-gathering activities involving customers, end-users and stakeholders will ensure a market-oriented and user-centric approach, substantially increasing the likelihood of commercial success of the developed solutions. Together, these mechanisms will establish an effective market feedback loop for our project.

In order to ensure that the that the innovations and market opportunities are optimally managed, the project has created two special roles. The **Innovation Manager** (Prof. Scholderer, NMBU) will be responsible for knowledge management within the consortium during the project's lifetime. He will lead the consortium in protecting the project results where appropriate, maintaining confidentiality of the results until the IP is protected to safeguard the innovation impacts, and developing the IP strategy of the consortium. The **Exploitation Manager** (Dr Haas, HS) will be responsible for the planning and initiation of the commercialisation and further business development activities after the end of the project's lifetime. The Innovation Manager and the Exploitation Manager will work in tandem, ensuring (a) that the interests of the consortium members are properly balanced against the interests of the parties which will commercialise the

technology, and (b) that the IP strategy of the consortium provides support and leverage to the further exploitation and business development plans. Both will be part of the Management Support Team, ensuring that innovation and exploitation are prioritised in all decisions made by the project.

Risk management
During the project, we will systematically analyse and manage risks that could jeopardise the implementation of the project, its scientific and technological outcomes, as well risks in relation to IPR, or the ability to deliver the projects innovation and long-term impact. The tasks and the organisational structure were analysed for their specific risks per WP. The severity of each risk was analysed and a plan for a solution was suggested. In case a risk does happen, the partner will immediately inform the work package leader who will inform the Coordinator. The Coordinator will directly inform the EU representative. Solutions will be discussed between the coordinator, the WP leader and the partner and proposed to the EU representative and only be implemented after discussion with and written approval of the EU representative. Consequences for distribution of budget will be identified and agreed upon with EU representative before the implementation of the proposed solution. At proposal stage, a number of critical risks have been identified and discussed in table 3.2b below including details on risk mitigation action to be taken.

Data management
A data management plan will be developed in WP8 according to the Guidelines on Data Management in Horizon 2020 (D8.7 M6). Although the DMP is not a fixed document and will evolve during the lifespan of the project. It covers the following elements: 1) Data set reference and name, 2) Data set description, 3) Standards and metadata, 4) Data sharing, 5) Archiving and preservation (incl. storage and backup). Details on the agreed standards for data collection, data exploitation (accessibility/shared), data preservation and dissemination of data results are provided in the Impact sections.

Table 3.2b: Critical risks for implementation

Description of risk (likelihood: Low/Medium/High)	WP(s)	Proposed risk-mitigation measures
Time Planned project activities cannot be realized in time as planned / M	All	Proper planning and tight regular follow up of the project work during the implementation phase. Several activities will start in parallel.
Cost Project activities cannot be completed within the budget / L	All	Careful planning and up-date follow up of the budget during the project period.
Management Key person(s) leave the partner organisations / M	All	Every key person has a competent named deputy in each WP.
Management One of the partners leaves the consortium / L	All	Motivation of partners by allocating suitable roles for them in the consortium. Re-allocating the partner's tasks to the other partners, or if not possible, recruiting a replacing partner according to the rule
Management Partners are reluctant to IP sharing / L	All	Carefully prepared Consortium Agreement that sets the IP sharing principles.
Implementation		
Technical User acceptance of the new photonic solution is low/L	WP1-WP5	User requirements and acceptance data from WP1 will be continuously feed to the WPs responsible for technology development to mitigate this risk
Technical Investment cost and unit cost of laboratory tests will exceed the estimated threshold / L	WP1-WP5	Alternative technical scenarios have been identified during the project proposal development to mitigate this risk. The HI-FI and MI-FI solutions all bear marketing opportunities that could be exploited.

Technical WP2: The desirable preconcentration clean-up effect is insufficient for the application/L WP2: The development of a fully integrated device could take long, which would result in delays in other WPs/L.	WP2	We have elaborated three strategies, based on existing research/concepts, for selective enrichment. In case of failure, there are back-up options readily available. The development of the different modules of the PHOTONFOOD solution is done in parallel. The modules will be combined in various stages of completion to ensure compatibility. Thus, if one modulus is delayed, the other WPs can still be achieved in time, whilst still assure that an integrated solution can be offered towards the end of the project.
Technical Delays due to unscheduled down-times of equipment/L Light sources do not meet expected out-put powers and wave-lengths/M	WP3	Standardized preventive maintenance plans for all critical machines and tools to minimize and avoid unscheduled down-times. establishment of alternative processing routes where possible, e.g. on 2nd tools 1) Usage of established simulation tools to start with refined structures and early tests of gain structures by fast test processes e.g. via broadboard area lasers to enable quick feedback for improvement. 2) Careful time-planning to allow for 2-3 iterations minimum
Technical Calibration models having not sufficient sensitivity/specificity/M.	WP5	Mitigation measure: more measurements required in WP6.
Management Recurring COVID-19 outbreaks in European countries result in restrictions on travel and of events and project meetings / M	WP7, WP8	Should it be required, project meetings can be held via video conferencing. The live demonstrations (D7.5) could be postponed or turned into short videos published online. The Final Scientific Symposium (D7.7) would have to be turned into an online event.
Technical Access to labs is restricted or limited as result of recurring COVID-19 outbreaks. / M	WP1-WP6	The possibility to carry out work in laboratories of another partner will be assessed. During periods of restricted access to labs, focus on theoretical work, documentation, reporting and communication tasks. If needed, seek solution with Project Officer, e.g. postponing tasks and extending the project.

3.3 Consortium as a whole

The PHOTONFOOD consortium has a wide geographical and cultural coverage across the Europe (Fig. 18). This naturally provides to the project an understanding on different business and healthcare practices, while enables to access several stakeholders and end-users providing wide feedback and promoting the project.

The Norwegian University of Life Sciences (NMBU) is located close to Oslo and will contribute with top expertise in spectroscopy, data analysis and experience in food science, microbiology, user acceptance and innovation. NMBU will provide the scientific manager, coordinate the project, lead WP8 and provide project manager and coordinator. NMBU has recently started Norway's first data science program and has top expertise in data analysis related to spectroscopy. NMBU will lead the data analysis and software development in WP5. NMBU has further top expertise in innovation and provide the innovation manager for the project.

The University of Ulm (UULM) located in the south of Germany, represented by the Institute of Analytical and Bio-analytical Chemistry is worldwide leading the development and application of innovative (micro- and nano-) analytical platforms providing advanced methods and technologies in sensing, diagnostics, and imaging of complex molecular processes relevant to environmental, industrial, and medical/biological applications. UULM will lead WP3 and be responsible for developing the prototypes of the HI-FI and HI-FI devices.

The University of Natural Resources and Life Sciences, Vienna (BOKU) located in Austria and represented by the Department of Agrobiotechnology (IFA-Tulln) is a leading institution in the area of agro-biotechnology with a special emphasis on mycotoxin control and reduction since 1994. BOKU will lead WP 6 on Demonstration and validation. BOKU will also summarize the feedback of WP1 on the usability of the

Organisation Short Name	NMBU	
Organisation Legal Name	Norwegian University of Life Sciences	
Country	NORWAY	
Website	https://www.nmbu.no/	

Description of the legal entity

The Norwegian University of Life Sciences (NMBU) in Ås is Norway's largest life sciences university with 5100 students; 1700 employees; 800 scientific staff; 580 PhD students; approx. 100 PhD degrees per year.

- The Faculty of Science of Technology (REALTEK) at NMBU has key expertise in biophysics, spectroscopy and data analysis. REALTEK has invested heavily in being at the forefront in advanced multivariate data analysis and vibrational spectroscopic techniques (NIR, FTIR and Raman *etc.*). Today "Biospectroscopy and data modelling" (BioSpec) is identified to be one of the main research areas at REALTEK. The BioSpec group at is at the forefront of application of vibrational spectroscopy techniques in life sciences ab and data science for vibrational spectroscopy. The group consists in average of 20 PhDs, PostDocs and Researcher. The academic staff of the group consist of Prof. Dr. Achim Kohler and Assoc. Prof. Dr. Volha Shapaval. The group has pioneered the development of algorithms for scatter correction and signal standardization in infrared spectroscopy. Over many years, the group has been developing so-called model-based pre-treatment approaches for spectral pre-treatment and data analysis methods for calibration and classification which are in use by a broad community in infrared spectroscopy in life sciences and biomedical vibrational spectroscopy.
- From the School of Economics and Business at NMBU, the Analytics, Innovation and Digital Business Transformation group will participate in the project. The group is led by Prof. Joachim Scholderer and consists of two professors, two associate professors, one postdoctoral fellow and four PhD students. The group cooperates closely with NMBU's TTO and the various incubators and accelerators in NMBU's innovation ecosystem. It will contribute with its expertise in the management of life science technology innovations, including IP strategy, market potential assessment, stakeholder and user acceptance along the value chain, forecasting, risk assessment and commercialization.

Main role and task in the project

The BioSpec group headed by **Prof. Achim Kohler (M)** will coordinate the project and lead the research related to the WP5. The BioSpec group will be strongly involved in spectroscopic measurements with benchmark and PHOTONFOOD devices in WP6. The group of **Prof. Joachim Scholderer** at the NMBU School of Economics and Business will be involved in the usability and technical-economic assessment activities (WP1), the demonstrations (WP6), the exploitation of the results (WP7) and the project management (WP8). Prof. Scholderer will be the Innovation Manager of the project.

Profiles of persons primarily responsible for carrying out the proposed activities

Prof. Dr. Achim Kohler (M) is Professor in Physics and an expert in spectroscopy and data science. Kohler leads the BioSpec group at NMBU-RealTek. He is an internationally recognized leader in data modelling for vibrational spectroscopy with 138 peer-reviewed papers, 10 book chapters, 3 patents in the field. His h-index is 40. Kohler has been initiation, leading and been participating in a high number of national and international projects. He has been participating in 22 projects funded by the Norwegian Research Council and project leader for 11 of them. Kohler has been participating in 6 EU-funded projects and been coordinating one of them. Kohler has been further participating in numerous Scandinavian projects. Kohler is the co-founder of three university spin-off companies. Further, Kohler has been participating in a course in Innovation management and Entrepreneurship at the Norwegian School of Management, Oslo and Stanford/Berkeley University, San Francisco.

Prof. Joachim Scholderer (M) is Professor of Innovation Management at NMBU. Previously, he was Research Director at University of Zurich (2015-2017), Professor of Quantitative Methods at Aarhus

University (2000-2015) and has had secondary affiliations at Agricultural University of Athens, ETH Zurich, University of Basel, University of Copenhagen and University of Tromsø. He has participated in altogether 17 EU projects (FP4-H2020) and numerous national ones in Denmark, Greece, Finland, Ireland, Germany, Hungary, Netherlands, Norway, Sweden, Switzerland, Tanzania, Uganda, Vietnam and the UK. Joachim is an expert in the management of life science technology innovations (incl. IP strategy, market potential assessment, stakeholder and user acceptance along the value chain, forecasting, risk assessment and commercialization), particularly in the food safety area (relevant previous EU projects include CADE-GENTECH, ENTRANSFOOD, EUFRAM, EUROPREVALL, NOFORISK, NOVEL-Q, PROSAFEBEEF, Q-PORKCHAINS, SAFECONSUME, SEAFOODPLUS). His h-index is 42.

Stine Brekke Vennemo (F) is a data engineer and working at the BioSpec group as data engineer and administrator. Stine Brekke Vennemo data science engineer will be responsible for the data management in the project and be responsible for the timely delivery of draft scientific and financial reports to the Coordinator.

Dr Volha Shapaval (F) is an Associate Professor in bioprocess technology, has a multidisciplinary background in the field of microbiology, biotechnology and biospectroscopy. Shapaval was among of the first ones introducing high-throughput analysis of food spoilage fungi by FTIR spectroscopy and developing fully automated sample preparation system for spectroscopic analysis of fungi.

Dr Boris Zimmermann (M) holds PhD degree in physical chemistry and has a strong background in analytical chemistry, organic chemistry and spectroscopy.

Dr Valeria Tafintseva (F) holds PhD degree in applied mathematics and data analysis and has a strong background and experience in data analysis of spectroscopic and other multivariate data. She will be WP5 leader and coordinate all activities related to data analysis in this WP.

Maren Anna Brandsrud (F) is a young researcher who holds a Master of Physics. In her PhD she is working on modelling light scattering.

Johanne Heitmann Solheim (F) holds a Master of Physics, and is a young researcher in the BioSpec group. Currently she is taking a PhD in modelling scattering and absorption in infrared spectroscopy.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. Y. Xiong, V. Shapaval, A. Kohler, PJ. From. A Laboratory-Built Fully Automated Ultrasonication Robot for Filamentous Fungi Homogenization. 2019. *SLAS Technology, 1-13 10.1177/2472630319861361*
2. Y. Xiong, V. Shapaval, A. Kohler, J. Li, PJ. From. A Fully Automated Robot for the Preparation of Fungal Samples for FTIR Spectroscopy Using Deep Learning. 2019. *IEEE Access, 7(1): 132763-132774. DOI 10.1109/ACCESS.2019.2941704*
3. J. Li, V. Shapaval, A. Kohler, R. Talintyre, J. Schmitt, R. Stone, A. J. Gallant, D. Zeze. A Modular Liquid Sample Handling Robot for High-Throughput Fourier Transform Infrared Spectroscopy. 2015. *In Book Advances in Reconfigurable Mechanics and Robotics II.*
4. V. Shapaval, J. Schmitt, T. Møretrø, H-P. Suso, I. Skaar, A.W. Åsli, D. Lilehaug, and A. Kohler. Characterization of food spoilage fungi by FTIR. 2013. *Journal of Applied Microbiology, 114 (3), 788-796.*
5. Bassan P, Kohler A, Martens H, Lee J, Byrne H J, Dumas P, Gazi E, Brown M, Clarke N, Gardner P (2010) Resonant Mie Scattering (RMieS) Correction of Infrared Spectra from Highly Scattering Biological Samples, Analyst **135**, 268–277.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. **2012-2014:** Coordinator for the EU project (FP7-SME-2012 'Research for the Benefit of SMEs') entitled '*Source tracking and monitoring of mould contamination in food production' with acronym 'FUST*' in European Union's Seventh Framework Programme managed by REA-Research Executive Agency (FP7-SME-2012-315271-FUST) Ranged as number 8 among approx. 150 funded projects. (EU funding: 1 130 050 Euros).

2. **2019-2022:** Coordinator for the NFR FriNaTek project entitled 'DeepHyperSpec -Combining spectral and image information in the analysis of hyperspectral imaging data' grant Nr. 289518 (Total grant: 12.684 Mio NOK).
3. **2016-2021:** Coordinator for the 'Multidisciplinary graduate and post-graduate education in big data analysis for life sciences' funded by SIU, the Norwegian Center For International Cooperation In Education, Eurasia 2016 - Long-term grant Nr. 257622 (Total grant: 3 Mio NOK).
4. **2018-2021:** Partner in the EU Horizon 2020 project: ICT-30-2017: Photonics KET 2017: Mid-infrared arthroscopy innovative imaging system for real-time clinical in depth examination and diagnosis of degenerative joint diseases. (EU funding: total budget: 6133635 Euro, NMBU part: 538431 Euros)
5. **2017-2022:** Safer food through changed consumer behaviour: Effective tools and products, communication strategies, education and a food safety policy reducing health burden from foodborne illnesses (SafeConsume). Source: European Commission, Horizon 2020. EU funding: EUR 9,500,000).

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

The BioSpec group is a world leader in the data analysis of spectroscopic data, developing specific protocols for the processing, modelling and integration of data from spectroscopy, chemical analysis and genetic analysis for data in life and biomedical sciences. The group has during the last decade developed an excellent data science infrastructure with in-house GPU computing facilities. The BioSpec group is world leading in modelling of infrared spectroscopic data for calibration and classification problems. The BioSpec group at NMBU has developed a vibrational spectroscopic database (BIOSDATA SQL) which contains a large number of spectra of biological samples and chemicals and which is used for calibration and classification model establishment. The group has long-standing experience in using national computing infrastructures such as Notur (high performance computing in Norway) and Orion at NMBU. The group has a newly established facility at NMBU, comprising a high-throughput vibrational spectroscopic screening platform for macro-scale measurements (two FTIR and one FT-Raman system), and a vibrational microspectroscopy platform (FTIR microscope with a wide variety of objectives for transmission/reflection chemical imaging; Purchase of Raman microscope is in process). The research groups involved in the project have a large spectrum of in-house developed modelling tools (in-house data science platform in Matlab and Python, data science platform Orange, numerical methods for electromagnetic field simulation such as integral methods and FDTD) and commercial software tools. TheBioSpec group has a well-established data management plan including a pipeline for storage and continuous documentation of experimental data, and a plan for archiving of data (see: https://www.nmbu.no/en/faculty/realtek/research/groups/biospectroscopy/data-management). For archiving data we use the Norwegian Centre for Research Data (NSD).

Organisation Short Name	UULM	
Organisation Legal Name	University of Ulm	
Country	GERMANY	
Website	https://www.uni-ulm.de	

Description of the legal entity

UULM is among the leading 'Golden Age Universities' ranked #3 in Germany, and #18 in the world (Times Higher Education ranking) educating 10,000+ students in (i) Natural Sciences, (ii) Mathematics & Economics, (iii) Engineering, Computer Science and Psychology, and (iv) Medicine. The Institute of Analytical and Bioanalytical Chemistry (IABC; uni-ulm.de/iabc) at UULM is chaired by Prof. B. Mizaikoff and comprises 3 lead Professors next to 8 permanent technical and scientific staff members. Embedded in the faculty of natural sciences, IABC is internationally acknowledged for its cross-disciplinary research programs, which are focused on the development and application of innovative (micro- and nano-) analytical platforms providing advanced methods and technologies in sensing, diagnostics, and imaging of complex molecular processes relevant to environmental, industrial, and medical/biological applications. On average, 20 Ph.D. students and 5 Postdoctoral Fellows are employed at IABC. IABC also operates the

Elemental Analysis Center and the *Focused Ion Beam Center UULM* for micro-/nano-fabrication/-characterization/-prototyping.

Main role and task in the project

Within PHOTONFOOD, UULM will be leading WP3, which is committed to developing, optimizing, calibrating, validating, and testing portable and hand-held mid-infrared laser- (HI-FI) and LED-coupled (MI-FI) sensing systems for analyzing fungal contaminations and mycotoxins in commodities. This will include close collaboration with all involved partners on tailoring the laser and LED technology, developing efficient thin-film waveguide transducers, and testing/validating of the established sensing systems in the field. Furthermore, UULM will be involved in activities related to data analysis in WP5, and in communication and dissemination activities in WP7.

Profiles of persons primarily responsible for carrying out the proposed activities

Prof. Boris Mizaikoff (M), Chaired Professor and Director of IABC since 2007. He received his Ph.D. in Analytical Chemistry in 1996, his Habilitation in 2000 – both at the Vienna University of Technology (Austria) - and has then been a faculty member at the Georgia Institute of Technology (Atlanta, USA) prior to joining UULM. He is author/co-author of 340+ peer-reviewed publications (h-index: 46) and 17 patents. Under his leadership, IABC has established an acknowledged track record of collaborations with small and large industrial partners including ExxonMobil, Agilent/Keysight, Boehringer-Ingelheim, Bruker Optics, Alpes Lasers, Sacher Laser, Nanoplus, OptoPrecision, etc. He has been elected *Fellow of the American Association for the Advancement of Science (AAAS)* for "building bridges with industry through academic research in sensors, and for facilitating technology transfer between academia and industry" in 2006. He is also Associate Editor of *Analytical Chemistry* (ACS), the highest ranked peer-reviewed journal in the analytical sciences. **Julian Haas (M)** is a research associate IABC, who has worked on integrated mid-infrared photonics since his MA and PhD. He has been a leading researcher developing thin-film waveguide technologies combined with laser spectroscopic methods, and their implementation in practical applications including chemical contaminants in water, mycotoxin contaminants in commodities, and has lately focused on the in-vivo analysis of cartilage damage via arthroscopic mid-infrared probes.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. R. Lu, G. Sheng, W. Li, H. Yu, Y. Raichlin, A. Katzir, **B. Mizaikoff**, *IR-ATR Chemical Sensors Based on Ethylene/Propylene Copolymer Coated Planar Silver Halide Waveguides for In-Situ and Simultaneous Detection of Multiple Organic Contaminants in Water*, Ange. Chem. Int. Ed. 52, 2265-2268 (2013).
2. B. Mizaikoff, *Waveguide-Enhanced Mid-Infrared Chem/Bio Sensors*, Chem. Soc. Rev. 42, 8683-8699 (2013).
3. M. Sieger, F. Balluff, X. Wang, S.-S. Kim, L. Leidner, G. Gauglitz, B. Mizaikoff, On-Chip Integrated Mid-Infrared GaAs/AlGaAs Mach-Zehnder Interferometer, Anal. Chem. 85, 3050-3052 (2013).
4. X. Wang, M. Karlsson, M. Sieger, L. Oesterlund, F. Nikolajeff, **B. Mizaikoff**, *Diamonds are a Spectroscopist's Best Friend: Thin-Film Diamond Mid-Infrared Waveguides for Advanced Chem/Bio Sensors*, Anal. Chem. 86, 8136−8141 (2014).
6. 5. J. Haas, R. Stach, M. Sieger, Z. Gashi, M. Godejohann, **B. Mizaikoff**, *Sensing Chlorinated Hydrocarbons via Miniaturized GaAs/AlGaAs Thin-Film Waveguide Flow Cells Coupled to Quantum Cascade Lasers*, Analyt. Methods 8, 6602-6606 (2016).

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. 2019 (EU): VOGAS - Screening of Gastric Cancer via Breath Volatile Organic Compounds by Hybrid Sensing Approach;
2. 2019 (EU): BREAK BIOFILMS - Breaking Bad Biofilms. Innovative Analysis and Design Rules for Next-Generation Antifouling Interfaces;

3. 2018 (EU): BRICS - Broadly Tunable Interband Cascade Laser Source for Gas Analyzers;
4. 2018 (EU): MIRACLE - Mid-infrared Arthroscopy Innovative Imaging System for Real-time Clinical in-depth Examination and Diagnosis of Degenerative Joint Diseases;
5. 2018 (DFG): SFB CATALIGHT - Light-driven Molecular Catalysts in Hierarchically Structured Materials;

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

IABC has developed a plethora of IR spectroscopic sensing systems for environmental/clinical/medical monitoring, as well as for food contamination analysis with specific focus on fungal contaminations and mycotoxins. Next to broad-band FT-IR based technologies, IABC has pioneered miniaturized sensing systems combining interband cascade and quantum cascade lasers (ICLs, QCLs) with thin-film waveguides providing superior sensitivity in environmental monitoring, biomedical/clinical diagnostics, and food safety applications. The latter have been in part commercialized as a main result of a preceding EU-project on IR-spectroscopic mycotoxin detection (MYCOSPEC; EU project #314048) and will contribute to the adaptable sensing interface in PHOTONFOOD.

Organisation Short Name	WU	
Organisation Legal Name	Wageningen University	
Country	NETHERLANDS	
Website	https://www.wur.nl/	

Description of the legal entity

Wageningen University (WU) provides world-class research to students from >100 different nationalities and has been awarded for the 15th time in a row as the best university by Dutch students. WU is heavily involved in EU framework research programmes, as coordinator or as core participant. The laboratory of organic chemistry is specialized in surface bound chemical synthesis and the physical and chemical characterization of surfaces. The embedded special chair on analytical chemistry is focussed on advanced diagnostics, biosensors and ambient mass spectrometry. The analytical group is strongly linked to the applied research institute Wageningen Food Safety Research (part of another legal entity, Wageningen Research). Consequently, the academic analytical research at WU is eventually aiming for innovative on-site applications in food quality and safety testing.

Main role and task in the project

WU will be leader of the work package on paper microfluidics (WP2) and contribute to the modelling and software development activities in WP5 and the dissemination of the project (WP7).

Profiles of persons primarily responsible for carrying out the proposed activities

Michel Nielen (M) is professor of analytical chemistry at WU. After a career at TNO and in chemical industry at AkzoNobel, he joined Wageningen University & Research in 2000 for a position at the RIKILT institute. Currently, he holds a part-time special chair on Analytical Chemistry at Wageningen University and is also part-time principal scientist at Wageningen Food Safety Research (WFSR). He is former president of the Dutch Society for Mass Spectrometry and former scientific director of TI-COAST, the Dutch public-private partnership on analytical science and technology. Nielen (co-)authored 200 peer-reviewed papers and book chapters (>8800 citations, H-index 53) covering a wide range of analytical technologies, including mass spectrometry, chromatography, capillary electrophoresis and biosensors.

Gert Salentijn (M) is assistant professor at WU and the recent recipient of a highly prestigious personal grant from the national science foundation. He has obtained his PhD in analytical chemistry with honors from the University of Groningen. In 2019 he was a visiting scientist at the world's leading MS group of Prof. Graham Cooks at Purdue in the USA. Salentijn is Work Package leader for detection technology in the Horizon 2020 ETN FoodSmartphone. He is co-founder of SG Papertronics, a spin-off company from the University of Groningen, developing platform technology for paper-based analyses, based on his patented technology for sample pre-concentration on paper.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. Ross GMS, Salentijn GIJ, Nielen MWF, Filippini D (**2020**) A 3D-micro-total analysis system for integrated multiplex immunosensor detection, Lab Chip, under review.
2. Ross GMS, Salentijn GIJ, Nielen MWF (**2019**) A Critical Comparison Between Flow-through and Lateral Flow Immunoassay Formats for Visual and Smartphone-based Multiplex Allergen Detection, Biosensors, 9, 1-21.
3. Santbergen MJC, van der Zande M, Bouwmeester H, Nielen MWF (**2019**) Online and in situ analysis of Organs-on-a-chip. Trends Anal. Chem., 115, 138-146.
4. Ross GMS, Bremer MGEG, Nielen MWF (2018) Consumer-friendly food allergen detection: moving towards smartphone-based immunoassays. Anal. Bioanal. Chem. 410, 5353–5371.
5. Salentijn GIJ, Grajewski M, Verpoorte E (**2018**) Reinventing (bio)chemical analysis with paper, Anal. Chem. 90, 13815–13825.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. 2017-2020 Nielen is coordinator of the EU H2020 Marie Curie Innovative Training Network on smartphone analyzers for on-site testing of food quality and safety, acronym FoodSmartphone (grant agreement 720325 – FoodSmartphone - H2020-MSCA-ITN, 2.8 M). Salentijn is Work Package leader for innovative detection mechanisms.
2. 2018-2020 Nielen is co-applicant and core-partner of a regional EFRD-project on Citizen Science, aiming for the design, development and validation of portable analytical instruments and devices in food and health sciences.
3. 2019-2023 Salentijn has received a prestigious personal grant from the Netherlands Organisation for Scientific Research, on the topic of *Surface-Induced Adaptive Switching in Paper Microfluidics.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Wageningen University has state-of-the-art analytical instruments, both within the laboratory of organic chemistry and on the Wageningen Campus. Moreover, full-time research technicians provide supports for undergrad and grad students in their research. The analytical segment of the lab of organic chemistry has expertise pertaining to relevant topics such as 3D-printing, paper microfluidics, on-site (bio)sensors, sample extraction and preparation, and a wide network of collaborators in these fields.

Organisation Short Name	BOKU	
Organisation Legal Name	University of Natural Resources and Life Sciences, Vienna	
Country	AUSTRIA	
Website	https://boku.ac.at/	

Description of the legal entity
The University of Natural Resources and Life Sciences, Vienna (BOKU) (founded in 1872) has more than 2,800 staff and 11,300 students, with 10 doctoral programmes and was ranked as a Top 25 GreenMetric World University for 7 years in a row. Its Department of Agrobiotechnology (IFA-Tulln) with some 200 staff is a leading institution in the area of agro-biotechnology with a special emphasis on mycotoxin control and reduction since 1994. The Institute for Bioanalytics and Agro-Metabolomics (iBAM; formerly Center for Analytical Chemistry) is well-known for the development of ground-breaking analytical methods for the determination of multiple mycotoxins and of novel LC-HR-MS-based metabolomics approaches.
Main role and task in the project
BOKU will coordinate WP6 on demonstration and validation of the photonics based detection systems. BOKU will also perform reference analysis of mycotoxins following a fully validated quantitative LC-MS/MS based multi-toxin approach.. Furthermore, BOKU will be involved in software development and models development activities in WP5.
Profiles of persons primarily responsible for carrying out the proposed activities
Prof. Rudolf Krska (M), full Professor of (Bio-)Analytical Chemistry at BOKU, has an impressive track record in the research area of biotoxins in food and feed. Since 2015 he has been recognised as highly cited (top 1%) researcher for the fourth successive time in the Web of Science. Prof. Krska (male) is the most cited author in the area of mycotoxins (>350 SCI papers, h-index: 62) of the last decade and has been coordinating several EU-funded projects incl. the current H2020 project MyToolBox dealing with integrated strategies to reduce mycotoxins in the food chain. He is also the President of the International Society for Mycotoxicology. **David Steiner (M)** is a research associate at the K1-competence center for Food and Feed Quality Safety and Innovation (FFoQSI) in Tulln, Austria. David Steiner was the leading researcher for mycotoxin method development and analysis following ISO 17025:2005 at the R&D department of the Austrian Competence Center for Food Safety (LVA GmbH). Furthermore, Steiner was responsible for the coordination of the accreditation processes of the IFA-Test Systems PT Scheme according to ISO 17043:2010. Within his Doctoral thesis in Food Chemistry and Biotechnology he expanded his knowledge in analytical chemistry and generated major experience in development of LC-MS/MS based methods for multi-class contaminants in complex feed within the FFoQSI work-package on-field/plant quality assurance.
List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content
1. Sulyok, M; Berthiller, F; **Krska, R**; Schuhmacher, R. Development and validation of a liquid chromatography/tandem mass spectrometric method for the determination of 39 mycotoxins in wheat and maize. Rapid Commun Mass Spectrom. 2006; 20(18):2649-2659. 2. R. Krska, M. de Nijs, O. McNerney, M. Pichler, J. Gilbert, S. Edwards, M. Suman, M, N. Magan, V. Rossi, HJ van der Fels-Klerx, F. Bagi, B. Poschmaier, M. Sulyok, F. Berthiller, H.P. van Egmond, Safe food and feed through an integrated toolbox for mycotoxin management: the MyToolBox approach. World Mycotoxin Journal, 2016, 9(4); doi.org/10.3920/WMJ2016.2136 3. Sulyok, M.; **Krska, R**.; Schuhmacher, R. A liquid chromatography-tandem mass spectrometric multi-mycotoxin method for the quantification of 87 analytes and its application to semi-quantitative screening of moldy food samples. ANAL BIOANAL CHEM, 389, 1505-1523; ISSN 1618-2642. 4. **Steiner, D**; **Krska, R**; Malachova, A; Taschl, I; Sulyok, M. Evaluation of Matrix Effects and Extraction Efficiencies of LC-MS/MS Methods as the Essential Part for Proper Validation of Multiclass Contaminants in Complex Feed. J AGR FOOD CHEM. 2020; 68(12): 3868-3880

5. Sulyok, M; Stadler, D; **Steiner, D**; **Krska**, **R**; Validation of an LC-MS/MS-based dilute-and-shoot approach for the quantification of >500 mycotoxin and other secondary metabolites in food crops: challenges and solutions, Analytical and Bioanalytical Chemistry 2020, 412, 2607; doi: 10.1007/s00216-020-02489-9
6.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. 2017-2021: Blue Area Leader for the project within a strategic participation (e.g. COMET) Feed and Food Quality Safety and Innovation (FFoQSI) funded by the Austrian Ministries BMVIT, BMWFW and the federal provinces of Niederösterreich, Upper Austria and Vienna. The programme COMET is administered by FFG.
2. 2019-2020: Project leader for the project Mycotoxin degradation mechanisms in mealworms (Tenebrionidae) funded by the federal province of Niederösterreich and the European Regional Development Funds (ERDF).
3. 2016-2020: Project leader for the project Safe Food and Feed through an Integrated Toolbox for Mycotoxin Management (MyToolbox) funded by the Commission of the European Communities, Rue de la Loi, Brussels, European Union.
4. 2016-2018: Project leader for the project Multidisciplinary approach to strengthen cooperation and establish novel platform for comprehensive assessment of food and feed safety (MultiCoop) funded by the Commission of the European Communities, Rue de la Loi, Brussels, European Union.
5. 2013-2016 Project leader for the project Novel infrared spectroscopic tools for mycotoxin determination in foodstuffs for increased food safety (MYCOSPEC) funded by the Commission of the European Communities, Rue de la Loi, Brussels, European Union.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

BOKU has already been involved in some 90 EC-funded projects within FP7 and H2020, co-ordinating 19 projects. Currently, BOKU is hosting three ERC Grants and is partner in nine H2020 projects. Prof. Krska coordinated 4 EU projects, and led WPs, e.g. the WP “Training activities” in the Horizon 2020 TWINNING project MultiCoop (GA no. 692195). BOKU was hosting the Special Research Programme of the Austrian Science Fund SFB-FUSARIUM, and had received some 10 Mio Euro of national funding. A major emphasis of the latter project was on metabolomics of Plant-Fusarium interactions.
Currently the BOKU/IFA-Tulln is the coordinating institution of the H2020-SFS-13-2015 project MyToolBox (Safe Food and Feed through an Integrated ToolBox for Mycotoxin Management)-coordinator Prof. Rudolf Krska. BOKU/IFA-Tulln is also beneficiary in the INTERFUTURE ITN (GA no. 722642) and CEREALPATHETN (GA no. 674964). BOKU/IFA-Tulln has also been receiving other major national funding for its research e.g. on biomarkers and the development of multi-toxin methods as part of the activities of the Austrian Competence Center for Feed and Food (FFOQSI) which is co-owned by BOKU and funded by the COMET programme.

Organisation Short Name	NEBIH	
Organisation Legal Name	National Food Chain Safety Office	
Country	HUNGARY	
Website	https://portal.nebih.gov.hu/	

Description of the legal entity

National Food Chain Safety Office (NEBIH) is the integrated food chain safety authority of Hungary with control responsibilities covering the whole food chain from agriculture and food processing to retail and catering. NEBIH is the official Hungarian contact point to the European Food Safety Authority, the Rapid Alert System for Food and Feed and the International Food Safety Authorities Network. In addition to the traditional authority measures – risk assessment and risk management – risk communication has become a very important focus area during the last 10 years. To ensure food chain safety, NEBIH operates the national reference laboratory network. Besides food safety, food quality is also addressed in NEBIH's activities. The KMÉ ("High-Quality Food") trademark system has been established in 2019, aiming to distinguish high-quality food on the product market, based on laboratory experiments, label check, sensory analysis and consumer preference.

Main role and task in the project

NEBIH is a key contributor in PHOTONFOOD project as the leader of WP1. They will lead the assessment of photonic sensor solutions for monitoring food quality. As a food-safety authority NEBIH has direct contact with more than 3000 small and medium sized farms in Hungary, and will contribute to the validation and demonstration in WP6. In addition, NEBIH will be involved in model validation and demonstration activities in WP5.

Profiles of persons primarily responsible for carrying out the proposed activities

Gyula Kasza, PhD (M) – *Leader of WP1*. Head of the Department of Risk Prevention and Education in NEBIH. Honorary professor of Szent István University, Faculty of Food Science and the University of Veterinary Medicine, Budapest. He gives lectures at Semmelweis University, Faculty of Health Sciences as well. He has an agricultural engineer and a law degree and a PhD in consumer research. He has experience in food related policy making and implementation of scientific findings into food control measures. His interests include food safety risk perception and risk communication. Author of more than 190 peer-reviewed publications.

Attila Nagy, DVM (M) – Director of NEBIH's Food Chain Safety Laboratory Directorate. He has a degree in veterinary medicine and law, and has a certificate as auditor of quality management system of food laboratories, based on EN ISO/IEC 17025:2001, internal auditor of EN ISO 22000:2005 and modern methods of sensory analysis certification. He is responsible for the supervision of the national laboratory network, including Food Analytical, Food Microbiological, Food Toxicological, Radioanalytical and Feed Laboratories.

Tekla Izsó (F) – Food safety referent at NEBIH. She has a master degree in Food Safety and Quality Engineering and currently is a PhD student in the Doctoral School of Food Sciences at Szent István University. She has experience in international projects (e.g. SafeConsume), risk assessment and risk communication. She has experience in qualitative and quantitative consumer research also. Author of several peer-reviewed publications.

Eszter Csenki (F) – Food safety referent at NEBIH. She has a degree in Food Engineering (BSc) and is currently studying Food Safety and Quality Engineering (MSc). She has experience in implementation of H2020 and national level ESF projects and has expertise in risk assessment and stakeholder coordination. She has been involved in several consumer awareness-raising communication campaigns in NEBIH, aiming to inform consumers about food-related risks.

Dávid Szakos (M) – Food safety referent at National Food Chain Safety Office. Dávid Szakos graduated as a chemical engineer (MSc) at Budapest University of Technology and Economics. Currently, he is a PhD student at the University of Veterinary Medicine. His PhD deals with novel functional foods. He has got expertise in qualitative and quantitative consumer research and multivariate statistical analysis. He has participated in national and international research projects.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. Kasza, G., Szabó-Bódi, B., Lakner, Z., & Izsó, T. (2019). Balancing the desire to decrease food waste with requirements of food safety. Trends in Food Science & Technology, 84, 74-76.
2. Vajda, Á., Kasza, G., & Mohácsi-Farkas, C. (2019). Estimation of the occurrence of foodborne salmonellosis in Hungary. Acta Alimentaria, 48(1), 96-104.
3. Süth, M., Mikulka, P., Izsó, T., & Kasza, G. Y. (2018). Possibilities of targeting in food chain safety risk communication. Acta Alimentaria, 47(3), 307-314.
4. Szabó-Bódi, B., Kasza, G., & Szakos, D. (2018). Assessment of household food waste in Hungary. British Food Journal.
5. Szabó-Bódi, B., Kasza, G., & Szakos, D. (2018). Assessment of household food waste in Hungary. British Food Journal.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. SafeConsume (H2020 Grant agreement No. 727580, 2017-2022): Safer food through changed consumer behaviour: Effective tools and products, information strategies, education and a food safety policy reducing health burden from foodborne illnesses (http://safeconsume.eu/) – stakeholder coordination activity within SafeConsume project;
2. LIFE-FOODWASTEPREV project (LIFE15 GIE/HU/001048, 2016-2020): Food waste prevention in the food chain to support the implementation of the 7th Environment Action Programme (http://maradeknelkul.hu/en/) – Good Practices Guides for catering, trade, food industry partners and community on food waste reduction;
3. Reinforced Consumer Rights for Enhanced Food Safety and Quality (RCR-EFSQ) (811235 - RCR-EFSQ – CONS-CPC-JA, 2018-2021) (https://portal.nebih.gov.hu/-/reinforced-consumer-rights-for-enhanced-food-safety-and-quality) - investigation and laboratory analysis of dual quality;
4. Nébih's Szupermenta (Super Mint) project: a national, complex product testing program that includes laboratory experiments and sensory analysis as well;
5. EU Menu Project (EFSA-2010-00877, 2011-): EFSA Comprehensive European Food Consumption Database for Exposure Assessment;

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Nébih has a dedicated directorate for food and feed laboratories, supervising 40 laboratories
Nébih has a dedicated Food Toxicology National Reference Laboratory, which is responsible for the analysis of mycotoxins, drug residues, PAH compounds, metals, hormones, food contact materials, and dioxin from food and feed. Nébih operates two external university departments: Department of Food Chain Risk Management with Faculty of Food Science, Szent István University and Department of Food Chain Safety with University of Veterinary Medicine, Budapest. Nébih operates a department for regular stakeholder forums.

Organisation Short Name	NANOPLUS	Nanosystems and Technologies GmbH nanoplus
Organisation Legal Name	Nanosystems and Technologies GmbH	
Country	GERMANY	
Website	https://nanoplus.com/	

Description of the legal entity

NP is an SME which was founded 1998 as spin off from Würzburg University. The company focuses on the development of customer specific optoelectronic devices for sensor applications. The company with currently 97 employees, has two clean room facilities with 250 m2 (Gerbrunn, Germany) and 650 m2 (Meiningen, Germany), respectively. A complete process line for

semiconductor laser fabrication is available, comprising mask aligners for optical lithography as well as systems for electron beam lithography, different dry etch systems, sputter systems, evaporators, laser mounting facilities, etc. Key products of NP are complex coupled distributed feedback (DFB) laser diodes. In these lasers complex coupling is obtained by combining a ridge waveguide with metal gratings on top of the waveguide layer on both sides of the laser ridge. DFB lasers realized by this technique in various material systems have a high single mode yield (>90%), high sidemode suppression ratio (>40dB) combined with low back reflection sensitivity and thresholds and efficiencies, which correspond to those of Fabry Perot lasers realized from the same layers.
NP manufactures and sells DFB and Fabry-Perot laser diodes in the wavelength range from 0.7 up to 3.0 µm for sensing and spectroscopic applications. As base material for the lasers GaAlAs/InGaAs, InGaAsP, GaInAsN and GaInAsSb heterostructures are used. In addition, NP offers interband cascade laser (ICL) DFBs from 3-6 µm and quantum cascade laser (QCL) DFBs at certain wavelengths between 5 and 14 µm for specific applications. Previous work within national and international cooperation have allowed NP to acquire significant experience in long wavelength device processing based on Quantum Cascade as well as Interband Cascade material. nanoplus has e.g. coordinated the European research projects SENSHY (www.senshy.eu) for the development of GaSb based devices in the 3.0 – 3.6 µm wavelength range for which the company was awarded the Prism Award for Green Photonics and Sustainable Energy in 2012.

Main role and task in the project

NP main task is the developments of cascaded MIR light sources in WP3. Cost effective LEDs operating in the 5-6.5 µm range are realized for the MI-FI device whereas high precision multi-wavelength lasers will be developed for the HI-FI device. NP will also support the WP5 Prototype development. NP will in particular contribute to the PHOTONFOOD project by bringing in capabilities and related expertise in the field of ICL and QCL technology. NP is a leading supplier of semiconductor laser sources for gas sensing applications and has already been able to provide a variety of innovations (as discussed above) into this field. Having a high interest to maintain a strategic position with respect to emerging technology and related market opportunities concerning infrared emitters in biomedical field of food contamination, nanoplus is fully committed to the project and the related investigations for future device applications in biophotonics.

Profiles of persons primarily responsible for carrying out the proposed activities

Dr. Robert Weih (Principal Investigator) **(M)** was born in Schwerin, Germany in 1985. He received his Diploma degree in nanotechnology engineering and PhD degree in Physics from the University of Würzburg in 2011 and 2018, respectively. His main research interests covered molecular beam epitaxy for mid infrared emitting and detecting devices in the GaSb material system. In 2015, he joined the epitaxy division at nanoplus Nanosystems and Technologies GmbH. He is responsible for the epitaxy and chip dev
elopment for innovative devices based on III-V semiconductors such as GaAs, InP and GaSb. **Dr. Caroline Kistner** (Co-Investigator) **(F)** was born in Würzburg, Germany in 1981. She graduated in Physics from the University of Konstanz, Germany in 2007. In 2011 she received her PhD degree for her thesis on the spectroscopy of electronically contacted micro-cavities from the University of Würzburg. Following her studies, C. Kistner joined the product development team of OSRAM Opto Semiconductors GmbH as Project Manager Engineering, where she led a variety of international product development projects concerning opto-electronical components. In 2016 C. Kistner joined the R&Ddivision of nanoplus as development engineer and is now heading the R&D department.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. N. Schäfer; J. Scheuermann; R. Weih; J. Koeth; S. Höfling “High efficiency mid-infrared interband cascade LEDs grown on low absorbing substrates emitting >5 mW of output power”, Optical Engineering 58(11), 117106 (2009).
2. M. v. Edlinger, J. Scheuermann, R. Weih, Ch. Zimmermann, L. Nähle, M. Fischer, J. Koeth,S. Höfling,and M. Kamp, "Monomode Interband Cascade Lasers at 5.2 µm for Nitric Oxide Sensing",Photonics Technology Letters, Volume 26 , Issue: 5 (2014);
3. L. Naehle, C. Zimmermann, S. Belahsene, M. Fischer, G. Boissier, P. Grech, G. Narcy, S. Lundqvist, Y.Rouillard, J. Koeth, M. Kamp and L. Worschech, “Monolithic tunable GaSb-based lasers at 3.3 µm“, Electronics Letters 47, 1092 (2011);
4. J. Koeth, M. von Edlinger, J. Scheuermann, L. Nähle, L. Hildebrandt, M. Fischer, R. Weih, M. Kamp, Distributed feedback interband cascade lasers for applications in research and industry“, Photonics West, 2015, February 7-12, 2015, San Francisco, USA.
5. P. Fuchs, J. Friedl, S. Höfling, J. Koeth, A. Forchel, L. Worschech, and M. Kamp, “Single mode quantum cascade lasers with shallow-etched distributed Bragg reflector”, Optics Express, Vol. 20, Issue 4, pp. 3890-3897 (2012).

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. **SENSHY** [FP7 EU]: develop high quality ICL material for device processing and since then commercially exploitation the project results (only commercial manufacturer worldwide to offer ICL based DFB lasers).
2. **MIRACLE** [Horizon 2020] development and integration of multiple technologies in the first mid-infrared attenuated total reflection (MIR-ATR) instrument for arthroscopic use based on DFB-QCL light sources.
3. **MIRPHAB** [Horizon 2020] European pilot line for single access point to all the Mid-IR technology and components to build customized chemical sensors: nanoplus provides ICL technology.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Nanosystems and Technologies GmbH, Germany is an ISO 9001:2015 and ISO 14001:2015 certified supplier of semiconductor devices and one of the leading manufacturers of lasers for gas sensing applications. NANOPLUS is currently supplying mono mode DFB lasers in different types of packages at various wavelengths. Some typical DFB devices and their performance figures based on different material systems can be found at www.nanoplus.com.

NP has significant experience with DFB lasers in the GaAs and InP, but also the GaSb material system and associated challenges like water-free chip processing. Based on this material system, NANOPLUS has e.g. supplied long-wavelength DFB lasers for the NASA Mars Science Laboratory MSL mission (isotope ratio measurements on-board the Mars Rover Curiosity). The company holds the current long-wavelength record for mono mode type I GaSb-based laser diodes, as well as the current long-wavelength record for mono mode Interband Cascade Lasers. MIR laser sources based on DFB and multi-section BSG ICL devices with performance defining the current state-of-the-art have been demonstrated. Researchers from NP also made various important contributions in the field of Quantum Cascade Laser device development. NP will in particular contribute to the PHOTONFOOD project by bringing in capabilities and related expertise in the field of ICL and QCL technology.

Organisation Short Name	ROMER	
Organisation Legal Name	Romer Labs	
Country	AUSTRIA	

Website	https://www.romerlabs.com/	

Description of the legal entity

Romer Labs is an Austrian company (founded in Washington, USA in 1982) and a leading global supplier of diagnostic solutions for mycotoxins, food pathogens, food allergens, GMOs, veterinary drug residues and other food contaminants. The main objective of the company is to provide scientifically sound, high quality products and services to make food safer. A broad range of innovative tests and services provide an important contribution to integrated food safety solutions also serving private and public laboratories. R&D facilities are operated in AT, UK, MY, and USA and the production plants are located in AT and USA.

Main role and task in the project

ROMER will contribute to the validation of the developed technology. As an analytical lab, Romer is in close contact with customers. Based on the end users requirements well known to Romer Labs, they will be able to provide valuable feedback on the strengths and weaknesses of the developed devices and suggest improvements.

Profiles of persons primarily responsible for carrying out the proposed activities

Dr. Kurt Brunner **(M)** is a Director Research and Development of the Romer labs. He led previously a research group for molecular diagnostics of food and feed at BOKU/IFA-Tulln, Austria. Since 20 years he is an active scientist in the field of food and feed safety.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. Products: more than 50 different immunoassays for food and feed safety testing;
2. Products: highly purified reference materials serving as analytical standards;
3. Patents: 6 patent families protecting Romer's core products;
4. Kluger B, Bueschl C, Lemmens M, Berthiller F, Haeubl G, Jaunecker G, Adam G, Krska R, Schuhmacher R. (2013) Stable isotopic labelling-assisted untargeted metabolic profiling reveals novel conjugates of the mycotoxin deoxynivalenol in wheat. Anal. Bioanal. Chem. 405(15): 5031-5036.
5. Ansari P, Haeubl G. (2016) Determination of cyclopiazonic acid in white mould cheese by liquid chromatography-tandem mass spectrometry (HPLC-MS/MS) using a novel internal standard. Food Chem. 211:978-982.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. Safe Food II; development of novel methods for food allergen testing, funded by the Austrian Research Promotion Agency;
2. FP7 project "MYCORED - Novel integrated strategies for worldwide mycotoxin reduction in the food and feed chains";
3. Austrian FFOQSI program; Integrated research from field to fork;
4. K1 Center of the Austrian Research Promotion Agency; "Competence Center for Electrochemical Surface Technologies";

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

As explained before, the farm is composed of three different production facilities and one processing facility. Each facility has specific technologies. In the greenhouses there are all the irrigation and heating systems with an automation system. In the fish farm, there are different technologies installed to be able to grow fish in a recirculating system: mechanical filtration, biological filtration, water oxygenation, fish purging system, feeding system, automation and alarm system, etc. We have a complex heating system with a heating pump and gas boilers.

Organisation Short Name	IRIS	
Organisation Legal Name	IRIS Technology Solutions	
Country	SPAIN	
Website	http://www.iristechnologygroup.com/	

Description of the legal entity

IRIS Technology Group (http://www.iristechnologygroup.com/) is an advanced engineering Group that was established in Barcelona in 2007. Since 2011, it also operates from Dublin (Ireland) whereas the Barcelona office is referred to as IRIS Technology Solutions S.L. It specializes in the manufacture and integration of Process Analytical Technology (PAT)-based real-time quality monitoring solutions for the process industries and IoT systems for smart manufacturing, predominantly targeted at the food, pharmaceutical and chemical industries. In particular, our product line comprises proprietary state-of-the-art IR analyzers for at-line and in-line process monitoring.

On the one hand, our solutions are based on the latest developments in photonics in the fields of MEMs, Near Infrared Spectroscopy, Hyperspectral Imaging, and Raman. On the other hand, we leverage ICTs, whereby we apply Machine learning, Artificial Intelligence and Data Mining techniques for predictive and active control, as well as for decision support systems. In addition to photonics-based analyzers, we also work with IoT sensors for enabling equipment and machinery for smart manufacturing and transitioning industry to the Industry4.0 paradigm. Our solutions are cloud-based for mobility, remote monitoring and multi-site applications.

Our Innovation Division which is governed by an Innovation Committee comprising Senior Management, and an Innovation Unit for managing and executing the company's innovation roadmap, has 3 core strategic lines, for shaping the Factory of the Future:

1. New optical photonics-based technologies and PAT systems for production process optimization
2. ICTs (Artificial intelligence / machine learning, IOT & cloud platforms)
3. Circular and Bio-based Economy: digitalization and waste valorization

In addition to running an internal R&D program, as well as carrying our various joint private projects with industry partners, our Innovation Division is highly experienced in working in collaborative R&D projects in H2020 (both as coordinators and partners), principally in the SPIRE, FoF, and BBI programs. The company was established in 2007 and currently employs 60+ staff with a turnover of € 5 M. It is equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. We are also certified by the local (Catalonian) government as a provider of technological R&D services (Tecnio Certification). IRIS is a member of the following sectorial networks: SPIRE, SPIE and SECPHO, in addition to regional industrial collectives such as Smart Space (association of companies and organizations that specialize in the 'smartification' of industry).

Main role and task in the project

IRIS has extensive expertise at this level, and in particular in the context of industry-driven FP7 and H2020 projects, and private contacts with the industry. They will contribute in WP5, whereby, they will be responsible for the development of the Global System Control, software and data management platform. IRIS will also have minor contribution in WP4 and WP6 during prototype phase.

Profiles of persons primarily responsible for carrying out the proposed activities

Dr. Ing. ELODIE BUGNICOURT (F). Innovation Unit leader. Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Leader of the EcoMaterials business unit in IRIS from 2009 till 2014 and now Head of the innovation Unit. Since at IRIS, Elodie coordinated several European projects in the fields of bio-based packaging materials development, process monitoring and surface treatment. She has an extensive previous R&D experience in the in chemical industry especially related to reactive polymer formulations: i) Arkema (ex. Cray Valley, 2002, Spain) in the field of polyester powder coating; Wacker Chemie (2002-2005, Germany and France), Silicone division that funded her PhD thesis regarding the development of nanocomposites based on an epoxy matrix and silica; iii) Ashland (2006-2009, located sequentially in USA, Spain and France) in the Composite Polymers division, focussing on bio-sourced resins, flame retardant resins, SMC resins, PU and epoxy structural adhesives. With a strong interest on innovation and sustainability, so far, Elodie led and participated in many R&D projects aimed at developing new formulations, understanding mechanisms and structure-properties relationships, developing new analytical and modelling methods (for improved quality management); as well as collaborated with academic labs and other players in the supply and value chain. Finally, Elodie is author of several papers in peer-reviewed and technical journals and has presented her work in numerous international conferences.

Dr. DAVID NETTLETON (M), Data Mining Analyst. He has obtained a PhD in Artificial Intelligence from the Universitat Politécnica de Catalunya, Barcelona, Spain, M.Sc. In Computer Software and Systems Design from the University of Newcastle upon Tyne, U.K., and B.Sc. (Hons) Computer Science from the University of Wolverhampton, U.K. Has over 20 years' experience of applying data mining algorithms (including neural networks, rule induction) and statistical techniques (including ANOVA, regression, principal components) in a diversity of data analysis and modeling projects, such as medical diagnosis/prognosis, fraud detection, customer loyalty and segmentation, online user web search behavior and data privacy. As well as commercial projects, he has conducted research and lectured at two universities. He has published a diversity of articles in specialized data analysis journals/congresses and three books. He has participated in several competitive multi-partner research projects and has a European patent application for a graph matching algorithm.

MSc. ALEJANDRO ROSALES (M) has a M. Sc. in Physics from the Universidad de La Habana (1986). His background comprises theoretical and technical skills in several scientific and technological areas: optics, lasers, high vacuum, colorimetry, UV-VIS-NIR-MIR spectroscopy, chemometrics and design of measurement instruments. He has extensive practical experience in programming, electronics and mechanical CAD design as well as R&D methodology and management. His career has included positions as researcher applications engineer, design engineer, business developer and R&D manager. In 2000s, he worked in the FTIR spectroscopy field applied to quality control in the agrofood (wine) industry, whereby he developed two new measurement instruments for wine analysis. His career as industrial physicist has been focused in applying his wide expertise to provide solutions for industrial problems. Since 2011, he has been the Technical Manager in IRIS, leading a staff of 20 engineers and researchers involved in R&D projects aimed at developing novel high-end solutions for industry. Currently, he is the Science & Technology Manager IRIS.

IVAN PAQUICO RODRIGUEZ (M) Telecommunications Degree in UPC (Barcelona) and a Joint Degree in Multimedia Development in LUAS (Finland). His final thesis, 'Augmented Reality Chess and Application to Detect Chess Moves', implemented by using Computer Vision, Augmented Reality and Artificial Intelligence techniques, has won several awards, such as the best final thesis of the UPC and the best final thesis of Spain (received from Vodafone Spain and Universia foundation). He is an official iOS and Android Developer since 2010, creating a social network for iPhone called: 'BeThere' developed by iOS SDK from Apple and used Couchbase technology for the database. He also has experience in different programming languages such as

C, C++, Objective C or Java, 8 bits embedded and real time systems. He has expertise in multimedia methodologies ranging from Virtual Reality programming techniques to 3D/2D modelling software such as OpenGL, Quest3D, Autodesk Maya, Autodesk 3DS Max or Adobe Photoshop.

TOMEU COLL (M). Project and Innovation Manager. Degree in telecommunication engineering (UPC). Master's degree in biomedical engineering (UB) and in project management (La Salle). 7 years experience in healthcare technology and biomedical engineering field, involved in different projects in collaboration with research centers, hospitals and technology companies. During this period he developed skills and deep knowledge to perform well in interdisciplinary environments involving engineering, medical field and life science. He has a solid academic background and proficiency to perform coordinate or interfacing functions that require a bridge between medical and engineering disciplines.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. EP18198458, 03/10/2018, Rosales Alejandro, Kehoe Timothy, Garrick Meabh. An integrated optical system for the examination of sample materials
2. EP19382101,14/02/2019, Alejandro Rosales, Adrián Manotas, Joan Puig. Device for detecting foreign matter in an edible substance or in a container containing the edible substance and method employing the device.
3. Rosales, O. Mc Nerney. Edited by Stefan Krämer and Sebastian Engell. Wiley-Vch. Print ISBN: 978-3-527-34074-3. Sensing Technology. Chapter 6. Resource Efficiency of Processing Plants. Monitoring and Improvement.
4. A.A. Gowen, E. Gaston, J. Burger (2014). C.P. O´Donnell, C. Fagan, P.J.Cullen (Eds.), Process Analytical Technology for the Food Industry, 199-216, Springer, ISBN 978-1-4939-0311-5. Hyperspectral imaging.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. Visum. SMEI-Phase 2, GA 731043. Cloud-enabled IR spectroscopy platform for the rapid integration and customization of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries.
2. Hypera. H2020- SMEInst, 2016-2017. Affordable hyperspectral Imaging System for on-line foreign body detection and chemical composition analysis.
3. ProPAT. SPIRE, GA 637232. Robust and affordable process control technologies for improving standards and optimizing industrial operations.
4. SPECTRAFISH. FP7, GA 605399. Hyperspectral imaging technology for the quality inspection of fish products.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.

Our optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").
Food and wet chemistry lab equipped with a DSC, a Surface Energy analyzer, an HPLC, a Cobb tester, microbiological equipment, microscope and microtome for structural analysis, migration cells, colorimeter (for color and opacity measurement), Ford cup viscometers, water vapor permeation cells, turbidimeter, climatic chambers, manual and automatic coaters, electrospray, different drying technologies, plasma surface activation, etc.

Organisation Short Name	accelCH	
Organisation Legal Name	Accelopment	
Country	SWITZERLAND	
Website	https://accelopment.com/	

Description of the legal entity

accelCH assists research institutions and companies in EU project management and the dissemination and exploitation of project results. The SME is specialised in European research, development and innovation support initiatives, such as Horizon 2020 and the SME-focused Eurostars programme. The core activities include professional project administration, consortium organisation and communication, the facilitation of dissemination of project results to external stakeholders, supporting the exploitation of project results and financial management and reporting. At present, accelCH is involved as a beneficiary, partner organisation or subcontractor in more than 25 H2020 projects. accelCH is ISO 9001 certified ensuring a professional quality management and internationally accredited by IQnet.

Main role and task in the project

accelCH will be responsible for dissemination and exploitation and will lead WP7.

Profiles of persons primarily responsible for carrying out the proposed activities

Dr Johannes Ripperger (M) is a Research & Innovation Manager | > 10 years experience in the management of national and EU funded collaborative research and innovation projects | Technology Transfer Manager of the Swiss national technology platform energie-cluster.ch, responsible for innovation projects linking renewable energy enterprises with academia | Administrative Project Manager of CHEOPS (perovskite photovoltaics) SOSLeM (fuel cells) and also the FET Quantum Technology Flagship project macQsimal (metrology and sensing) | Dissemination Manager of other projects.

Dr. Emily Rose Ciscato (F) is Project Manager in Communication at accelCH. She has experience in communication and research dissemination from her involvement in the ideation, organisation and management of working groups, dissemination activities, seminar series, workshops and conferences at ETH Zürich. She is involved in the communication and dissemination of various EU funded projects, including MOSAICS, CLASSY, EXIMIOUS and HEAT-INSYDE.

Dr Jeanette Mueller (F): founder and CEO of accelopment Schweiz AG. She is an Economic Geography graduate from the University of Bayreuth, Germany and the Université de Perpignan, France and received her Ph.D. on European funding programmes from the University of Wales Aberystwyth (UK). She has more than 20 years of experience in the acquisition and management of publicly funded projects (ICT, life sciences, social sciences and renewable energies) and is specialised in R&D and innovation projects and previously hold several executive positions in IT and other companies. Alongside that, she has been the founder of several enterprises and a board member of a commercial and non-profit organisation. She is the Administrative Project Manager

and Dissemination (co-)leader of the EU Health projects EURO SHOCK and IMPROvED (both clinical studies). Additionally, she has been Work Package Dissemination Leader of Supporting LIFE (eHealth), EAGLE (ICT) and Robo-Mate (health support device) and Exploitation Manager of Robo-Mate.

Julia Götz (F): works in project communications at accelCH. Having previously worked as marketing project leader, she has experience in project management, marketing and design with a B.Sc. in International Management from the Zurich University of Applied Sciences. At accelCH, she has coordinated the dissemination work package for Mat4Rail, XoSoft, and PEARRL amongst others. She will be responsible for communication and dissemination activities of the EXIMIOUS project, a role she has previously held for the ClickGene, Robo-Mate and SOSLeM EU projects.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

1. Ortiz, J., Rocon, E., Power, V., de Eyto, A., O'Sullivan, L., Müller, J., ... & Teeuw, W. B. (2017). XoSoft-A Vision for a Soft Modular Lower Limb Exoskeleton. In Wearable Robotics: Challenges and Trends (pp. 83-88). Springer International Publishing.
2. Power, V., O'Sullivan, L., de Eyto, A., Schülein, S., Nikamp, C., Müller, J., ... & Ortiz, J. (2016, June). Exploring User Requirements for a Lower Body Soft Exoskeleton to Assist Mobility. In Proceedings of the 9th ACM International Conference on Pervasive Technologies Related to Assistive Environments (p.69). ACM.
3. Neville, K., Doyle, C., Sugrue, A., & **Müller, J.** (2013). "Supporting cross border emergency management decision-making". Paper no. 1023 at the *21st European Conference on Information Systems*.
4. Kaar, M., & **Müller, J.** (2011). *The Impact of Participation in Publicly Funded R&D Projects on Firm Competitiveness: Benefits and Barriers to the Use of National and EU Funding Programmes on the Example of Swiss SMEs in the Field of Renewable Energy*. München: GRIN Verlag GmbH.
5. **Müller, J.**, & Spengler, C. (2008). Welcher Touchpoint zählt? In H. Kaul & C. Steinmann (Eds.), *Community-Marketing: Wie Unternehmen in sozialen Netzwerken Werte schaffen* (pp. 217-234). Stuttgart: Schäffer-Poeschel.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. XoSoft: Soft modular biomimetic exoskeleton to assist people with mobility impairments. GA No. 688175, H2020-ICT-2015, 2016-2019;
2. MORE: Educating Europe`s Future Engineers in Next Generation Heavy Duty Mobile Machinery: Artificial Intelligence-driven Robotisation, Energy Efficiency and Process Optimisation. GA No. 858101, H2020-MSCA-ITN, 2020-2023.
3. VANGUARD: New Generation Cell Therapy: Bioartificial Pancreas to Cure Type 1 Diabetes. GA no. 874700, H2020-SC1, 2020-2024;
4. EXIMIOUS: Mapping Exposure-Induced Immune Effects: Connecting the Exposome and Immunome. GA no. 874707, H2020-SC1, 2020-2024;
5. FRINGE: Fluorescence and reactive oxygen intermediates by neutron generated electronic excitation as a foundation for radically new cancer therapies. GA no. 828922, H2020-FETOPEN, 2019-2023.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

A secure and central online storage for project files will be provided by accelopment. This Document Management System will allow for notifications on changed files, version control and will be regularly backed up. All files are hosted within Europe. An extensive set of tools for administration, reporting, communication, dissemination and exploitation, e.g. tracking of

deliverables, communication tools and templates, contact lists, etc. will be provided to all consortium members by accelCH. Communication tools for webinars, video conferences and phone conferences will be in place for the project start.

Organisation Short Name	BIGH	
Organisation Legal Name	BIGH Anderlecht SPRL (BIGH = Building Integrated Greenhouses)	
Country	BELGIUM	
Website	https://bigh.farm/	

Description of the legal entity

BIGH is a company that was founded in 2016 by Steven Beckers, a belgium architect expert in circular economy who works on the topic of "sustainable cities". The aim of it is to develop urban aquaponics farms in Europe. The first, and so far only farm, is "Ferme Abattoir", located in the center of Brussels on the roof of the Foodmet, a local market hall, on the historical Abattoir site in the heart of Brussels canal priority development area. It has been built in 2017 and productions started in 2018. This aquaponics farm, a combination of hydroponics and aquaculture, is the biggest urban farm located on a roof in Europe. The team consists of 7 employees plus a few interns and students who help us and in exchange of gaining experience in the aquaponics sector. The total area of the farm is around 4000 m2 with the following production facilities:

- An outside garden of 2000 m2
- A tomatoes greenhouse (770 m2)
- An aromatic herb greenhouse (600m2)
- A recirculating aquaculture system (RAS) of 500 m2

The farm uses two closed-loop recirculating systems where fish and plants are linked by a biological filter. The filter with billions of microorganisms purifies the fish water, which in turn serves to feed the plants in the greenhouse and the outdoor garden.

We have a capacity production of around 25 tons of Hybrid Stripped Bass in our pisciculture, a popular fish similar to sea bass that can grow healthily in fresh water. We process our fish directly on the farm. In 2020 we are planning a few changes in the RAS system to improve the production. The goal of a RAS is to optimize the water using, by renewing only 10% of the water volume everyday instead of 100% in traditional aquaculture.

Regarding the horticulture production, we produce 6 different types of aromatic herbs in pots, for a weekly production of 2700 pots. The production occurs during all the year and it has an organic certification. We also produce around 10 tons of tomatoes during 8 months along a year, and in 2020 we will also try to grow eggplant and pepper. These hydroponics production are irrigated by fishwater and rainwater. All the production facilities are heated by a heat pump, which uses the energy losses from the cold chambers of the building.

All our products are sold, for half to direct customers (individuals, restaurant, caterers) and for the other half to supermarkets or grocery stores, all in and around Brussels, offering customers the chance to know where their food comes from.

Our goal is to grow healthy and sustainable products, and sell them in a local market to promote a circular economy. We also want to reach zero losses so we transform products that were not sold (for example damaged basil leaves transformed in pesto, damaged tomatoes that are dried), to show that everything can be used and consumed.

Finally, we also organize different events, workshops and visits in the farm to share our experience in urban farming and more specifically in aquaponics.

Main role and task in the project

BIGH will contribute to testing, validation and demonstration for the detection of fungal contamination of herbs in WP6.

Profiles of persons primarily responsible for carrying out the proposed activities

Mrs Gwenn GUILLAUMIE (Principal Investigator) **(F)** was born in Plouaret, France in 1992. She graduated in Environmental and Sustainable Development Economics from the University of Paris-Saclay, France in 2016. She also has an engineering degree in agronomy and environmental sciences from the engineering school of AgroParisTech. She has been working for 2 years in Mozambique in the French Embassy as a Civil Society Cooperation Officer until mid-2019. She entered the BIGH team in October 2019, as the French company Veolia became one of the investors of the company, to bring a technical assistance to optimize the functioning of the farm regarding different aspects (energy and water consumptions, production, sales, communication) but mostly focused so far on the aquaculture production.

Mr Mathias DE VOS (Co-Investigator) **(M)** was born in Eupen, Belgium in 1991. He graduated in Bio-engineering from the University of UC Louvain in Belgium in 2014. He is currently the manager of BIGH since early 2018 but participated to the project since the beginning, helping with the design and construction of the farm. Previously to this experience he also worked for Urban farmers in Bern, Switzerland and at ECF Farmsystems in Berlin, Germany, which also developed aquaponics farms. He has therefore gained a wide experience on urban farming in general and more specifically on aquaponics farms, regarding all the aspects: development, management, sales, production, communication, quality management, but also the main issues we can face while managing an urban farm.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

Not relevant

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

Not relevant

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

As explained before, the farm is composed of three different production facilities and one processing facility. Each facility has specific technologies. In the greenhouses there are all the irrigation and heating systems with an automation system. In the fish farm, there are different technologies installed to be able to grow fish in a recirculating system: mechanical filtration, biological filtration, water oxygenation, fish purging system, feeding system, automation and alarm system, etc. We have a complex heating system with a heating pump and gas boilers.

Organisation Short Name	Seeberger	Seeberger
Organisation Legal Name	Seeberger GmbH	
Country	Germany	
Website	https://www.seeberger.de/	

Description of the legal entity

As a medium-sized family business that has been in existence since 1844, we know everything about tradition. Nevertheless, all our efforts are directed towards the future. As an aspiring company with growth in mind, we are always coming up with new ideas. We design our own products, open up new markets and guarantee maximum enjoyment and the very best quality. Thanks to decades of experience, we are at home in today's markets around the world. Despite being open-minded and globally active, our top priority is still to ensure the future of our family

business for our employees, customers and shareholders on the long-term. We have close ties to our location and bear responsibility via diverse commitments in the area. As one of the oldest coffee roasting companies in Germany, our range of products not only includes nuts and dried fruit but also coffee specialties. Whether its dried fruits and nuts for a quick snack, high-quality ingredients for cooking and baking or the finest coffee beans from the best altitudes of the world, Seeberger guarantees incomparable enjoyment on every occasion from the very best nature has to offer. Therefore, food/commodity quality and safety along the entire value chain is of prime importance to us.
Seeberger GmbH is specialized on the production of nut (roasted and plain), dried fruits, soft fruits and cereals, regularly packed under protective atmosphere. Furthermore we have facilities for roasting coffee and peanuts, to caramelize nuts and seeds and to coat different products with chocolate. In this regard organoleptic, wet-chemical, physical and chemical analyses are implemented in the company's own food chemistry laboratory for the quality control of raw and finished goods. Decades of trading with nut kernels, dried fruit and coffee. State-of-the-art packaging technology. Routine laboratory analysis of aflatoxins B1, B2, G1 and G2, and ochtratoxin A by imunoaffinity chromatography and HPLC with fluorescence detection.
NO. OF EMPLOYEES: approx. 700
NO. OF LOCATIONS: 1 (Ulm, Germany)
MAIN FOCUS OF BUSINESS ACTIVITIES: Dried fruits, nuts, quality coffee

Main role and task in the project

Participation in the testing, validation and demonstration of PHOTONFOOD technology.

Profiles of persons primarily responsible for carrying out the proposed activities

Dipl.-Ing. (FH) Heidi Wöhrlin (F) was born in Singen a. Htw., Germany in 1968. She graduated in food technology from the University of Trier, Germany in 1991. She started her career in the chocolate industry followed by the pizza industry, where she was establishing and heading chemical, physical and microbiological laboratories, as well as implementing the associated analytical methods and the QA-System. She was then head-hunted to an animal feed manufacturer, first joining the R&D division where she was in charge of project management for extruder technology, and for developing snacks and treats for cats and dogs in the respective European Competence Center. In the following, she accepted an offer from the European Quality Management Division of Nestlé being responsible for DACH-region (i.e., Germany, Austria, Switzerland) to be in charge of implementing Nestlé Quality Systems for all branches of the region, and served as its European representative. During maternity leave, she worked as an author for a distance learning university (Fernuniversität) writing learning material for the Food Technology Department. Finally, 10 years ago she received the offer to work as Head of Quality Management and Quality Assurance at Seeberger GmbH.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

- numerous packing systems (linear, duplex or triplex) with multipoint weighers under protective atmosphere
- packing and pasteurization line for soft fruits
- drum roaster for coffee beans
- belt roaster for peanuts
- equipment for the caramelizing of nuts

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

None

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Food chemistry laboratory following QM-QA guidelines for routine analysis of e.g. mycotoxins, caffeine, sorbic acid, Sulphur dioxid and moisture. Equipment includes predominantly HPLC (high performance liquid chromatography) systems with fluorescence detectors. The laboratory is specialized on the analysis of incoming goods and commodities. Although not accredited, the proof of suitability for the various analyses is regularly provided by participation in national and international proficiency tests.

Organisation Short Name	BAMA
Organisation Legal Name	BAMA Gruppen AS
Country	Norway
Website	https://bama.no/

Description of the legal entity

BAMA is the Norwegian market leader within fresh fruit and vegetables. The company is also gaining a steadily increasing market share within other fresh products, including flowers, drinks and sandwich products. While this presents opportunities, it also confers significant social responsibility in the markets where we operate. BAMA is one of Norway's oldest privately-owned retail companies and has been in operation for 133 years. BAMA's business strategy provides a basis for sustainable development of the company. Our vision is: Making Norway fresher and healthier. Our aim is to increase sales and stimulate greater interest in our products through inspiration and extra knowledge. BAMA shall be a growth driver in the Norwegian market, working in close collaboration with customers in all market segments. BAMA shall deliver products to its customers with an absolute focus on quality in all stages of the value chain and possess unique knowledge about suppliers and goods supply. Growth shall be based on sustainable choices along two dimensions: The first dimension is growth in the value chain through product development, research and innovation. The second involves an increased focus on a healthy diet and physical activity. Availability, taste, knowledge and inspiration are important for work to fulfil our societal remit in this area. BAMA shall be a driver of and model business for sustainable development with regard to the environment and social responsibility in the company's product areas.

CORE VALUES

Creating value for the entire value chain is the driving force behind all our activities,
to which our employees proudly contribute each day while displaying:

- Team spirit
- Integrity
- Consideration of others

NO. OF EMPLOYEES: approx. 2700

NO. OF LOCATIONS: Norway, Netherlands, Poland, Sweden and Finland

MAIN FOCUS OF BUSINESS ACTIVITIES: Fresh fruits, berries, vegetable, potatoes, dried nuts, juice and flowers.

Main role and task in the project

Participation in the testing, validation and demonstration of PHOTONFOODA technology.

Profiles of persons primarily responsible for carrying out the proposed activities

Harald Osa (M) is Research and Development manager, Department of Sustainability & Innovation. Mr. Osa is a renowned Norwegian chef that now focuses on research and innovation in fruits, berries and vegetables in BAMA.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

None
List of up to 5 relevant previous projects or activities, connected to the subject of this proposal
None
Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
BAMA is active in many countries. In Norway, we have local divisions across the country, in addition to packaging plants and processing production. This provides us with proximity to Norwegian producers and customers. In Sweden, we carry out processing production and produce and sell meal solutions for the Swedish market. We also carry out processing production in Finland, while our Polish company delivers fruit and vegetables to Norway and the European market. Our Netherlands-based operations deliver berries, exotic produce and flowers to the European market, including Norway.

Organisation Short Name	HS	
Organisation Legal Name	Hahn-Schickard-Gesellschaft	
Country	Germany	
Website	https://www.hahn-schickard.de/	

Description of the legal entity
The Hahn-Schickard-Gesellschaft (HS) is a private, non-profit research organization that works in close partnership with the industry, providing innovative products and technologies. HS's turnover in 2018 was €32m, with €10.2m from direct industry contracts. HS currently has >250 staff members and it is one of the leading institutes in Europe in the fields of microsystems engineering and microfluidics (https://www.hahn-schickard.de/en/). With its interdisciplinary staff, HS covers the whole range of lab-on-a-chip research and development, from computer-aided design (CAD) and simulation of fluid dynamics in microfluidics, fast prototyping and fluidic testing to the development and integration of biochemical assays and production scale-up. The three HS institutes are located in southern Germany (State of Baden-Württemberg) and are dedicated to: (i) manufacturing (Stuttgart), (ii) sensors and MEMS (Villingen-Schwenningen) and (iii) microanalysis systems (Freiburg and Ulm) with the most recently founded location in Ulm; the latter participates in this project. The HS location in Ulm has access to the exquisitely equipped laboratories, and closely collaborates with Ulm University and the research group of B. Mizaikoff at the Institute of Analytical and Bioanalytical Chemistry (IABC). Thus, a strong collaboration and complementary interconnection of capabilities and expertise of both project partners is ensured. Strong connections to stakeholders from industry and research facilities, as well as insights into recent developments, technological advancements, and market demands render HS ideally suited for exploitation management and dissemination within the project.
Main role and task in the project
HS will contribute to the dissemination, the exploitation and the innovation management of the project (WP7 and WP8).
Profiles of persons primarily responsible for carrying out the proposed activities
Dr. Julian Haas (julian.haas@hahn-schickard.de) is a group leader within the HS location in Ulm. He has obtained his PhD in chemistry with honors from Ulm University in 2020. In 2017, he was a visiting scientist in the Materials science group at Uppsala University in Sweden. During his PhD at Ulm University, he was involved in the development of infrared waveguide technology and

infrared spectroscopy of complex biological tissues within the Horizon 2020 project MIRACLE. He further contributed to the projects MYCOSPEC (EU), IQST (BMBF) and TROPSENSE (EU). His research focuses on advancing infrared spectroscopy, attenuated total reflection spectroscopy, quantum cascade laser technology, thin-film waveguide technology, and on-chip integrated photonics.
List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content
Haas, J. & Mizaikoff, B. Advances in Mid-Infrared Spectroscopy for Chemical Analysis. Annu. Rev. Anal. Chem. 9, 45–68 (2016). Haas, J., Wilk, A., Kokoric, V. & Mizaikoff, B. Next-Generation Mid-Infrared Chemical and Biological Sensors: Combining Quantum Cascade Lasers with Thin-Film and Hollow Waveguides. Spectrosc. Online 32, 22–30 (2017). Haas, J. et al. Polycrystalline Diamond Thin-Film Waveguides for Mid-Infrared Evanescent Field Sensors. ACS Omega 3, 6190–6198 (2018). Haas, J., Catalán, E. V., Piron, P., Karlsson, M. & Mizaikoff, B. Infrared spectroscopy based on broadly tunable quantum cascade lasers and polycrystalline diamond waveguides. Analyst 143, 5112–5119 (2018). Haas, J.; Stach, R.; Kolm, C.; Krska, R.;Mizaikoff B. Gallium Arsenide Waveguides as a Platform for Direct Mid Infrared Vibrational Spectroscopy. Analytical and Bioanalytical Chemistry 2020, 412, 3447-3456.
List of up to 5 relevant previous projects or activities, connected to the subject of this proposal
None
Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
HS occupies approx. 160 sqm of highest quality office and laboratory space at the TFU Biotechnology Center in Ulm. Facilities include modelling work stations for optical design and system development, optoelectronic workbenches for photonic system prototyping, spectroscopic equipment, high-end 3D printing, and wet-chemistry laboratory facilities.

4.2. Third parties involved in the project (including use of third party resources)

NEBIH have subcontracting for 20 000 EUR. The subcontracting will be related to the activities in WP1 and the subcontractor will be selected according to the "best value for money" principle of the EU. This will be done by collecting at least three independent offers

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	Yes
Does the participant envisage that part of its work is performed by linked third parties	No
Does the participant envisage that part of the work is performed by International Partners (Article 14a of the General Model Grant Agreement)?	No

Other PHOTONFOOD partners do not have any third party involved.

Section 5: Ethics and Security

5.1 Ethics

PHOTONFOOD project does not have any ethical issues as it was reported in the ethical issue table in the administrative proposal forms. The evaluation of the ethical issues was performed according to the guidance 'How to complete your ethics self-assessment'.

5.1.1 Protection and privacy of personal data

The mechanisms to protect the personal data will be organized according to the GDPR (https://gdpr.eu/). The contact with project partners and external stakeholders, identified as data subjects will be performed as (written) informed consent. The data controller will be Data Protection Officer (Dr. V. Tafintseva) from NMBU.

Rights of the data subjects:

- Right to be informed, link it to art.12 GDPR

The data controller will take appropriate measures to provide any information and any communication relating to processing to the data subject in a concise, transparent, intelligible and easily accessible form, using clear and plain language. The information will be provided in writing, or by other means, including, where appropriate, by electronic means.

- Right to access, link it to art. 15 GDPR

The data controller will provide a copy of the personal data undergoing processing. For any further copies requested by the data subject, the controller may charge a reasonable fee based on administrative costs. Where the data subject makes the request by electronic means, and unless otherwise requested by the data subject, the information will be provided in a commonly used electronic form.

- Right to rectification, link it to art.16 GDPR

The data subject will have the right to obtain from the controller without undue delay the rectification of inaccurate personal data concerning him or her. Taking into account the purposes of the processing, the data subject will have the right to have incomplete personal data completed, including by means of providing a supplementary statement.

- Right to be forgotten, link it to art.17 GDPR

The data subject will have the right to obtain from the controller the erasure of personal data concerning him or her without undue delay and the controller shall have the obligation to erase personal data without undue delay where one of the following grounds applies: (i) the personal data are no longer necessary in relation to the purposes for which they were collected or otherwise processed; (ii) the data subject withdraws consent on which the processing is based according to point (a) of Article 6(1) GDPR, or point (a) of Article 9(2) GDPR , and where there is no other legal ground for the processing; (iii) the data subject objects to the processing pursuant to Article 21(1) GDPR and there are no overriding legitimate grounds for the processing, or the data subject objects to the processing pursuant to Article 21(2) GDPR; (iv) the personal data have been unlawfully processed; (v) the personal data have to be erased for compliance with a legal obligation in Union or Member State law to which the controller is subject; (vi) the personal data have been collected in relation to the offer of information society services referred to in Article 8(1) GDPR.

- Right to restriction of processing (art. 18-2 GDPR)

Where processing has been restricted under paragraph 1 art. 18 GDPR, such personal data will, with the exception of storage, only be processed with the data subject's consent or for the establishment,

exercise or defence of legal claims or for the protection of the rights of another natural or legal person or for reasons of important public interest of the Union or of a Member State.

- Right to data portability, link it to art. 20 GDPR

The data subject will have the right to receive the personal data concerning him or her, which he or she has provided to a controller, in a structured, commonly used and machine-readable format and have the right to transmit those data to another controller without hindrance from the controller to which the personal data have been provided.

- Right to object, link it to art. 21 GDPR

The data subject will have the right to object, on grounds relating to his or her particular situation, at any time to processing of personal data concerning him or her which is based on point (e) or (f) of Article 6(1) GDPR, including profiling based on those provisions. The controller will no longer process the personal data unless the controller demonstrates compelling legitimate grounds for the processing which override the interests, rights and freedoms of the data subject or for the establishment, exercise or defence of legal claims.

- Stakeholder engagement (art. 35-9 GDPR)

Where a type of processing in particular using new technologies, and taking into account the nature, scope, context and purposes of the processing, is likely to result in a high risk to the rights and freedoms of natural persons, the controller will, prior to the processing, carry out an assessment of the impact of the envisaged processing operations on the protection of personal data. The controller will seek the advice of the data protection officer, where designated, when carrying out a data protection impact assessment.

- Right to a judicial remedy and the right to receive compensation (art. 79 GDPR)

Without prejudice to any available administrative or non-judicial remedy, including the right to lodge a complaint with a supervisory authority pursuant to Article 77 GDPR, each data subject will have the right to an effective judicial remedy where he or she considers that his or her rights under this Regulation have been infringed as a result of the processing of his or her personal data in non-compliance with this Regulation.

Proceedings against a controller or a processor will be brought before the courts of the Member State where the controller or processor has an establishment. Alternatively, such proceedings may be brought before the courts of the Member State where the data subject has his or her habitual residence, unless the controller or processor is a public authority of a Member State acting in the exercise of its public powers.

Detail data management approach in the project, including:

- Record of processing activities (art. 13, 14 GDPR)

Where personal data relating to a data subject are collected from the data subject, the controller will, at the time when personal data are obtained, provide the data subject with all of the following information. In addition, the controller will, at the time when personal data are obtained, provide the data subject with the following further information necessary to ensure fair and transparent processing. Where personal data have not been obtained from the data subject, the controller shall provide the data subject with the requested information.

- Data security technical and organizational measures (art. 29, 32-2 GDPR)

The processor and any person acting under the authority of the controller or of the processor, who has access to personal data, shall not process those data except on instructions from the controller, unless required to do so by Union or Member State law.

In assessing the appropriate level of security account will be taken in particular of the risks that are presented by processing, in particular from accidental or unlawful destruction, loss, alteration, unauthorised disclosure of, or access to personal data transmitted, stored or otherwise processed.

- Personal data breach notification (art. 33 GDPR)

In the case of a personal data breach, the controller will without undue delay and, where feasible, not later than 72 hours after having become aware of it, notify the personal data breach to the supervisory authority competent in accordance with Article 55 GDPR, unless the personal data breach is unlikely to result in a risk to the rights and freedoms of natural persons. Where the notification to the supervisory authority is not made within 72 hours, it shall be accompanied by reasons for the delay.

- Transfer of data.

The data transfer between controller and data object will be performed on the request and according to the described above procedure.

5.2 Security

Please indicate if your project will involve:

- activities or results raising security issues: (YES/NO)
- 'EU-classified information' as background or results: (YES/NO)